

APPENDIX E - DEVELOPMENT ASSURANCE CONTIGUOUS EXAMPLE

NOTE: The main body of this document contains information which places the information in this appendix in context. This appendix should be used in conjunction with the main body of the document.

TABLE OF CONTENTS

E.1	INTRODUCTION.....	104
E.2	DOCUMENT GUIDANCE	106
E.3	AIRPLANE DEVELOPMENT PROCESS	106
E.3.1	Airplane Introduction	106
E.3.2	Airplane Planning Documents.....	106
E.3.3	Airplane Concept.....	108
E.3.4	Airplane Functions	110
E.3.5	Airplane Functional Decomposition	110
E.3.6	Aircraft Functional Hazard Assessment (AFHA) Interactions	111
E.3.7	Initial Airplane Architecture Development and Function Allocation	112
E.3.8	Airplane Requirements Development	114
E.3.9	Preliminary System-Level Architecture Development for “Decelerate on Ground” Function	116
E.3.10	Preliminary Aircraft Safety Assessment Interactions	117
E.3.11	Follow-On Airplane-Level Architecture Development	119
E.3.12	Airplane Requirements Validation and Assumption Management	120
E.4	WHEEL BRAKE SYSTEM DEVELOPMENT PROCESS	124
E.4.1	Wheel Brake System Introduction.....	124
E.4.2	Wheel Brake System Planning	124
E.4.3	Wheel Brake System Description	124
E.4.4	Wheel Brake System Functions	126
E.4.5	System Functional Hazard Assessment	127
E.4.6	Wheel Brake System Requirements	128
E.4.7	Wheel Brake System Requirements Validation and Assumption Management	152
E.5	WHEEL BRAKE SYSTEM (WBS) IMPLEMENTATION AND VERIFICATION	159
E.5.1	Wheel Brake System Implementation.....	159
E.5.2	Brake System Control Unit Verification.....	159
E.5.3	Wheel Brake System Verification.....	162
E.5.4	S18 Airplane Verification.....	164
E.6	WHEEL BRAKE SYSTEM CONFIGURATION MANAGEMENT	167
E.6.1	Configuration Identification and Baseline Establishment.....	167
E.6.2	Change Control and Problem Reporting.....	169
E.7	PROCESS ASSURANCE	169
E.7.1	S18 Process Assurance Section of the Development Assurance Plan	170
E.7.2	Process Assurance Audit of Configuration Management Activities	170
E.7.3	Configuration Management Audit Results	171
Figure E1	ARP4754B Appendix E and ARP4761A Appendix Q example flow	105
Figure E2	Airplane concept drawing.....	109
Figure E3	Decomposition of Decelerate on Ground function	111
Figure E4	Initial airplane-level Wheel Brake System architecture schematic	113
Figure E5	Airplane-level Wheel Brake System architecture schematic	120
Figure E6	WBS mode transitions.....	125
Figure E7	Decomposition of Wheel Brake System functions	127
Figure E8	High-level interface diagram	130
Figure E9	High-level Wheel Brake System architecture.....	132

Figure E9	High-level Wheel Brake System architecture (continued)	133
Figure E10	Wheel Brake System architecture trade study summary	135
Figure E11	WBS architecture 3 with BSCU architecture iteration 1	136
Figure E12	BSCU architecture update iteration: addition of monitor functions	146
Figure E13	BSCU functional mapping	148
Figure E14	BSCU requirements traceability	149
Figure E15	Example Problem Report during validation	153
Figure E16	BSCU System Integration Test procedure	161
Figure E17	Open BSCU verification Problem Report description (example)	162
Figure E18	Wheel Brake System integration test procedure form	164
Figure E19	Airplane-level test procedure form	166
Figure E20	Wheel Brake System engineering change notice	169
Figure E21	Requirements change management	170
Table E1	Example Development Assurance Plan table of contents	107
Table E2	AFHA assumptions	112
Table E3	Airplane function allocation (excerpt)	114
Table E4	Initial S18 Airplane Requirements Specification (excerpt)	115
Table E5	FDAL assignment (PASA outputs)	117
Table E6	Proposed safety requirements (PASA output)	118
Table E7	Proposed safety requirements (PRA output)	119
Table E8	Airplane Requirements Validation Matrix (sample)	121
Table E9	Airplane-level safety assumptions	123
Table E10	SFHA assumptions	128
Table E11	Initial Wheel Brake System requirements WBS-0001-103 - Version 1.0 (excerpt)	128
Table E12	Proposed independence requirements (WBS PSSA outputs)	137
Table E13	Proposed safety requirements (WBS PSSA outputs)	137
Table E14	Safety assessment assumptions (WBS PSSA outputs)	137
Table E15	Wheel Brake System supplier derived requirements	138
Table E16	WBS requirements for dual channel BSCU and two hydraulic systems architecture	138
Table E17	Wheel Brake System interface requirements	140
Table E18	Wheel Brake System intra-system interface requirements WBS-0001-103 - Version 4.0	141
Table E19	Wheel Brake System Requirements Specification WBS-0001-103 - Version 5.0	142
Table E20	Wheel Brake System requirements allocated to BSCU development	145
Table E21	IDAL assignments (BSCU PSSA outputs)	149
Table E22	Proposed safety requirements (BSCU PSSA outputs)	150
Table E23	BSCU PSSA assumptions communicated to WBS level	150
Table E24	Brake System Control Unit Requirements Specification BSCU-0001-100 - Version 1.0	150
Table E25	Wheel Brake System requirement(s) correctness check (excerpt)	152
Table E26	Wheel Brake System requirements set completeness check (excerpt)	154
Table E27	Wheel Brake System Validation Matrix (excerpt)	156
Table E28	WBS level safety assumptions	157
Table E29	Brake System Control Unit Verification Matrix (excerpt)	159
Table E30	Wheel Brake System Verification Matrix (excerpt)	162
Table E31	S18 Airplane Verification Matrix (excerpt)	165
Table E32	Wheel Brake System Configuration Index WBS-0001-001 - Version 1.0	167
Table E33	BSCU COM Configuration Index - Version 4.0	168

E.1 INTRODUCTION

This appendix provides a detailed example of the aircraft and systems development for a function of a hypothetical S18 airplane. Because the aircraft selected for this example is an airplane, “S18 airplane” is the terminology used.

This appendix describes, in detail, a contiguous example of the design development process for the S18 airplane Wheel Brake System (WBS), with interactive exchanges including handoffs to and from the safety analysis contiguous example in ARP4761A/ED-135 Appendix Q. The goal is to show the interactive relationship between ARP4754B/ED-79B and ARP4761A/ED-135. In order to present a clear picture, some of the airplane-level tasks are provided to better understand potential interactions that may drive requirements down to the WBS level. The function “Decelerate on Ground” is broken down into a single system and its items. This function is chosen because it has sufficient complexity to allow use of all the methodologies, yet is simple enough to present a clear picture of the process flow.

The methodologies applied here are an example of one way to utilize the principles defined in the main body. Other formats may be used to accomplish the documentation, as long as the principles outlined in the main body are followed.

It should be stressed here that the content of this appendix is an illustrative and informative example, and is not meant to be comprehensive in its discussion of the development activity that a program to develop a WBS would entail, or fully represent the outputs that could be created, either in scope or content. This example is not intended to establish any additional recommended practices beyond those described in the main body of the document.

This example contains references to documentation that a company may use to assure itself of the safety of its products, but does not include the documentation that the Original Equipment Manufacturer (OEM) would be required to submit at the airplane level for airplane certification. Some of these documents are submitted to the Certification Authorities for the purpose of certification (e.g., the WBS Functional Hazard Assessment (FHA)). Other documents are internal to the company and not required to be submitted for certification. No implication is made that these documents should be submitted to a Certification Authority, although all documents should be available for submission if requested by the Certification Authority. Safety and certification are not synonymous terms. The example shows the systems engineering process as applied to the development of an airplane, including some processes that are beyond certification requirements. This development process in itself does not constitute a comprehensive process for guaranteeing certification but provides a significant input to the overall certification process success.

In a real development program, the development process is usually far more complex. For example, in a real development program, development of the airplane, airplane systems and items often occurs concurrently, rather than serially as depicted in example flow.

In a real development program, the safety process occurs concurrently with the development processes, constantly receiving inputs from and providing feedback to development processes.

[Figure E1](#) depicts the interaction of activities within this example and in ARP4761A/ED-135 Appendix Q. The blue-shaded rectangles represent the development activities. The yellow-shaded octagons represent the safety activities described in ARP4761A/ED-135 Appendix Q. This figure provides a guide to the structure of the examples and should allow the reader to quickly find specific areas within the example using the cross references. Where iterations are indicated, the directionality of these interactions is not shown; the nature of the interactions in these cases is described in the main body of this document and/or in this example.

In addition to the activities shown in [Figure E1](#), there are two additional integral processes of aircraft and systems development described in this example: configuration management (Section [E.6](#)) and process assurance (Section [E.7](#)).



E.2 DOCUMENT GUIDANCE

This section is intended for guidance to be used throughout the example. This section includes definitions, explanation of sections, and thoughts from the authors.

Editor's Notes: Editor's Notes are enclosed in parentheses and provided in italics. Where necessary, the reader is directed to the appropriate section of ARP4754B/ED-79B or ARP4761A/ED-135 for further information on the method or process involved. Editor's Notes may also provide clarification to the reader, e.g., when the scope of the example is limited for brevity.

The phrase "main body": When the reader sees the phrase "main body," it implies the main body of ARP4754B/ED-79B and the section called out in reference to the example given.

Blue table borders: The use of blue borders indicates information that was received from ARP4761A/ED-135 Appendix Q. The system development process and safety assessment process are concurrent processes with data interfaces. This system development process example contains a number of data inputs (documents, tables, etc.) that are produced by following the safety assessment process in ARP4761A/ED-135 and documented in ARP4761A/ED-135 Appendix Q. Inputs from the safety assessment process are enclosed in boxes to distinguish the data inputs from explanatory text. An example of such a data input is:

Title	
Object Identifier	Object Content

E.3 AIRPLANE DEVELOPMENT PROCESS

E.3.1 Airplane Introduction

The airplane development process included the development of the airplane-level functions, requirements, and architecture for the S18 airplane. The purpose was to develop a list of airplane requirements of all classes as shown in main body [5.3.1](#), determine a proposed airplane design that can reasonably satisfy the requirements, and develop the lower-level requirements to be considered in the design of the airplane systems, lower-level systems, and items.

E.3.2 Airplane Planning Documents

The airplane Development Assurance Plan defined the development assurance process activities including integral processes of the airplane development life cycle. Also included were the life cycle environment, including the methods and tools to be used for the activities of each life cycle process, and the development standards. The plan was communicated to all S18 program stakeholders.

(Editor's Note: Early coordination with the Certification Authorities for development assurance planning as described in main body [3.3](#) has not been included in this example. This example does not include any planning documents.)

An example Development Assurance Plan table of contents is included in [Table E1](#). This table is an example of how all planning elements from the main body can be included in a single plan. In some cases, similar content is combined into a single plan section, such as key individual responsibilities, with references to multiple main body sections which call for responsibilities to be defined.

The left-hand side of the table provides a notional plan's section numbers and headers. The right-hand side of the table represents where the reader can find relevant information in the main body. The section numbers have no correlation to the example in this appendix.

Table E1 - Example Development Assurance Plan table of contents

Plan Table of Contents		ARP4754B Reference Section
1	Introduction	
1.1	Scope	
1.2	Applicability	
1.5	Definitions and Acronyms	
2	Applicable Documents	
3	Development Assurance Plan	Section 3
3.1	Development Assurance	1.2
3.1.1	Scope of Development Activity	Section 4, 4.1, 4.2, 4.3, 4.4, 4.5, 4.6, 4.7
3.1.2	Plan Tailoring Rationale	3.3(a.1)
3.1.3	Design Description	3.2.1
3.1.4	Organizational Structure with Roles and Responsibilities, Including Independence	3.2(d), 3.3(c.6), 5.1.7(c), 5.4.7.1(g), 5.5.4(a), 5.6.2.1, 5.7
3.1.5	Complexity Management (Aircraft- or System-Level)	3.2, 4.1.1(a), 4.1.2
3.1.6	Deviations from Plan	3.2.3, 4.7, 5.4.7.3, 5.5.6.4, 5.7.2, 5.7.3.2
3.1.7	Top-Level Processes and Methods	3.1, 3.2, 4.1, 5.1, 5.4.1, 5.4.2, 5.4.6, 5.5.1, 5.5.2, 5.5.5, 5.6.2
3.1.7.1	Architecture Development	4.3, 4.5, 4.6.1.1, 4.6.1.2(b)
3.1.7.2	Integration	4.6.3
3.1.7.3	Implementation	4.2, 4.6, 5.5.3
3.1.7.4	Relationships between Integral Processes	Section 3, 4.1
3.1.7.5	Interactions between Requirement, Design, Safety Activities	4.1.3
3.1.8	Plan to Re-Use Processes, Methods, or Data	3.3(c.4), 5.2.3.3.2, 5.3.1.11
3.1.9	Rationale and Plan to Address “Fully Analyzable and Fully Testable” Systems	3.3(c.5), 4.6.4, 5.5.4, 5.5.5
3.1.10	Key Development Cycle Milestones and Events	3, 3.2(c), 5.1.7
3.1.11	Transition Criteria	3.2.2
3.1.12	Development Assurance Process Outputs	3.3, 4.2, 4.5, 4.6.2, 4.7, 5.1.7, 5.4.7, 5.5.6
3.2	Safety Process	
3.2.1	Scope and Content of Safety Activities (Aircraft- or System-Level)	5.1
3.2.2	Input Requirements for Safety Design and Analysis	5.1, 5.1.7(a)
3.2.3	Applicable Safety Standards	5.1.7(b)
3.2.4	Safety Organization Relationship with Partners or Suppliers	3.3(c.6), 5.1.7(c)
3.2.5	Safety Assessment Activities and Outputs	5.1.7(d)
3.2.6	Validation and Verification of Safety Requirements	5.1.7(f)
3.2.7	Safety Process Interaction with Development Assurance Planning Elements	5.1.7(g)
3.3	Requirements Management	3.3 (c.2)
3.3.1	Requirements Capture	3.1, 5.3
3.3.2	Requirements Management	3.3(c.2)
3.3.3	Management of Safety Requirements	3.3(c.2), 4.1.3, 5.1.2, 5.1.4, 5.1.7(f), 5.3.1.1, 5.3.2
3.3.4	Derived Requirement Evaluation for Safety Impact	5.3.1.10, 5.3.3
3.4	Requirement Validation Process	5.4.1, 5.4.2, 5.4.2.5, 5.4.5
3.4.1	Validation Methods	5.4.5, 5.4.6
3.4.2	Correctness and Completeness Checks	5.4.2.3, 5.4.3, 5.4.4
3.4.3	Management of Assumptions	5.4.2.4
3.4.4	Validation Matrix	5.4.2.5, 5.4.7.2
3.4.5	Validation Summary	5.4.2.5, 5.4.7.2
3.4.6	Validation of Requirement Changes	5.4.7.1(f)
3.5	Implementation Verification	5.5, 5.5.2, 5.5.3, 5.5.4, 5.5.5, 5.5.6
3.5.1	System or Item Configuration to be Verified	5.5.4(b)
3.5.2	Verification Methods	5.5.4(c), 5.5.5, 5.5.6.1(c)

Plan Table of Contents	ARP4754B Reference Section
3.5.3 Verification Success Criteria	5.5.4(d)
3.5.4 Verification Credit from Hardware or Software Activities	5.5.4(e), 5.5.4(f), 5.5.6.1(g)
3.5.5 Verification Activities	5.5.4(g)
3.5.6 Verification Procedures	5.5.3, 5.5.5.3, 5.5.6.3(d)
3.5.7 Verification Data	5.5.6, 5.5.6.1, 5.5.6.2, 5.5.6.3, 5.5.6.4
3.5.8 Sequence of Dependent Verification Activities	5.5.4(g), 5.5.6.1(e)
3.5.9 Verification Test Environment	5.5.4(i), 5.5.5.3.1, 5.5.6.1(h)
3.6 Configuration Management Process	5.6
3.6.1 Configuration Management Methods	5.6.2.1
3.6.2 Configuration Management Environment: Procedures, Tools, Methods, Standards	5.6.2.1
3.6.3 Configuration Identification	5.6.2.2, 5.6.3.1
3.6.4 Configuration Baseline Establishment	5.6.2.3
3.6.5 Change Control and Problem Reporting	5.6.2.4
3.6.6 Archival and Retrieval	5.4.7.1(d), 5.6.2.5
3.6.7 Configuration Data	5.6.3, 5.6.4
3.7 Process Assurance Process	5.7.2
3.7.1 Project Plan Reviews	5.3.7.1
3.7.2 Activity, Data, and Report Reviews	5.7.3.2
3.7.3 Process Assurance Data	5.7.4

E.3.3 Airplane Concept

The S18 airplane was developed by finalizing different high-level airplane operational profiles. The airplane operational profiles were developed from the marketing and business decisions made during these initial marketing efforts, such as independent market surveys, customer requests, and initial design and trade studies. The airplane concept development was the finalization of the different high-level airplane operational profiles, some of which include:

- High-level airplane requirements (e.g., payload, range)
- Tradeoff between an airplane that flies faster versus an airplane that consumes less fuel
- Tradeoff between an airplane that predominately supports hub-and-spoke versus point-to-point operations

The S18 airplane program operating characteristics were established based on customer interface during the initial market research. The airplane concept of operations led to the airplane mission specification, which then led to preliminary airplane sizing and preliminary airplane design. This information was used to create an operational profile for the airplane and was flowed from the airplane level to all systems for use in safety assessments, including the single event effects analysis.

The operational profile for the S18 airplane includes:

- Two engine passenger airplane for 300 to 350 passengers
- Maximum range of 5000 nautical miles at 0.86 Mach
- Altitude ceiling of 41000 feet
- An average flight duration of 5 hours
- An airplane life of 100000 flight hours
- A power on time of 100 operating hours

A concept drawing developed for the S18 airplane is shown in [Figure E2](#).

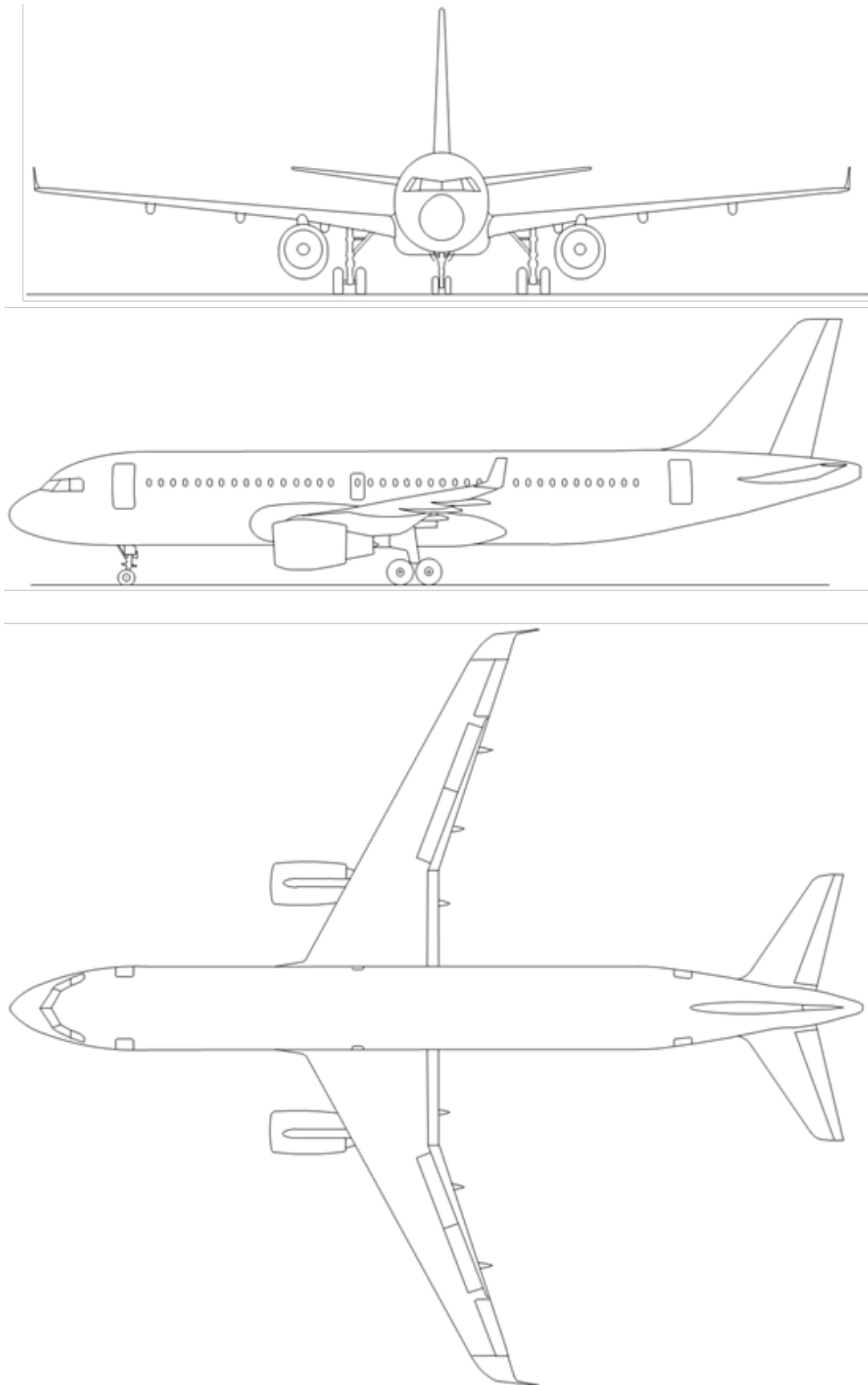


Figure E2 - Airplane concept drawing

Once the airplane concept was firmed up, the development of its functions, architecture, functions allocation, and requirements was initiated, with the process conducted in an almost concurrent, iterative sequence.

E.3.4 Airplane Functions

The airplane functions for the S18 airplane program were initially developed based on prior airplane program history, marketing specifications, and certification requirements for the intended operation of the airplane.

The S18 airplane was designed to perform the following functions:

- a. Provide Aerodynamic Performance
- b. Control Airplane Trajectory
- c. Control Airplane Energy
 - 1. Maintain or Increase Airplane Energy
 - 2. Reduce Airplane Energy
 - i. Provide Controlled Aerodynamic Drag
 - ii. **Decelerate on Ground**
 - 3. Provide High Lift Capability
- d. Provide Survivable Environment
- e. Provide Crew Situational Awareness
- f. Maintain Structural Integrity
- g. Provide Emergency Services
- h. Provide Passenger/Cargo Services

(Editor's Note: "Decelerate on Ground" function is in bold text because it is the focus of the example.)

(Editor's Note: The airplane function list shows that there is typically a large context and hierarchy of functions provided to other processes. This example provides enough functional definition to support the system development and safety assessment process examples, but is not a complete functional definition for the S18 airplane.)

E.3.5 Airplane Functional Decomposition

The S18 airplane-level functions as shown in [E.3.4](#) were further defined.

(Editor's Note: To the maximum extent possible, the functions do not imply a specific design/implementation. For example, there are several different design solutions which could be used to decelerate on the ground. The determination of the allocation of functions to systems was completed as part of a trade study not included in this example.)

The S18 airplane-level function "Decelerate on Ground" was decomposed to the system level as shown in [Figure E3](#).

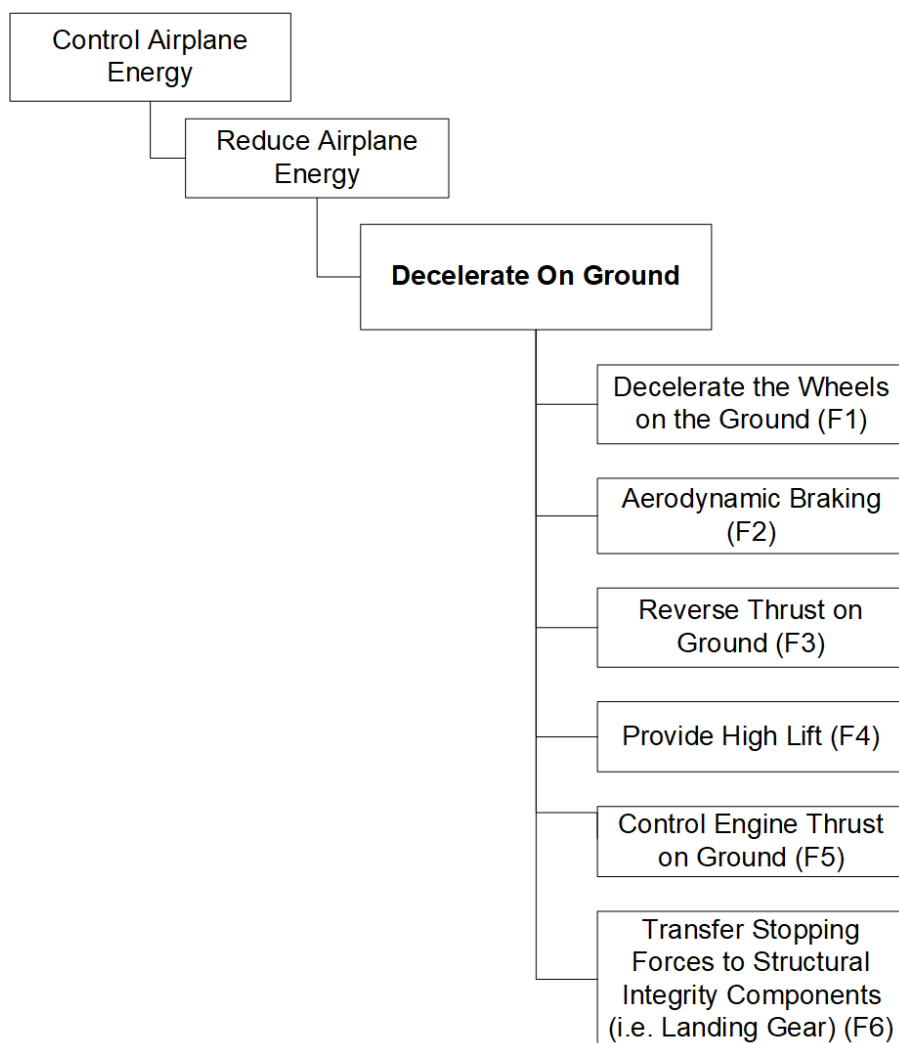


Figure E3 - Decomposition of Decelerate on Ground function

When used, Fx notation represents the identification of a function that is considered to implement the aircraft function “Decelerate on Ground.”

E.3.6 Aircraft Functional Hazard Assessment (AFHA) Interactions

Using the airplane operational profile from [E.3.3](#) and functional failures for each of the airplane functions identified in [E.3.4](#), the AFHA determined airplane-level hazards as part of the safety assessment process.

The AFHA provided hazards associated with the function “Decelerate on Ground.” The failure condition severity classification assigned in the AFHA (e.g., Major, Hazardous, Catastrophic), along with conditions assigned that moderate the hazard (e.g., runway departure speeds), was a factor in determining the airplane architecture required for the “Decelerate on Ground” function.

The AFHA hazards for the “Decelerate on Ground” function are found in ARP4761A/ED-135 Appendix Q.

(Editor’s Note: The AFHA activity was conducted in accordance with the guidance of ARP4761A/ED-135 Appendix A. Although this example focuses on the “Decelerate on Ground” function, other airplane functions might have been allocated to the WBS, such as “Control Direction on Ground.”)

E.3.6.1 AFHA Assumptions

Assumptions made while accomplishing the effect evaluation of each failure condition have been captured and numerically identified for reference. [Table E2](#) presents the assumptions analysis made during the development of this functional hazard assessment. These assumptions were confirmed through the development process and the confirmation communicated back to the AFHA process.

Table E2 - AFHA assumptions

Assumption Identifier	Description
ASMP 3.2.2-1	Overrunning the runway length above "XYZ" knots is considered a high speed overrun.
ASMP 3.2.2-2	The directional aspect of asymmetric failures of deceleration systems are functionally addressed by failure conditions of the "Control direction on ground" airplane function (see function 2.3 AFHA).
ASMP 3.2.2-3	The flight crew will divert to a suitable airfield if aware of a condition that renders the airplane incapable of landing at the originally intended destination.
ASMP 3.2.2-4	Crew awareness is not a factor for the identified malfunction, as these are immediately evident due to aircraft behavior and do not have their effects intensified or mitigated by crew awareness features.
ASMP 3.2.2-5	The flight crew will not initiate a Rejected Takeoff (RTO) in response to an annunciated failure of deceleration features during takeoff due to alert suppression.
ASMP 3.2.2-6	Taxi is performed at groundspeeds below 30 knots.
ASMP 3.2.2-7	Landings with failure condition in combination with environmental factors have been assessed.
ASMP 3.2.2-8	Failures of deceleration capability will be detected and annunciated by on-board systems.

E.3.7 Initial Airplane Architecture Development and Function Allocation

After airplane function decomposition in [Figure E3](#), the airplane architecture development was initiated with supporting systems identified, and airplane sub-functions were allocated concurrently to the airplane's supporting systems.

E.3.7.1 Initial Airplane Architecture

Airplane architectures were developed for each airplane function, given airplane requirements and AFHA categorizations that were developed. The airplane architecture development determined resulting airplane systems included on the airplane, and was an iterative process with the airplane function allocation process and the Preliminary Aircraft Safety Assessment (PASA) process.

Review of the S18 airplane architecture showed that the "Decelerate on Ground" function was intended to be accomplished using two independent systems:

- Wheel brakes. The S18 airplane has two main landing gear struts with four wheels each for a total of eight wheels. Each wheel is equipped with a brake.
- Thrust reversers. The S18 airplane is equipped with a thrust reverser on each engine. The thrust reversers are intended to aid deceleration, especially in conditions where friction-based deceleration is ineffective (wet or iced runways).

The following airplane features might have affected the "Decelerate on Ground" function:

- Engines. The S18 airplane has two under-wing mounted turbofan engines in order to produce forward thrust. The engine thrust is reduced during deceleration on ground, to maximize deceleration and allow deployment of the thrust reversers.
- Flaps. The S18 airplane's wings are each equipped with two flap panels. The flaps are extended to increase the wing's lift and drag coefficients. The flaps are extended to allow lower takeoff and landing speeds, which facilitates deceleration on ground.
- Spoilers. The S18 airplane's wings are each equipped with two spoiler panels. The spoilers are intended to be deployed on landing, to reduce lift and increase the effectiveness of the wheel brakes.

The initial airplane-level architecture of the WBS, as shown in [Figure E4](#), was provided to the PASA process.

(Editor's Note: Interfaces to the hydraulics systems and the Electric Brake Unit (EBU) are shown, but detailed development and safety analysis are outside of the scope of this example.)

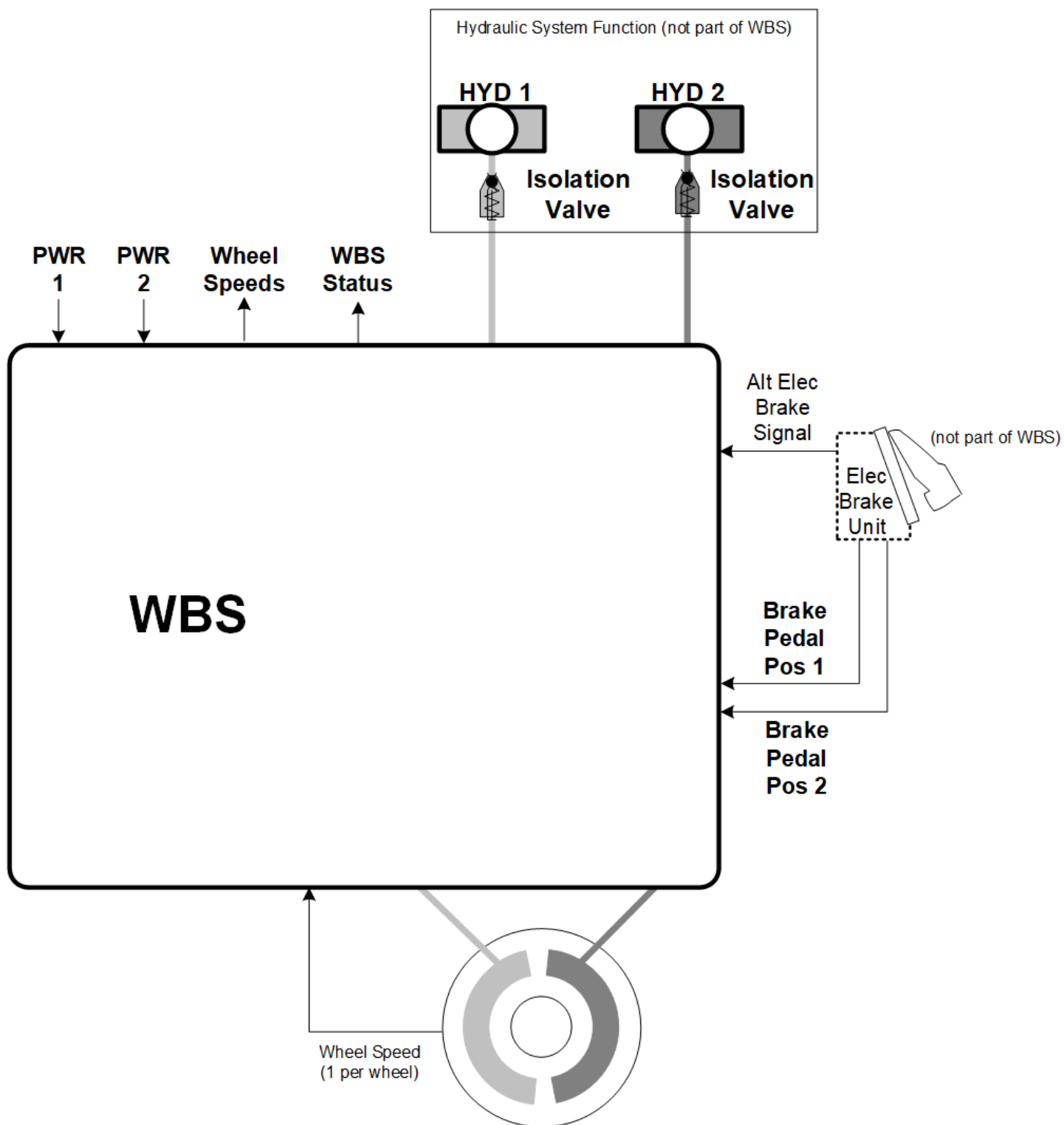


Figure E4 - Initial airplane-level Wheel Brake System architecture schematic

E.3.7.2 Airplane Function Allocation

The S18 airplane-level functions as shown in [E.3.4](#) were defined during the airplane architecture development process. The next process step was to allocate functions to the supporting systems.

The S18 airplane-level function allocation for “Decelerate on Ground” is shown in [Table E3](#). The initial WBS requirements, presented later in [E.4.5](#), were outputs of the allocation activity, along with the functional requirements for the other systems’ functions shown in [Table E3](#). [Table E3](#) content assumes some systems were present on the S18 airplane, as shown in the top row (Wheel Brake System, thrust reverser, etc.). These systems were identified from the development of the initial high-level architectures. This allocation was an important input into the PASA. The purpose of the function allocation was to identify the system(s) which implement the respective function(s).

(Editor’s Note: Function allocation was a recursive, iterative process. After airplane functions were allocated to systems by the design process, each system which integrates multiple airplane functions was re-examined using the System Functional Hazard Assessment (SFHA) process.)

Table E3 - Airplane function allocation (excerpt)

Function	S18 Airplane	Wheel Brake System	Thrust Reverser	Spoilers	Flaps	Engine Controls	Structural Integrity (Landing Gear, Fuselage, etc.)
Decelerate on Ground	X	X	X	X	X	X	X
Decelerate the Wheels on the Ground		X					
Reverse Thrust on Ground			X			X	
Provide Aerodynamic Braking				X			
Provide High Lift					X		
Control Engine Thrust on Ground						X	
Transfer Stopping Forces to Structural Integrity Components				X			X

E.3.8 Airplane Requirements Development

The initial development of airplane requirements included market/technology research and certification requirements. These requirements established preliminary requirement sets. Usage of “know how” from previous programs expanded this preliminary requirement set to a degree sufficient for initial architecture trade studies.

(Editor’s note: For the sake of brevity, not all airplane requirements are discussed.)

E.3.8.1 Sizing Requirements Derived from Airplane Runway Operating Requirements

The S18 airplane and its systems were “sized” for the intended mission as defined by market requirements. The intended mission included requirements to operate from specific runway types and lengths. In addition, regulations 14 CFR/CS 25.125(c) and 14 CFR/CS 25.735 specified requirements applicable to this function. Factors that translated the function “Decelerate on Ground” into quantified requirements for the WBS included the following:

- Airplane landing weight
- Airplane takeoff weight
- Takeoff and approach speeds
- Flight characteristics of the airplane
- Specified airplane operating environmental conditions

- Assumed failure conditions, including those specified by certification requirement 14 CFR/CS 25.735

The sizing requirements for the WBS was developed as a function of required energy absorption, and deceleration requirements.

E.3.8.2 Airplane Requirements Specification

[Table E4](#) is an excerpt from the initial airplane requirements specification that showed a subset of the requirements which were used to trace to WBS requirements. Additional airplane requirements might have been obtained from other sources, such as regulatory material. It should be noted that those requirements required to comply with 14 CFR/CS 25.735 and sub-sections were further decomposed once allocated to the WBS.

Table E4 - Initial S18 Airplane Requirements Specification (excerpt)

Requirement Number	Requirement	Source	Additional Information
S18-ACFT-R-1000	The S18 airplane shall have a means to decelerate on ground.	Airplane concept	Must comply with 14 CFR/CS 25.735.
S18-ACFT-R-1100	The S18 airplane shall have a means to decelerate the wheels on ground.	Airplane concept	Must comply with 14 CFR/CS 25.735.
S18-ACFT-R-1110	The S18 airplane shall have pilot-controlled wheel braking capability.	Airplane concept	Conventional pedal based controls scheme was confirmed by airframe design team in coordination meeting ABC123. Must comply with 14 CFR/CS 25.735.
S18-ACFT-R-1120	The S18 airplane shall have an autobraking capability during landing and RTO.	Business case trade study	Business case trade study established the need for technological improvements in CAT IIIb auto-landing capability. (Report MRS18-XXX) Must comply with 14 CFR/CS 25.735(c)(2).
S18-ACFT-R-1130	The S18 airplane shall have anti-skid braking.	Business case trade study	Business case trade study established the need for all weather operation and stability of the airplane during runway runs. (Report MRS18-XXX). Must comply with 14 CFR/CS 25.735 (e).
S18-ACFT-R-1140	The S18 airplane shall provide interface for Wheel Brake System status and annunciations.	Business case trade study	Business case trade study TS18-XXXX determined the need to provide WBS information to the crew.
S18-ACFT-R-0181	The S18 airplane shall decelerate the wheels on gear retraction.	Business case trade study	Business case trade study TS18-XXXX determined the need to prevent high speed wheel rotation after the gear are stowed.
S18-ACFT-R-0182	The S18 airplane shall be capable of decelerating the wheels differentially.	Business case trade study	Business case trade study TS18-XXXX determined that nose gear alone are not sufficient to steer the aircraft under all environmental and operational conditions.
S18-ACFT-R-0183	The S18 airplane shall have a parking brake to prevent airplane motion on the ground.	Airplane concept	The regulation defines the need for parking brake control and defines performance requirements. Must comply with 14 CFR/CS 25.735(d).
S18-ACFT-R-0184	The S18 airplane shall have hydraulically driven braking.	Business case trade study	Business case trade study TS18-XXXX determined that the hydraulic drive of WBS is more economically feasible than electrical systems given the reuse of hydraulic systems from previous S18 airplane. Once the mechanism has been determined that will power the wheel brake, then that power system must be compliant with 14 CFR/CS 25.735 (b).

Requirement Number	Requirement	Source	Additional Information
S18-ACFT-R-0185	The S18 airplane shall have an autobrake override that can be initiated by the flight crew.	14 CFR/CS 25.735(c)(2)	N/A
S18-ACFT-R-0186	The S18 airplane shall meet safety requirements while operating in an average atmospheric radiation environment per the IEC 62396 standard with an altitude of 40000 feet and a latitude of 45 degrees.	S18 airplane internal documentation	IEC 62396, referenced in AIR6219/ER-008, is the industry standard for analyzing single event effects. For the operational mission selected, the industry average for the radiation environment is appropriate.

E.3.9 Preliminary System-Level Architecture Development for “Decelerate on Ground” Function

Once the airplane-level function decomposition and allocation to systems (shown in [Figure E3](#) and [Table E3](#)) and architecture matured in development, development of system-level architecture was pursued further to fulfill airplane-level requirements. At the same time, common resource systems required for the airplane systems to perform their functions were identified and defined. System-level architecture and system-level common resources that supported the “Decelerate on Ground” function are described in [E.3.9.1](#) and [E.3.9.2](#), respectively.

E.3.9.1 Architecture for “Decelerate On Ground” Function

- a. **Wheel Brake System (WBS):** The WBS actuates all eight brakes on the main gear wheels. An EBU provides brake pedal position inputs to the WBS. The WBS is hydraulically actuated and powered by hydraulic system 1 (HYD 1) and hydraulic system 2 (HYD 2). The WBS is electrically controlled. The WBS uses redundant electrical buses such that no single electrical bus loss results in loss of the WBS. The WBS uses the ground detection information as an input. The WBS implements the function “Decelerate the Wheels on the Ground (F1).”
- b. **Ground Spoiler System:** The ground spoiler system actuates all four spoilers on the wings. The symmetric spoilers on the left and right wings are controlled in pairs. The two pairs of spoilers may be commanded symmetrically in response to pilot manual commands. There is no automatic spoiler command in the S18 airplane. There is no other method of controlling or actuating the spoilers. The ground spoiler system is hydraulically actuated and powered by HYD 1 and hydraulic system 3 (HYD 3). The ground spoiler system is electrically controlled by an Electronic Flight Control Unit (EFCU). The ground spoiler system uses redundant electrical buses such that no single electrical bus loss results in loss of any EFCU channel. The ground spoiler system uses the ground detection information and wheel speed data as inputs. The ground spoiler system implements the function “Aerodynamic Braking (F2).”
- c. **Thrust Reverser System:** The thrust reverser system controls and actuates the thrust reversing mechanisms on each engine. Each reversing mechanism is controlled independently in response to pilot manual commands. There is no automatic thrust reverser command in the S18 airplane. The thrust reverser system is hydraulically actuated and powered by HYD 1 and HYD 2. The thrust reverser system is electrically controlled by an Electronic Engine Control Unit (EECU). The thrust reverser system uses redundant electrical buses such that no single electrical bus loss results in loss of any EECU channel. The thrust reverser system uses the ground detection information as an input. The thrust reverser system implements the function “Reverse Thrust on Ground (F3).”
- d. **Flap System:** The flap system actuates the multiple flap surfaces on the wings. All flap surfaces are controlled simultaneously in response to pilot manual commands. There is no automatic flap command in the S18 airplane. There is no other method of controlling or actuating the flaps. The flap system is hydraulically actuated and powered by HYD 1 and HYD 3. The flap system is electrically controlled by the EFCU. The EFCU uses redundant electrical buses such that no single electrical bus loss results in loss of any EFCU channel. The flap system implements the function “Provide High Lift (F4).”
- e. **Propulsion System:** The propulsion system controls forward thrust on each engine in response to pilot manual commands. There is no automatic propulsion command in this S18 airplane. The propulsion system is electrically controlled through 2 EECUs. The left-hand engine EECU controls the left-hand engine and the right-hand engine EECU controls the right-hand engine. The propulsion system relies on ground detection information from other airplane systems as an input. The propulsion system implements the function “Control Engine Thrust on Ground (F5).”

E.3.9.2 Common Resources for “Decelerate On Ground” Function

- a. R1 Hydraulic System: The hydraulic system provides power to multiple airplane systems. There are three hydraulic systems on the S18 airplane. HYD 1 is driven by engine 1 (left hand). HYD 2 is driven by engine 2 (right hand). An additional hydraulic system, HYD 3, provides minimal flight control capability in case of the loss of all engines in flight; refer to 14 CFR/CS 25.671(d).
- b. R2 Electrical System: The electrical system provides power to multiple airplane systems. There are three major electrical buses distributing power to airplane systems. Each major electrical bus can be powered by one or more engine driven generators. The system architecture was conceived to provide power to multiple systems such that no single electrical system failure causes loss of any essential loads.
- c. R3 Ground Detection Information System: The ground detection information system provides information to multiple airplane systems. The in-air or on-ground status of the airplane is determined by detecting compression of the left and right main landing gear shock absorbers and its information is consolidated using both signals. Whenever the signals mismatch, an additional input from wheel speed is used to accommodate this failure. The air/ground system is electrically powered.

E.3.10 Preliminary Aircraft Safety Assessment Interactions

In parallel with the airplane development process, the PASAs were performed with the design configuration identified early in the airplane development. This was an iterative process between development and safety assessment, with the objective to define an airplane architecture that met the safety requirements.

E.3.10.1 Provide Airplane Information to PASA Process

The development process provided airplane functions, airplane requirements, airplane architecture, and airplane function allocation to the PASA process. The airplane functions, architecture, function allocation, and requirements are shown in [E.3.4](#), [E.3.7.1](#), [E.3.7.2](#), and [E.3.8](#), respectively.

E.3.10.2 Assignment of Function Development Assurance Level

The AFHA was conducted as part of the safety assessment process described in ARP4761A/ED-135 Appendix Q. The outputs of the AFHA process, which provided inputs to the PASA process, were the failure conditions, effects, classification, and safety objectives. Based on those safety objectives, the PASA process assigned the Function Development Assurance Levels (FDALs) as shown in [Table E5](#). The Development Assurance Plan included the means to develop functions as FDAL A and C. Further development assurance level details were expected from the Preliminary System Safety Assessment (PSSA).

When used, Fx notation represents the identification of a function, while [FFx] is a general function failure of Fx, and [FFx.y] is a detailed functional failure of function Fx. While used in the aircraft-level safety assessments, this notation is not otherwise carried forward into system development activities or data.

Table E5 - FDAL assignment (PASA outputs)

Function Development Assurance Level	
PASA-SR-01	[F1] Decelerate wheels function shall be developed FDAL A
PASA-SR-02	[F2] Ground spoiler function shall be developed FDAL C (as minimum)
PASA-SR-03	[F3] Thrust reverser function shall be developed FDAL C (as minimum)
PASA-SR-04	[F4] Flap function shall be developed FDAL C (as minimum)

(Editor's Note: All PASA FDAL assignments are shown in [Table E5](#) for consistency with ARP4761A/ED-135 Appendix Q. However, only the Decelerate Wheels FDAL assignment, PASA-SR-01, is discussed later in this appendix. The other FDAL assignments shown in [Table E5](#) apply to systems outside the focus of this example. PASA-SR-01 is addressed through the generalization of representing a single FDAL A function validated and allocated to the systems level for further decomposition. This is captured with an FDAL attribute for each requirement at the lower levels. [Table A1](#) provides the applicability of the objectives for each FDAL and process. Safety requirements were developed by the PSSA and required validation and verification commensurate to their failure condition.)

The Development Assurance Plan addressed the generation of evidence and data that supported the development of functions with different FDALs. Development activities were performed, and associated evidence and data was generated as defined in this plan. The Development Assurance Plan outlined the means used to perform, review, and approve development activities for varying FDAL levels. The Development Assurance Plan defined the appropriate process required for the FDAL for a particular function or item (see main body [3.2](#)).

(Editor's Note: FDALs do not imply particular random failure probabilities. Probabilities are allocated separately from the FDAL assignment process.)

E.3.10.3 Proposed Safety Requirements from PASA and Safety Methods

The proposed safety requirements in [Table E6](#) were inputs received from ARP4761A/ED-135 Appendix Q, Preliminary Aircraft Safety Assessment.

Table E6 - Proposed safety requirements (PASA output)

SR#	Proposed Safety Requirements
PASA-SR-05	[FF1.1] Complete loss of wheel brake shall be less than 1.0E-07 for a landing.
PASA-SR-06	[FF2.2] Symmetrical partial loss of ground spoiler shall be less than 1.0E-03 for a landing.
PASA-SR-07	[FF3.2] Loss of one thrust reverser shall be less than 1.0E-03 for a landing.
PASA-SR-08	[FF4.2] Symmetrical partial loss of flaps shall be less than 1.0E-03 for a landing.
PASA-SR-09	No single failure or event shall result in the complete loss of wheel brake and the symmetrical partial loss of ground spoiler
PASA-SR-10	No single failure or event shall result in the complete loss of wheel brake and the loss of one thrust reverser.
PASA-SR-11	No single failure or event shall result in the complete loss of wheel brake and the symmetrical partial loss of flap.
PASA-SR-12	Loss of power from both hydraulic subsystems powered by the engines shall not lead to complete loss of wheel braking.
PASA-SR-13	The Alternate/Emergency Brake System hydraulic equipment and piping shall be installed aft of the engine 1 UERF trajectory envelope.
PASA-SR-14	Two redundant control lanes shall be provided between the Electric Brake Unit and each of the two Alternate/Emergency Meter Valves.
PASA-SR-15	Two redundant control lanes defined in the proposed requirement [PASA-SR-14] shall use vertically separated routes in the portion of the fuselage crossing the UERF area so that no engine 1 UERF debris can affect both lanes together.
PASA-SR-16	At least one of the two redundant control lanes between the Electric Brake Unit and each of the two Alternate/Emergency Meter Valves shall allow control of the corresponding valve until complete stop of the airplane in case of loss of power from both engine driven generators.
PASA-SR-17	The control lane between the EBU and each of the two Alternate/Emergency Meter Valves allowing control of the corresponding valve until complete stop of the airplane in case of loss of power from both engine driven generators shall be routed in the cabin ceiling area in the portion of the fuselage crossing the UERF area.
PASA-SR-18	Flailing shaft of the flap hydraulic motor moving parts shall not impact on brake pressure valve.

(Editor's Note: PASA proposed safety requirements are shown in [Table E6](#) for consistency with ARP4761A/ED-135 Appendix Q. However, only requirements PASA-SR-05, PASA-SR-10, PASA-SR-12, PASA-SR-13, and PASA-SR-14 are discussed later in this appendix. The other proposed safety requirements shown in [Table E6](#) apply to systems outside the focus of this example. UERF acronym stands for Uncontained Engine Rotor Failure. The acronym is left in [Table E6](#) because that is how it is received from safety as shown by the blue box.)

While most proposed safety requirements were expected to be generated by the PASA, safety analysis methods generated proposed safety requirements and provided them to the development process. The proposed safety requirement as shown in [Table E7](#) was received from the ARP4761A/ED-135 Appendix Q, Particular Risk Analysis (PRA).

Table E7 - Proposed safety requirements (PRA output)

Reference	Proposed Requirement	Rationale	Allocated to
PRA-UERF-FUEL-01-01	The fuel tank design shall include outer wing tanks which alone retain sufficient fuel reserves to ensure completion of the flight or a safe diversion, and are located outside the UERF trajectory envelope.	In case of large external fuel leak as a result of damage to fuel tanks caused by a UERF, the airplane's flying range may be affected to the point that the aircraft will not be able to reach its destination airport or any diversion runway, with potentially Catastrophic consequences.	Structure Design

(Editor's Note: This proposed safety requirement from the PRA is shown in this example to illustrate this source of safety requirements. However, the proposed safety requirement shown in [Table E7](#) is outside the focus of this example.)

E.3.11 Follow-On Airplane-Level Architecture Development

The PASA process provided the following architecture feedback:

The hydraulic common power source analysis described in ARP4761A/ED-135 Appendix Q pointed out that because thrust reversers and wheel brakes use only HYD 1 and HYD 2 power sources, the airplane does not satisfy the safety objective for loss of deceleration capability if the probability of loss of both HYD 1 and HYD 2 was not extremely improbable ($>1.0E-09$).

Furthermore, the PRA showed that the current airplane design was not sufficient to satisfy an independence principle because a single UERF event could cause loss of wheel brakes and thrust reversers caused by the simultaneous loss of both HYD 1 and HYD 2 power sources.

Therefore, PASA recommended an additional means of ensuring sufficient power for airplane deceleration other than using HYD 1 and HYD 2 powers only, according to PASA-SR-12. The development process captured this requirement in S18-ACFT-R-1550 and considered all possible solutions. Based on a Hydraulic-Electric Brake Study, the design team chose to incorporate a hydraulic accumulator (i.e., emergency accumulator) capable of providing sufficient energy to assure safe airplane deceleration by braking the wheels during all airplane operational modes including the HYD 1 and HYD 2 loss case (S18-ACFT-R-1551).

Furthermore, the PRA showed that the current airplane design was not sufficient to satisfy an independence principle because a single UERF event could cause loss of wheel brakes caused by the simultaneous loss of both NORMAL Mode and the single alternate brake signal between the EBU and the WBS.

Therefore, the PASA proposed a dual redundant command path from the EBU to the Alternate/Emergency Meter Valves, according to PASA-SR-14.

[Figure E5](#) reflects the design changes resulting from architecture feedback.

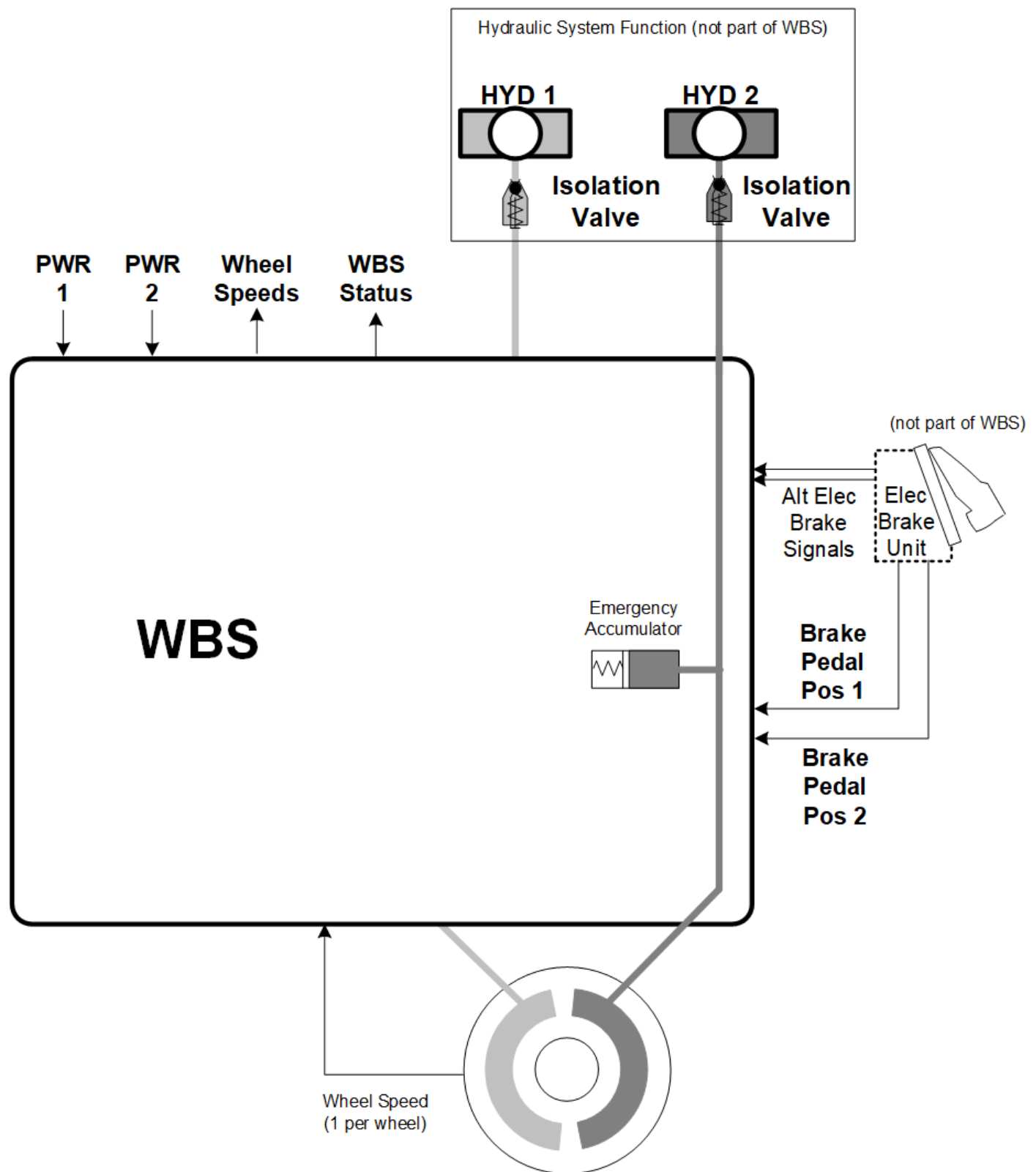


Figure E5 - Airplane-level Wheel Brake System architecture schematic

E.3.12 Airplane Requirements Validation and Assumption Management

The requirements validation and assumption management processes of main body [5.4](#) were continuous throughout the development life cycle.

E.3.12.1 Airplane Requirements Validation

At the airplane level, the main goal was to ensure that the set of requirements was complete and that each requirement was correct. A critical aspect was to ensure airplane-level requirements were both necessary and sufficient to meet the needs of operators, maintainers, and Certification Authorities, as well as airplane and system developers. A validation matrix, shown in [Table E8](#), was used to illustrate the outcome of the validation process. In this case, the three validation methods determined to be applicable were engineering review, analysis, and traceability. The review included evaluation of questions as described in main body [5.4](#).

(Editor's Note: For the sake of brevity, only a sample of requirements are shown in the validation matrix, [Table E8](#), and the actual evidence articles; e.g., the simulation methods, results, and the correctness and completeness checklist are not shown. In this example, the FDALs associated with functions/requirements are captured as separate requirements and/or documented as attributes as shown in the associated function column in [Table E8](#).)

Following the guidance of main body [5.4.7.2](#), [Table E8](#) was created to track the status of validating each requirement and the set of requirements as a whole including additional requirements resulting from architecting decisions and PASA activity. The categorization as a safety requirement resulted from the PASA activity. (See main body [4.1.3](#), [5.3.1.1](#), and [5.3.2](#) for discussion about identifying safety requirements through safety assessments.) The FDALs shown in [Table E8](#) came from the PASA as shown in [Table E5](#).

Table E8 - Airplane Requirements Validation Matrix (sample)

Requirement ID and Requirement	Safety Req	Source of Requirement	Associated Function (if Applicable) and the FDAL		Validation Method	Validation Evidence	Validation Conclusion
S18-ACFT-R-1000 The S18 airplane shall have a means to decelerate on ground.	No	Airplane concept	Decelerate on Ground	A	Engineering review	Correct & Completeness Checklist 1234	Valid
S18-ACFT-R-1100 The S18 airplane shall have a means to decelerate the wheels on ground.	No	Airplane concept	Decelerate the Wheels on Ground	A	Engineering review Traceability	Correct & Completeness Checklist 1234	Valid
S18-ACFT-R-1110 The S18 airplane shall have pilot-controlled wheel braking capability.	No	Airplane concept	Decelerate the Wheels on Ground	A	Engineering review	Correct & Completeness Checklist 1234	Valid
S18-ACFT-R-1120 The S18 airplane shall have an autobraking capability during landing and RTO.	No	Business case trade study	Decelerate the Wheels on Ground	A	Engineering review	Correct & Completeness Checklist 1234	Valid
S18-ACFT-R-1130 The S18 airplane shall have anti-skid braking.	No	Business case trade study	Decelerate the Wheels on Ground	A	Engineering review	Correct & Completeness Checklist 1234	Valid
S18-ACFT-R-1140 The S18 airplane shall provide interface for WBS status and annunciations.	No	Business case trade study	Decelerate the Wheels on Ground	A	Engineering review	Correct & Completeness Checklist 1234	Valid
S18-ACFT-R-0181 The S18 airplane shall decelerate the wheels on gear retraction.	No	Business case trade study	Decelerate the Wheels on Gear Retraction	A	Engineering review	Correct & Completeness Checklist 1234	Valid

Requirement ID and Requirement	Safety Req	Source of Requirement	Associated Function (if Applicable) and the FDAL		Validation Method	Validation Evidence	Validation Conclusion
S18-ACFT-R-0182 The S18 airplane shall be capable of decelerating the wheels differentially.	No	Business case trade study	Decelerate the Wheels Differentially for Directional Control	A	Engineering review	Correct & Completeness Checklist 1234	Valid
S18-ACFT-R-0183 The S18 airplane shall have a parking brake to prevent airplane motion when parked.	No	Airplane concept	Control Airplane Energy	A	Engineering review	Correct & Completeness Checklist 1234	Valid
S18-ACFT-R-0184 The S18 airplane shall have hydraulically driven braking.	No	Business case trade study	Decelerate the Wheels on Ground	A	Engineering review	Correct & Completeness Checklist 1234	Valid
S18-ACFT-R-0185 The S18 airplane shall have an autobrake override that can be initiated by the flight crew.	No	14 CFR/CS 25.735(c)(2)	Decelerate the Wheels on Ground	A	Engineering review Traceability	Correct & Completeness Checklist 1234	Valid
S18-ACFT-R-0186 The S18 airplane shall meet safety requirements while operating in an average atmospheric radiation environment per the IEC 62396 standard with an altitude of 40000 feet and a latitude of 45 degrees.	No	S18 airplane internal documentation	Protect against Single Event Effects (SEE)	A	Engineering review	Correct & Completeness Checklist 1234	Valid
S18-ACFT-R-0933 The Wheel Brake System decelerate the wheels on the ground function shall be developed as FDAL A.	Yes	PASA-SR-01	Decelerate the Wheels on Ground	A	Analysis	PASA	Valid
S18-ACFT-R-1322 No single failure or event shall result in the complete loss of wheel brake and the loss of one thrust reverser.	Yes	PASA-SR-10	Decelerate on Ground	A	Engineering review Traceability Analysis	Correct & Completeness Checklist 1234	Valid
S18-ACFT-R-1385 Complete loss of wheel braking shall be less than 1.0E-07 for a landing.	Yes	PASA-SR-05	Decelerate the Wheels on Ground	A	Engineering review Traceability Analysis	Correct & Completeness Checklist 1234	Valid
S18-ACFT-R-1550 Loss of power from both hydraulic subsystems powered by the engines shall not lead to complete loss of wheel braking.	Yes	PASA-SR-12	Decelerate on Ground	A	Engineering review Traceability Analysis	Validation Checklist 1234	Valid
S18-ACFT-R-1551 Wheel Brake System shall include an emergency accumulator to supply hydraulic power to the wheel brakes.	No	Hydraulic-electric brake study	Decelerate the Wheels on Ground	A	Engineering review	Correct & Completeness Checklist 1234	Valid

Requirement ID and Requirement	Safety Req	Source of Requirement	Associated Function (if Applicable) and the FDAL		Validation Method	Validation Evidence	Validation Conclusion
S18-ACFT-R-1552 The Alternate/Emergency Brake System hydraulic equipment and piping shall be installed aft of the engine 1 UERF trajectory envelope.	Yes	PASA-SR-13	Decelerate on Ground	A	Engineering review Traceability Analysis	Correct & Completeness Checklist 1234	Valid
S18-ACFT-R-1600 Two redundant control lanes shall be provided between the Electric Brake Unit (EBU) and each of the two Alternate/Emergency Meter Valves.	Yes	PASA-SR-14	Decelerate on Ground	A	Engineering review Traceability Analysis	Correct & Completeness Checklist 1234	Valid

(Editor's Note: S18-ACFT-R-0186 is included here to provide visibility of an aircraft-level overarching requirement that is later allocated to aircraft systems and items. This is the introduction of the function "Protect against Single Event Effects (SEE).")

E.3.12.2 Airplane Assumption Management

For confirmation tracking purposes, assumptions have been summarized in [Table E9](#). For tracking purposes, the assumptions received from the AFHA, as shown in [Table E2](#), have had identifier prefix "AFHA" added. PASA and PRA assumptions were identified as received from the PASA and PRA.

Table E9 - Airplane-level safety assumptions

Assumption Identifier	Description	Confirmation
AFHA ASMP 3.2.2-1	Overrunning the runway length above "XYZ" knots is considered a high speed overrun.	Coord Memo S18a-CM00Y1
AFHA ASMP 3.2.2-2	The directional aspect of asymmetric failures of deceleration systems are functionally addressed by failure conditions of the "Control direction on ground" airplane function. (See function 2.3 AFHA.)	Coord Memo S18a-CM00Y1
AFHA ASMP 3.2.2-3	The flight crew will divert to a suitable airfield if aware of a condition that renders the airplane incapable of landing at the originally intended destination.	Coord Memo S18a-CM00Y1
AFHA ASMP 3.2.2-4	Crew awareness is not a factor for the identified malfunction, as these are immediately evident due to aircraft behavior and do not have their effects intensified or mitigated by crew awareness features.	Coord Memo S18a-CM00Y1
AFHA ASMP 3.2.2-5	The flight crew will not initiate an RTO in response to an annunciated failure of deceleration features during takeoff due to alert suppression.	Coord Memo S18a-CM00Y1
AFHA ASMP 3.2.2-6	Taxi is performed at groundspeeds below 30 knots.	Coord Memo S18a-CM00Y1
AFHA ASMP 3.2.2-7	Landings with failure condition in combination with environmental factors have been assessed.	Coord Memo S18a-CM00Y1
AFHA ASMP 3.2.2-8	Failures of deceleration capability will be detected and annunciated by on-board systems.	Coord Memo S18a-CM00Y1
PASA-ASMP-01	It is assumed that the high speed overrun is above 30 knots and low speed overrun is below (or equal) 30 knots. This assumption has been derived from the AFHA assumption ASMP 3.2.2-1 and ASMP 3.2.2-6 for the establishment of the criteria and terms of "high speed overrun" and "low speed overrun."	Coord Memo S18a-CM00Y2
PASA-ASMP-02	The degraded state of systems considered in the CoFFE analysis has been defined such as half functional capability.	Coord Memo S18a-CM00Y2
PASA-ASMP-03	Wheel Brake System uses ground detection information together with the wheel speed information. These functions are developed independently so that even if Erroneous Ground Detection Information (AGS.MF) is provided on ground, i.e., with false in-flight status, wheel braking function is available if there is correct wheel speed information.	Coord Memo S18a-CM00Y2
PRA-UERF-ASSUMPTION-01	The BSCU is installed in the avionics compartment located in the nose fuselage section of the airplane, forward of the UERF area.	PRA
PRA-UERF-ASSUMPTION-02	The electrical lines transmitting the brake control signals from the Electric Brake Unit to the BSCU are routed "to the shortest" from one end to the other in the nose fuselage section of the airplane, thus are kept forward of the UERF area.	PRA

Assumption Identifier	Description	Confirmation
PRA-UERF-ASSUMPTION-03	The hydraulic reservoirs and the high pressure manifolds of the hydraulic subsystems 1 and 2 are located in the fuselage, aft of the UERF area.	PRA
PRA-UERF-ASSUMPTION-04	The hydraulic power distribution lines from the engine driven pumps to the respective hydraulic reservoirs and high pressure manifolds exit the engine pylons aft of the wing rear spars, then run along, aft of, the wing rear spars to the fuselage, and are kept aft of the UERF area inside the fuselage.	PRA
PRA-UERF-ASSUMPTION-05	The hydraulic distribution lines routed to the forward part of the fuselage to supply hydraulic equipment located forward of the UERF area are fitted with appropriate isolation means located aft of the UERF area.	PRA
PRA-UERF-ASSUMPTION-06	The power supply lines from the electrical power center(s) to the EBU are routed "to the shortest" from one end to the other in the nose fuselage section of the airplane.	PRA
PRA-UERF-ASSUMPTION-07	In case of cabin depressurization the airplane altitude will be limited by procedure to 10000 feet.	PRA

The assumptions as shown in [E.3.6.1](#) were confirmed through the development process and the confirmation was communicated back to the AFHA in Coord Memo S18a-CM00Y1 and to the PASA process in Coord Memo S18a-CM00Y2. The PRA assumptions were reviewed as part of the final design review for confirmation by the PRA.

E.4 WHEEL BRAKE SYSTEM DEVELOPMENT PROCESS

E.4.1 Wheel Brake System Introduction

This section describes, in detail, the system architecture development for the WBS. The WBS is developed to implement the "Decelerate the Wheels on the Ground" sub-function of the "Decelerate on Ground" airplane-level function (see [E.3.8](#)).

The purpose of this section is to illustrate that development of a system architecture involves taking system functions, requirements, results of the safety assessment analyses, interfacing systems, and other activities into account.

E.4.2 Wheel Brake System Planning

At the systems level, the WBS Development Assurance Plan was the top-level planning document which is similar to the airplane Development Assurance Plan (see [E.3.2](#)).

E.4.3 Wheel Brake System Description

The S18 WBS is comprised of the brake pedal control system, brake hydraulic system, anti-skid and autobrake systems, brakes/wheels/tires, brake temperature monitor system, brake cooling fans, tire pressure indication system, and tire and brake monitoring system.

(Editor's Note: For brevity, the example will only show certain aspects of the systems architecture in detail.)

The airplane has two main landing gear attached to the wings and a nose gear. The WBS is installed on the two main landing gear. The nose gear wheels are unbraked. The eight main gear wheels have multi-disc carbon brakes.

Brakes on the main gear wheels are used to provide safe retardation of the airplane during taxiing and landing phases and in the event of a Rejected Takeoff (RTO). The wheel brakes also prevent unintended airplane motion when parked, and may be used to provide differential braking for airplane directional control. A secondary function of the WBS is to stop main gear wheel rotation upon gear retraction.

Braking on ground is commanded manually via brake pedals, or automatically (autobrake) without the need for pedal application. The autobrake functionality allows the pilot to pre-arm the decelerate function prior to takeoff or landing and is only available in the NORMAL Mode.

Brake application is controlled by left and right meter valves located in the wheel wells. The meter valves are operated via electrical signals, through the EBU, from toe pedals integral to the rudder pedal assembly. Differential control of the left and right brakes is available to both the captain and first officer. The parking brake handle is used to set the parking brake. To set the parking brake, the brake pedals are fully depressed. The parking brake handle can be pulled up and will latch when the pedals are released. The WBS maintains brake clamping force without further flight crew action once the parking brake has been set. The brake pedals are depressed to unlatch the parking brake handle.

The brake pedal position is also electrically fed to a brake computer. This in turn produces corresponding control signals to the brakes. In addition, this computer monitors various signals which denote certain critical airplane and system states, to provide correct brake functions and improve system fault tolerance, and generate warnings, indications, and maintenance information to other systems. This computer is accordingly named the Brake System Control Unit (BSCU).

The WBS includes a Normal Brake System and Alternate/Emergency Brake System which operate in NORMAL, ALTERNATE, or EMERGENCY Modes.

- a. NORMAL Mode is operated with BSCU and HYD 1 hydraulic system. NORMAL Mode includes either autobrake or manual braking. Autobrake is only available in the NORMAL Mode.
- b. ALTERNATE Mode is on standby and is selected automatically when the Normal Brake System fails. It is operated with HYD 2 hydraulic system.
- c. EMERGENCY Mode is selected when the Normal Brake System has failed and HYD 2 hydraulic system is lost. It is operated with an emergency hydraulic accumulator.

Switch-over of the mode can be automatic under defined failure conditions or manually selected. Mode transitions are depicted in [Figure E6](#).

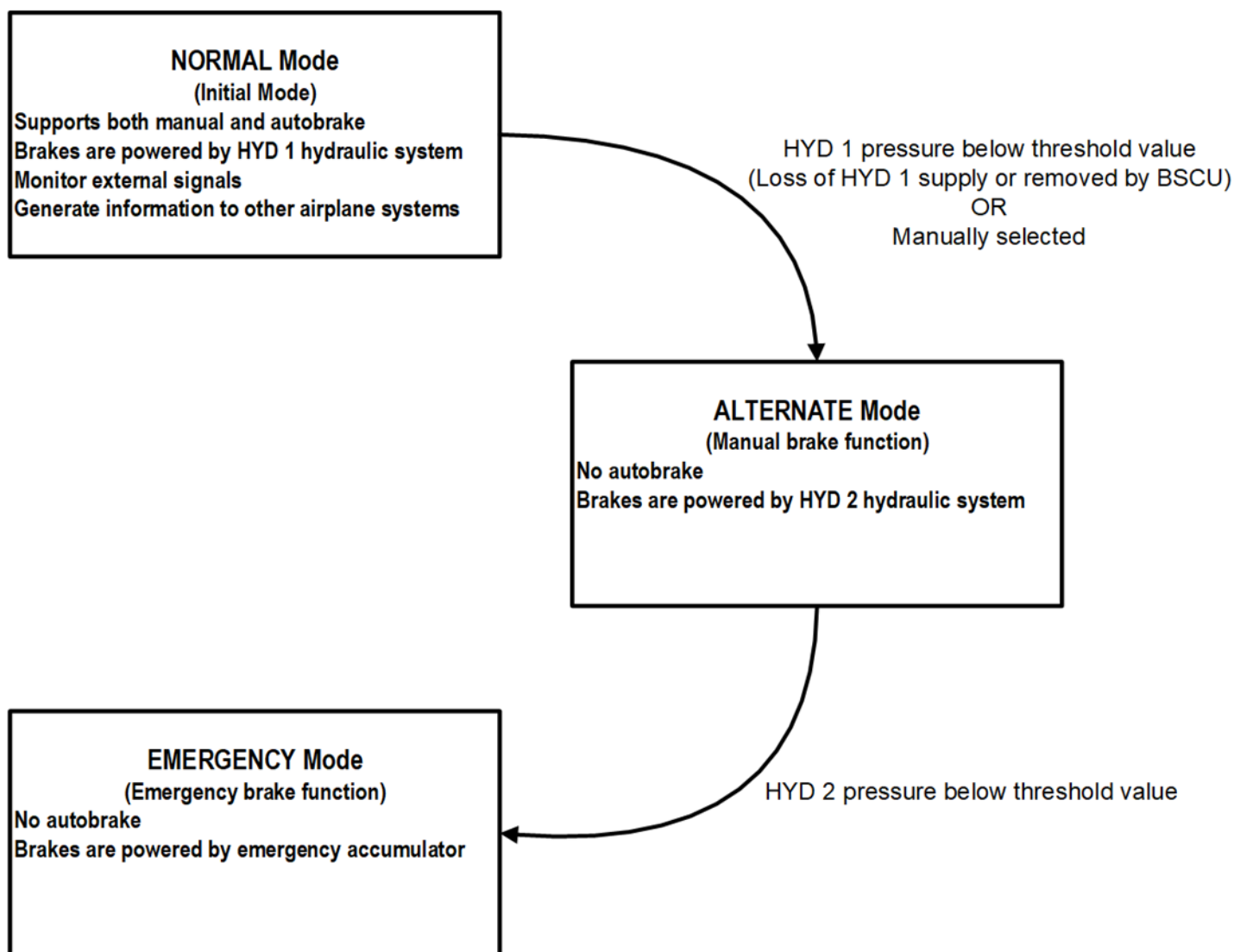


Figure E6 - WBS mode transitions

E.4.4 Wheel Brake System Functions

The WBS is assigned to perform the following functions:

- a. Decelerate the wheels on the ground.
 - 1. Decelerate the wheels on command (manual or automatic).
 - 2. Automatically command wheel deceleration on landing and Rejected Takeoff (RTO).
 - 3. Prevent tire skidding during wheel deceleration.
 - 4. Provide WBS annunciation.
- b. Decelerate the wheels on gear retraction.
- c. Decelerate the wheels differentially for directional control.
- d. Prevent airplane motion when parked.
- e. Provide wheel speed data to the airplane.

The wheel brake function is also supported by and/or provides support to a number of functions from different airplane systems. The interfacing functions associated with the WBS identified for this example are:

- a. Provide electrical power.
- b. Provide hydraulic power.
- c. Display system annunciation.

[Figure E7](#) depicts how the airplane-level interfacing functions, “Provide Electrical Power” and “Provide Hydraulic Power,” support the WBS functions. Likewise, the WBS function “provide WBS annunciation” supports the interfacing function “Display System Annunciation” which provides crew operational awareness to the WBS status.

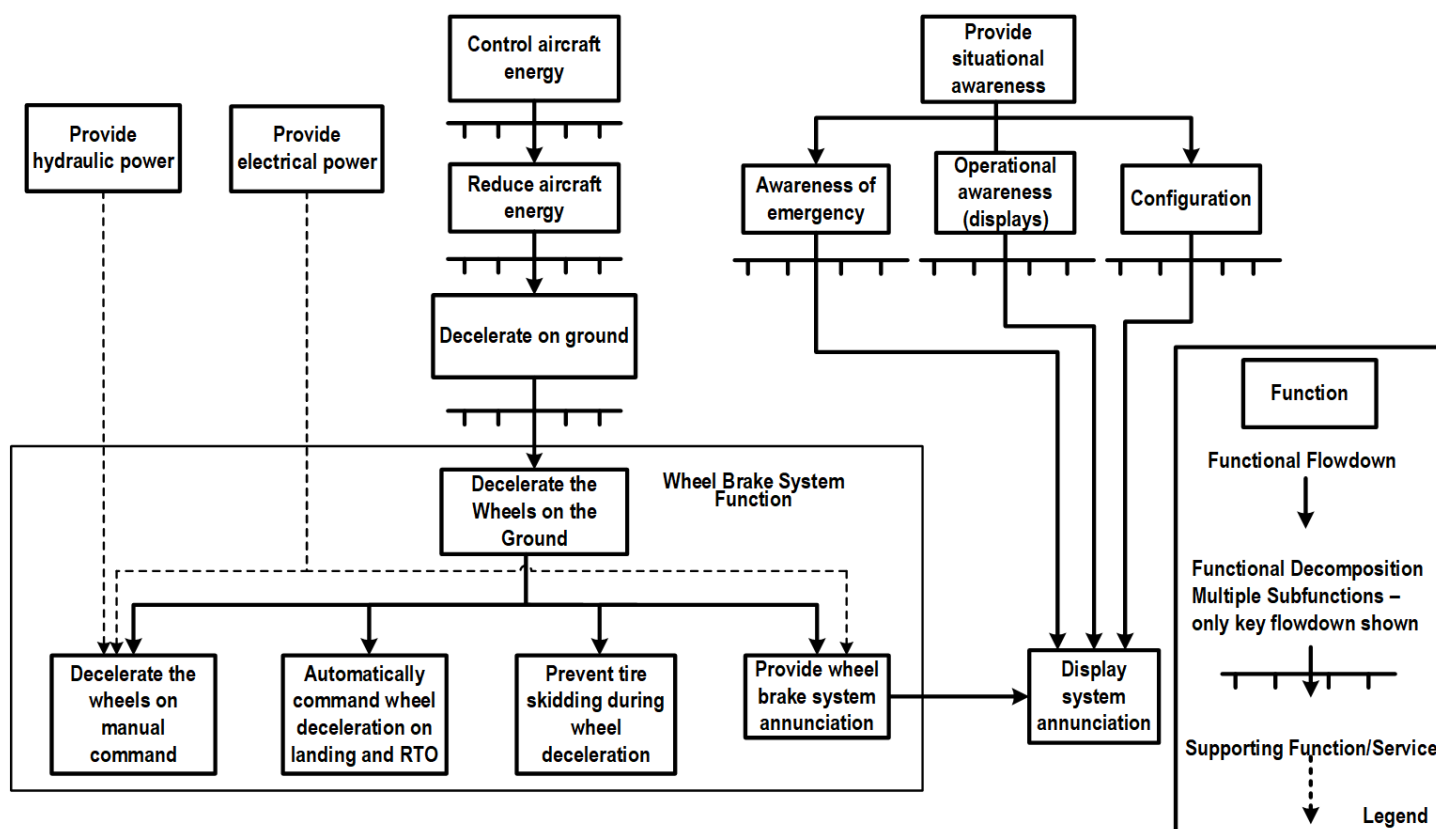


Figure E7 - Decomposition of Wheel Brake System functions

E.4.5 System Functional Hazard Assessment

The SFHA was prepared through the safety assessment process. Using the WBS description from [E.4.3](#) and functional failures for each of the WBS functions in [E.4.4](#), the SFHA determined failure conditions.

The SFHA provided failure conditions associated with the function “Decelerate the Wheels on the Ground.” The severity classifications assigned in the SFHA (e.g., Catastrophic, Hazardous, Major), along with conditions assigned that further defined the failure condition (e.g., loss of less than 50% wheel deceleration) were a factor in determining the WBS architecture required for the “Decelerate the Wheels on the Ground” function and other associated functions.

The SFHA failure conditions for the “Decelerate the Wheels on the Ground” function are found in ARP4761A/ED-135 Appendix Q.

E.4.5.1 SFHA Assumptions

Assumptions made while accomplishing the effect evaluation of each failure condition have been captured and numerically identified for reference. [Table E10](#) presents the assumptions analysis made during the development of this functional hazard assessment. These assumptions were confirmed through the development process and the confirmation communicated back to the SFHA process.

Table E10 - SFHA assumptions

Assumption Identifier	Description
SASP 1.1-1	Loss of 80% or more of deceleration capability is considered a total loss of the deceleration means.
SASP 1.1-2	Failure Conditions are assessed on wet runway conditions.
SASP 1.1-3	RTO is considered an operational condition caused either by an external event or by a system failure (annunciated or perceived by the flight crew) during takeoff run.
SASP 1.1-4	The flight crew will not initiate an RTO due to an annunciated failure of the deceleration function after V1.
SASP 1.1-5	Taxi is performed at groundspeeds below 30 knots.
SASP 1.1-6	Overrunning the runway length at or above "XYZ" knots is considered a high speed overrun.
SASP 1.1-7	Overrunning the runway length below "XYZ" knots is considered a low speed overrun.
SASP 1.1-8	Failures of deceleration capability will be detected and annunciated by on-board systems.

E.4.6 Wheel Brake System Requirements

Initial WBS requirements were captured by the allocation of the WBS-related airplane functional requirements, the derived requirements associated with these functions, and the associated functional interfaces to the WBS. The certification requirements were assigned to the WBS either by decomposing airplane-level certification requirements or directly from the regulations.

[Table E11](#) illustrates the functional and certification requirements allocated to WBS in order to define initial system architecture for use in the safety analysis defined in ARP4761A/ED-135 Appendix Q. The table also contains design decisions from past experience and additional functions derived from the certification requirements in order to set an initial WBS architecture. The source column provides additional context for the traced from content. The validation for these requirements is described in [E.4.7](#).

At this stage of requirements development, not all certification requirements were agreed upon with the Certification Authority. The certification requirements were finalized after the WBS architecture was agreed to by the OEM and applicable subcontractor.

The WBS requirement set was provided at this point in sufficient detail so that the SFHA process could commence. The requirements allocation and derivation process continued parallel to the safety analysis.

Table E11 - Initial Wheel Brake System requirements WBS-0001-103 - Version 1.0 (excerpt)

Requirement Number	Requirement	Associated Function (if Applicable) and the FDAL		Traced From	Source
S18-WBS-R-0020	The Wheel Brake System shall have a means to decelerate the wheels on the ground.	Decelerate the wheels on ground	A	S18-ACFT-R-1100	S18 Airplane Requirements Specification
S18-WBS-R-0021	The Wheel Brake System shall be capable of decelerating the S18 airplane to a complete stop/to taxi speed in 2000 feet when wheel brakes, high lift speed brakes and reverse thrust are available, including when at maximum landing weight.	Decelerate the wheels on ground	A	Derived	S18 airplane trade study
S18-WBS-R-0041	The Wheel Brake System shall provide directional control on ground by a differential braking function.	Decelerate the wheels differentially for directional control	A	S18-ACFT-R-0182	S18 Airplane Requirements Specification
S18-WBS-R-0042	The Wheel Brake System shall provide a parking brake to prevent airplane motion on the ground.	Prevent airplane motion when parked	A	S18-ACFT-R-0183	S18 Airplane Requirements Specification
S18-WBS-R-0043	The Wheel Brake System shall provide autobraking capability during landing and RTO.	Decelerate the wheels on ground	A	S18-ACFT-R-1120	S18 Airplane Requirements Specification
S18-WBS-R-0044	The Wheel Brake System shall provide anti-skid braking.	Decelerate the wheels on ground	A	S18-ACFT-R-1130	S18 Airplane Requirements Specification
S18-WBS-R-0045	The Wheel Brake System shall provide hydraulic brake control.	Decelerate the wheels on ground	A	S18-ACFT-R-0184	S18 Airplane Requirements Specification

Requirement Number	Requirement	Associated Function (if Applicable) and the FDAL		Traced From	Source
S18-WBS-R-0046	The Wheel Brake System shall override the autobrake when commanded by the flight crew.	Decelerate the wheels on ground	A	S18-ACFT-R-0185	S18 Airplane Requirements Specification
S18-WBS-R-0047	The Wheel Brake System shall meet safety requirements while operating in an average atmospheric radiation environment per the IEC 62396 standard with an altitude of 40000 feet and a latitude of 45 degrees.	Protect against Single Event Effects (SEE)	A	S18-ACFT-R-0186	S18 Airplane Requirements Specification
S18-WBS-R-0049	Wheel Brake System shall be controlled and monitored by a computer system called the Brake System Control Unit.	Decelerate the wheels on ground	A	Derived	Design decision made to decrease the pilot workload and fulfill the brake functions like autobrake, annunciation, anti-skid
S18-WBS-R-0050	Each wheel shall have its own hydraulic circuit.	Decelerate the wheels on ground	A	Derived	Design architecture decision
S18-WBS-R-0052	Each hydraulic Wheel Brake System circuit shall have meter valves, anti-skid valves and hydraulic fuses.	Decelerate the wheels on ground	A	Derived	Design architecture decision
S18-WBS-R-0055	Anti-skid system shall be capable to prevent skidding of the tires by reducing the pressure applied to the brakes.	Decelerate the wheels on ground	A	Derived	Design architecture decision
S18-WBS-R-0062	Wheel Brake System shall include an emergency accumulator to supply hydraulic power to the wheel brakes.	Decelerate the wheels on ground	A	S18-ACFT-R-1551	S18 Airplane Requirements Specification
S18-WBS-R-0065	The emergency accumulator shall be capable of providing 2000 psi.	Decelerate the wheels on ground	A	Derived	Design architecture decision
S18-WBS-R-0066	The Alternate/Emergency Brake System hydraulic equipment and piping shall be installed aft of the engine 1 UERF trajectory envelope.	Decelerate the wheels on ground	A	S18-ACFT-R-1552	S18 Airplane Requirements Specification
S18-WBS-R-0120	No single failure or event shall cause the complete loss of hydraulic power for both the WBS Normal and Alternate/Emergency Brake Systems.	Decelerate the wheels on ground	A	S18-ACFT-R-1550	S18 Airplane Requirements Specification
S18-WBS-R-0100	The Wheel Brake System decelerate the wheels on the ground function shall be developed as FDAL A.	Decelerate the wheels on ground	A	S18-ACFT-R-0933	S18 Airplane Requirements Specification
S18-WBS-R-0150	Complete loss of decelerate the wheels on the ground function shall be less probable than 1.0E-07 for a landing.	Decelerate the wheels on ground	A	S18-ACFT-R-1385	S18 Airplane Requirements Specification
S18-WBS-R-0200	Two redundant control lanes shall be provided between the Electric Brake Unit (EBU) and each of the two Alternate/Emergency Meter Valves.	Decelerate on ground	A	S18-ACFT-R-1600	S18 Airplane Requirements Specification

(Editor's Note: Due to the complexity of the requirement capturing process, not all requirements are shown here but some requirements related to the "Decelerate the wheels on the ground" function are shown to explain the process.)

(Editor's Note: The generic wording of requirement S18-WBS-R-0047 does not imply nor assume that the WBS is commanding braking during flight at high altitude. The expressed need is that the WBS should withstand the specified environment without impacting compliance to its safety requirements, considering that atmospheric radiation encountered during flight may affect WBS equipment with lasting effects on braking functions up to and including the ground phase.)

With the initial requirements set, the architecture studies and the safety assessment process (commencing with the SFHA) began. The requirements development process was a coupled process where the system designers deriving the requirements worked closely with the safety engineer. The results of the requirements development and safety assessment processes were uploaded to the requirement database. Through the architecture development and the SFHA studies the interface requirements were clarified between the WBS and other airplane supporting functions. Section [E.4.6.6](#) describes how these interface requirements were derived from these studies.

The WBS shall be controlled and monitored by a computer system called the Brake System Control Unit (BSCU).

In addition to a BSCU, the WBS consists of:

- a. **Shutoff Valve:** The Shutoff Valve (SOV) responds to commands from the BSCU to apply hydraulic pressure to the brake discs. The SOV is used to switch from the Normal Brake System to the Alternate/Emergency Brake System, when the BSCU becomes unable to control the Normal Brake System via the Normal Meter Valve (NMV).
- b. **Selector Valve:** The Selector Valve connects the HYD 2 supply to the Alternate/Emergency Brake System and removes the HYD 1 supply to the Normal Brake System when the HYD 1 pressure falls below a threshold value, either from loss of HYD 1 supply itself or from its removal by the BSCU due to the presence of faults.
- c. **Normal Meter Valves:** The function of the NMVs is to control pressure in the Normal Brake System to the demanded level and to provide regulation for the anti-skid function while the WBS is in the NORMAL Mode. The NMVs are controlled by the BSCU, which receives electrical brake pedal position inputs from the EBU.
- d. **Emergency Accumulator:** The Emergency Accumulator provides an emergency reserve of hydraulic pressure for the Alternate/Emergency Brake System. A fully charged emergency accumulator provides enough pressure to apply pressure for the required number of presses of the brake pedals while the WBS is in the EMERGENCY Mode. The emergency accumulator also supplies hydraulic pressure for the parking brake function.
- e. **Alternate/Emergency Meter Valves:** The function of the Alternate/Emergency Meter Valve is to control pressure in the Alternate/Emergency Brake System to the level demanded by the brake pedals while the WBS is in the ALTERNATE or EMERGENCY Modes. The Alternate/Emergency Meter Valves receive alternate electrical brake signals from the EBU and have no interface with the BSCU.
- f. **Shutoff/Anti-Skid Valves:** The Shutoff/Anti-Skid Valves (S/ASVs) follow BSCU commands to control hydraulic pressure to the brake pads in the Alternate/Emergency Brake System. These valves perform two functions: (1) they are the shutoff valves for Alternate/Emergency Brake System when the WBS is in the NORMAL Mode, and (2) they are used to restrict the hydraulic line pressure to the brakes in order to prevent locking of the wheels when the WBS is in the ALTERNATE or EMERGENCY Modes.

Other components associated with wheel braking are:

- a. **Electric Brake Unit:** The EBU provides the mechanical interface to the flight crew and electrical interface to the WBS. The EBU provides brake pedal positions to the BSCU. The BSCU uses these inputs to calculate commands to the SOV and NMVs in the NORMAL Mode, and the Shutoff/Anti-Skid Valves in the ALTERNATE and EMERGENCY Modes. The EBU also provides dual-redundant alternate electrical brake signals to control the Alternate/Emergency Meter Valves in the ALTERNATE and EMERGENCY Modes.
- b. **Wheel Brakes:** The wheel brakes will provide friction force to the wheel. The brakes will also house temperature sensors which will provide brake temperature to the BSCU.
- c. **Parking Brake:** The parking brake handle is used to set the parking brake system. The parking brake has been intentionally not detailed in [Figure E9](#) to simplify the schematic.
- d. **Wheel Speed Sensor:** A wheel speed sensor is used to sense wheel speed on each wheel.

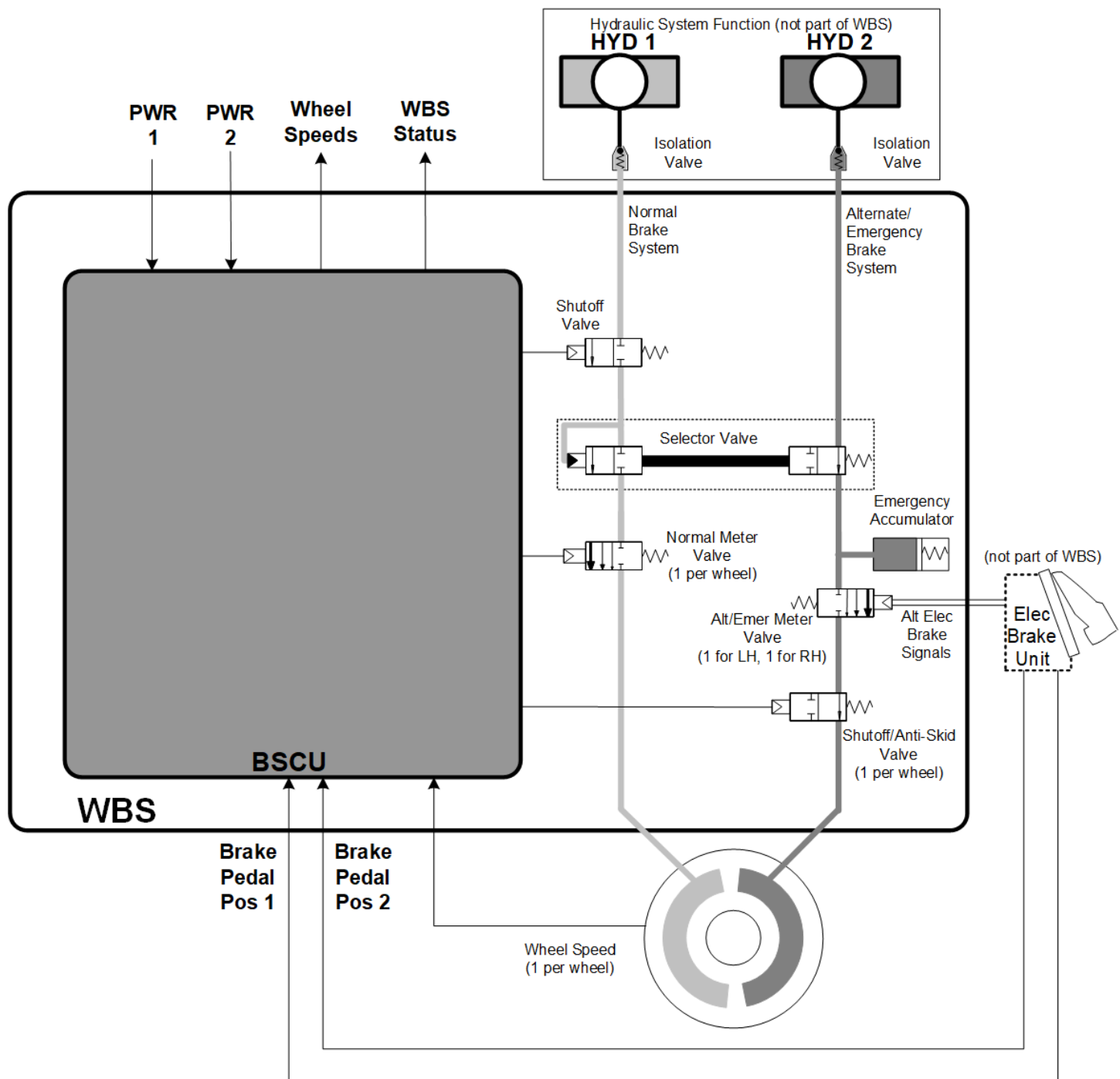


Figure E9 - High-level Wheel Brake System architecture

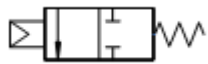
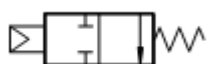
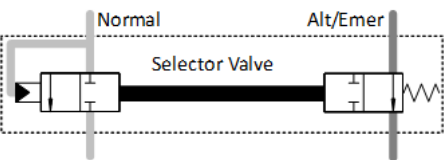
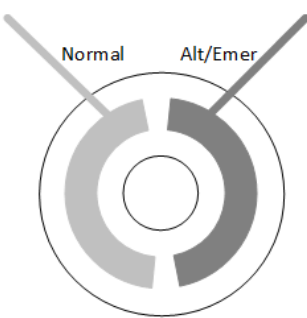
Legend	
Electro-Hydraulic Component	Description
	Normally closed valve; open when signal drives input.
	Normally open valve; closed when signal drives input.
	Normally closed valve; open when signal drives input. Opens with increasing flow rate with increasing signal.
	Selector Valve normally closed for Normal, open for Alt/Emer; opens for Normal, closes for Alt/Emer when pressure is applied from Normal. In the example, the spring-loaded Selector Valve automatically opens Alt/Emer if normal pressure falls below a threshold or is shut off via the Shutoff Valve.
	Wheel brakes, either input (Normal or Alt/Emer) provides hydraulic pressure to apply 100% of braking.

Figure E9 - High-level Wheel Brake System architecture (continued)

Since the proposed high-level architecture has been established, it was re-evaluated against the system-level functional, operational, and other classes of requirements that have been identified thus far. This architecture was passed to the system safety team to initiate the PSSA process for initial evaluation from a safety perspective.

E.4.6.1.1 Wheel Brake System Architectural Decisions Based on Safety Requirements

The WBS safety assessment process began as soon as the systems design team had an initial architecture concept (WBS ARCH 1). The team analyzed the initial architecture concept against the requirements input from the airplane level and the safety objectives from the AFHA.

The proposed safety requirement from PASA “[FF1.1] Complete loss of wheel brake shall be less than 1.0E-07 for a landing” was based on PASA results that the loss of brake commands needs to have a probability lower than 1.0E-07. Safety analysis of the initial architecture relative to this safety requirement identified a need for architectural mitigation. The system design was modified to include two redundant BSCU commands in order to meet the loss of function requirement.

As a result of the analysis, the following requirement was added to the requirements database along with justification information: “S18-WBS-R-0509: The WBS shall have dual BSCU command functions.”

E.4.6.1.2 Modified Wheel Brake System Description

Braking on ground is commanded manually, via brake pedals, or automatically (autobrake) without the need for pedal application. The autobrake functionality allows the pilot to pre-arm the deceleration rate prior to takeoff or landing. Autobrake is only available in NORMAL Mode.

Based on the requirement (S18-ACFT-R-1385) that loss of all wheel braking was less probable than $1.0E-07$ for a landing, a design decision was made that each wheel has a brake assembly operated by two independent sets of hydraulic pistons. One set is operated from the HYD 1 hydraulic supply and is used in the NORMAL Mode. The Alternate/Emergency Brake System is on standby and is selected automatically when the Normal Brake System fails. It is operated independently using the HYD 2 hydraulic power supply and is backed by an emergency accumulator which is also used to drive the parking brake. The emergency accumulator supplies the Alternate/Emergency Brake System in the EMERGENCY Mode, when the HYD 2 supply is lost and the NORMAL Mode is not available. Switch-over is automatic under various failure conditions, or can be manually selected. Reduction of HYD 1 pressure below a threshold value, either from loss of HYD 1 supply itself or from its removal by the BSCU due to the presence of faults, causes an automatic selector valve to connect the HYD 2 supply to the Alternate/Emergency Brake System. An anti-skid function is available in both the normal and alternate/emergency brake systems, and operates at all speeds greater than 2 m/s.

In the Normal Brake System, all eight wheels are individually braked from their own servo valves, which are also used to apply anti-skid. In the Alternate/Emergency Brake System, a dual meter valve provides a low pressure hydraulic braking input via four servo valves which provide the anti-skid function to four pairs of wheels. Operation of the Alternate/Emergency Brake System is precluded when the Normal Brake System is in use to maintain independence between the two systems.

In the NORMAL Mode, the brake pedal position is electrically fed to the BSCU. This in turn produces corresponding control signals to the brakes. In addition, the BSCU monitors various signals which denote certain critical airplane and system states, to provide proper brake system functionality and improve fault tolerance, and generates warnings, indications and maintenance information to other systems.

E.4.6.2 Wheel Brake System Architecture Trade Studies

(Editor's Note: The main body does not include guidelines to perform trade studies during the development of an airplane. However, since trade studies can often help to find and optimize solutions to design problems, the S18 WBS team has decided to perform one to help refine the system architecture. There are many methods and procedures for performing trade studies and the main body does not prescribe or recommend any particular method or procedure. Therefore, this example does not show the details of the trade study. Instead this example shows only the results and how the results were utilized to finalize the architecture with which the development will continue. It should be noted that the parameters in the trade study results are examples of possible parameters. The inputs to and the outputs of a real trade study depend upon the system at hand, environment in which the development is taking place, the business, and numerous other factors.)

A trade study ensued to evaluate options for achieving dual BSCU command functions. Considerations for the trade study included:

- a. Where to install two Line Replaceable Units (LRUs)
- b. Cost of two LRUs versus cost of one LRU with two systems
- c. Potential common causes of two LRUs versus one with two systems
- d. How will the two systems coordinate with each other

The safety and development teams examined architecture options in a trade study, summarized in [Figure E10](#), that helped determine the optimized architecture. Once the architecture was selected, system development continued the requirements development process, allocating system-level requirements to system components. This architecture continued to be analyzed and had the potential to change until the requirements were validated. To ensure a successful development life cycle, it was important that architecture at this point was firm enough to ensure that future changes have little or preferably no impact on other systems. Therefore the validation of interface requirements was thorough and completed fairly early in the requirements validation process.

After two candidate architectures (WBS ARCH 2 and WBS ARCH 3) that met safety objectives were downselected, the third WBS architecture (WBS ARCH 3) from the trade study was selected and specified to the BSCU supplier. The associated BSCU architecture, referenced as “iteration 1” in [Figure E11](#), was carried forward in this example. This BSCU architecture consists of one BSCU, which houses two independent command channels.

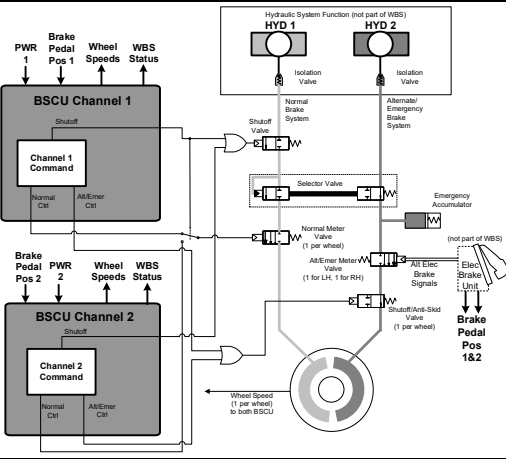
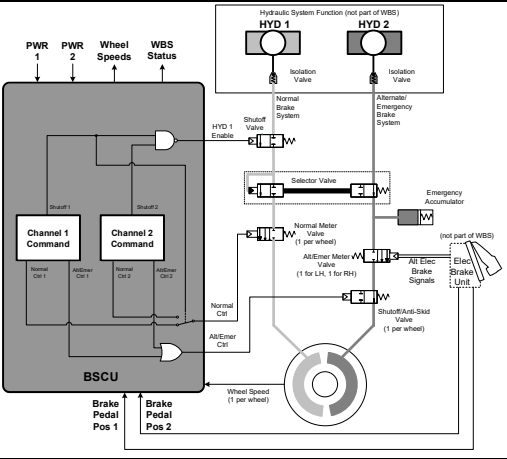
System Architecture	WBS ARCH2 (2 BSCU)	WBS ARCH3 (1 Dual BSCU)
Uses the following facts: 1) All costs are based on system supplier rates 2) Arch 2 is based on reuse of previous S-18 wheel braking system design		
Cost	WBS ARCH2 (2 BSCU)	WBS ARCH3 (1 Dual BSCU)
OEM Cost		
Non-Recurring Cost	\$X	\$Y
Recurring Cost		
Price per Aircraft	\$X1	\$Y1
Manufacturing per Aircraft (ROM)	\$X2	\$Y2
Total Recurring	\$X3	\$Y3
System Characteristics	WBS ARCH2 (2 BSCU)	WBS ARCH3 (1 Dual BSCU)
Number of Line Replaceable Units	X	Y
System Weight in kg	X	Y
in lbs	X	Y
Power Consumption in W		
Installation Hours (Hrs)	X	Y
Difficulty of installation	X	Y
(1 Difficult - 10 Easy)	3 - Note due to location of second BSCU and interconnect switches	6
Technical		
FAR 25.735 Compliant	Yes	Yes
Meets Safety Objectives	Yes	Yes
Certification Risk	Low	Low
Reliability / Maintainability	WBS ARCH2 (2 BSCU)	WBS ARCH3 (1 Dual BSCU)
MTBF		
System in flight hours	X	Y

Figure E10 - Wheel Brake System architecture trade study summary

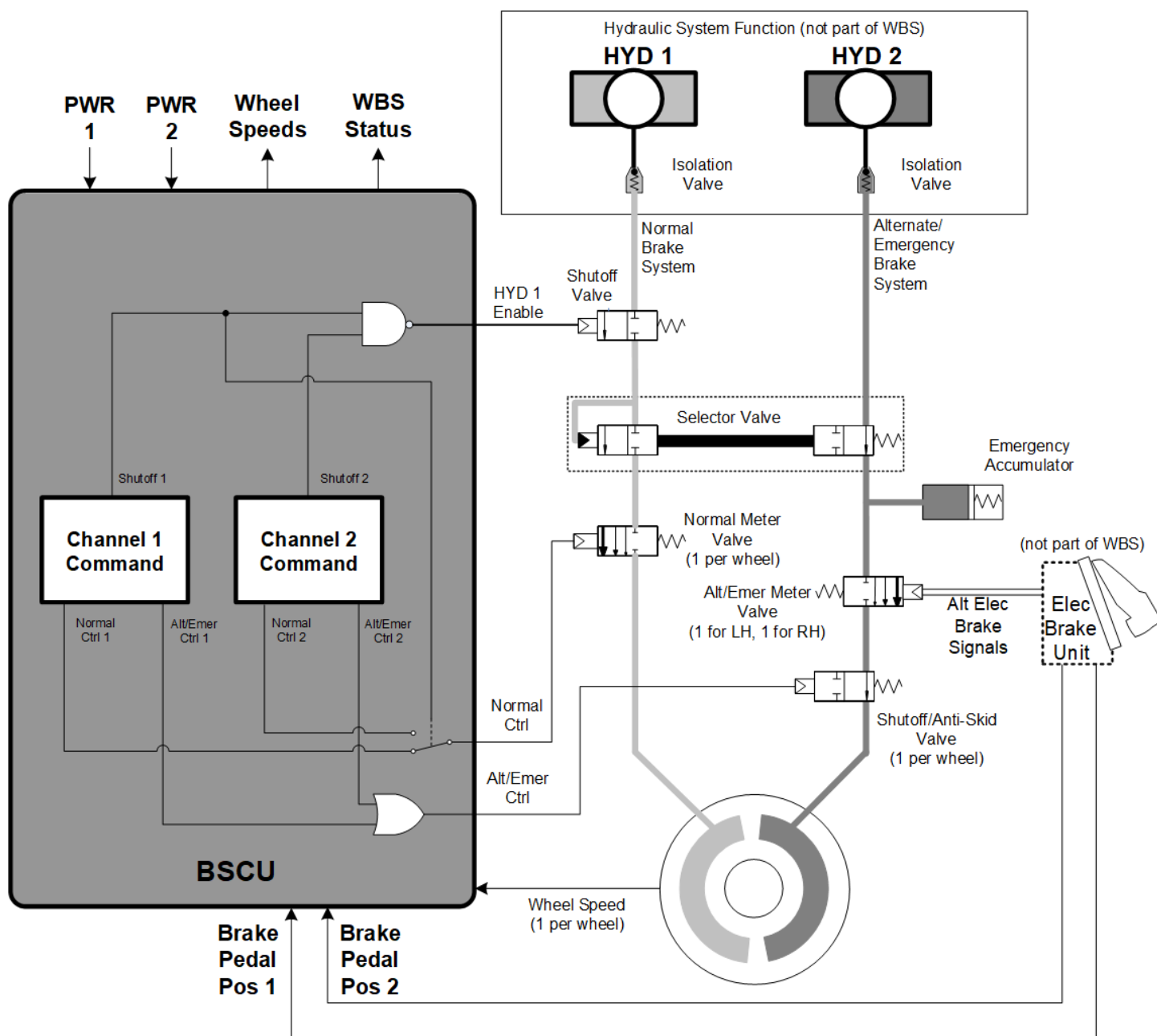


Figure E11 - WBS architecture 3 with BSCU architecture iteration 1

E.4.6.3 Provide Outputs to Wheel Brake System PSSA Process

The development process provided WBS functions, WBS requirements, WBS architecture, and WBS function allocation to the WBS PSSA process. The WBS functions, requirements, architecture, and function allocation are shown in [E.4.4](#), [E.4.6](#), [E.4.6.2](#), and [E.4.4](#), respectively.

E.4.6.4 Wheel Brake System PSSA Assumptions and Proposed Safety Requirements

As described in main body [5.2](#), the WBS functions were assigned FDALs applied to the various development processes associated with each function. The required FDALs for the WBS functions were derived from failure conditions and classifications provided by the WBS SFHA. Assignment of FDAL to functions within the WBS was done as part of the ARP4761A/ED-135 PSSA process.

The WBS PSSA process determined that the WBS functions responsible for “Decelerate the Wheels on the Ground” required FDAL A.

Lower-level proposed safety requirements were identified through the analyses carried out during the WBS PSSA process. The requirements, shown in [Tables E12](#) and [E13](#), were identified and passed down for assessment at the BSCU level. The reassessment at the BSCU level confirmed the proposed design satisfied the lower-level safety requirements identified.

Table E12 - Proposed independence requirements (WBS PSSA outputs)

Ref No.	Proposed Independence Requirement
1	Airplane electrical power 1 independent from airplane electrical power 2
2	NORMAL Mode braking function independent from ALTERNATE Mode braking function, for total loss failure condition.
3	The NMVs and their control are independent from the SOV and its control such that no single failure results in uncommanded braking.

Proposed independence requirements 2 and 3 were applicable to the WBS. Therefore, safety requirements S18-WBS-R-0120 and S18-WBS-R-0326 were created in coordination with the safety team (see [Table E19](#)). Proposed independence requirement 1 was outside the scope of the WBS and airplane requirements flowdown was not traceable to this proposed requirement. This was an assumption communicated to the airplane development process. In addition, this independence requirement was decomposed into WBS requirement S18-WBS-R-0130 to maintain independence with respect to electrical power.

Table E13 - Proposed safety requirements (WBS PSSA outputs)

Ref No.	Proposed Safety Requirement
1	The probability of BSCU failure resulting in loss of a valid braking command output to the NMV shall not exceed 2.0E-04 per flight.
2	The probability of BSCU failure resulting in unannounced erroneous braking command to the NMV shall not exceed 2.0E-04 per flight.
3	The probability of BSCU failure resulting in the loss of command to open the SOV shall not exceed 2.0E-04 per flight.
4	The probability of BSCU failure resulting in unintended closure of the S/ASV shall not exceed 2.0E-04 per flight.
5	The SOV and NMV commands shall be provided by the BSCU upon loss of either airplane electrical power input.
6	When “HYD 1 Enable” output is enabled, then “Alt/Emer Ctrl” output shall be disabled.
7	When “HYD 1 Enable” output is disabled, then “Alt/Emer Ctrl” output shall be enabled.
8	No single failure shall cause erroneous NMV command and inhibit the SOV function.
9	The wheel brake command function of the BSCU shall be developed to FDAL A.

The WBS PSSA process also identified assumptions shown in [Table E14](#) used in the safety assessment that were managed by the system development process.

Table E14 - Safety assessment assumptions (WBS PSSA outputs)

Ref No.	Safety Assessment Assumption
1	The probability of “Loss of Normal Braking System Hydraulic Equipment” will be less than 3.3E-05 per flight.
2	The probability of “Loss of Alternate Braking System Hydraulic Equipment” will be less than 3.3E-05 per flight
3	The probability of 7 or more wheel speed sensors erroneous or inoperative will be less than 1.0E-07 per flight.
4	The probability of loss of an airplane electrical power bus will be less than 1.0E-04 per flight.
5	The probability of loss of a Left brake pedal position input will be less than 1.0E-04 per flight.
6	Airplane electrical power bus 1 is independent from airplane electrical power bus 2.
7	HYD 1 hydraulic system is independent from HYD 2 hydraulic system.

Assumptions 1 and 2 were applicable to the WBS, therefore safety requirements S18-WBS-R-6111 and S18-WBS-R-6112 were created to capture assumptions 1 and 2. Assumptions 3 through 7 were outside the scope of the WBS; these assumptions were communicated to the airplane development process.

E.4.6.5 Derived Wheel Brake System Requirements

The safety, reliability, and design teams' discussions about the safety requirements resulted in a new architecture. The BSCU supplier declared that past experience with BSCU design showed that the failure rate of a single control system could not satisfy the safety objectives of the SFHA. In response to the BSCU declaration, the WBS supplier documented changes in Coord Memo S18-CM00XX. The requirements in [Table E15](#) were inserted into the WBS specification.

Table E15 - Wheel Brake System supplier derived requirements

Requirement Number	Requirement	Source
S18-WBS-R-0508	The Wheel Brake System shall have at least two independent hydraulic pressure sources.	Coord Memo S18a-CM00XX
S18-WBS-R-0509	The Wheel Brake System shall have dual BSCU command functions.	Coord Memo S18a-CM00XX
S18-WBS-R-0510	The rudder and nose wheel steering functions shall not be implemented by the Wheel Brake System.	Coord Memo S18a-CM00XX

The [Table E15](#) requirements imposed a new architecture and the architecture trade studies continued until all the requirements were validated.

The requirements in [Table E16](#) applied to the new architecture with a dual channel BSCU and two hydraulic systems as described in [E.4.6.2](#). Within the new architecture, new requirements were entered into the database from safety assessments, certification requirements and design decisions.

Table E16 - WBS requirements for dual channel BSCU and two hydraulic systems architecture

Requirement Number	Requirement	Source
S18-WBS-R-0631	The Wheel Brake System shall indicate individual brake temperature.	14 CFR/CS 25.735(j)
S18-WBS-R-0632	The Wheel Brake System shall control individual wheel pressure.	14 CFR/CS 25.735(j)
S18-WBS-R-2971	The Wheel Brake System shall have one NORMAL operating mode.	Design architecture decision
S18-WBS-R-2972	The Wheel Brake System shall have one ALTERNATE operating mode.	Design architecture decision
S18-WBS-R-2973	Operation of the Alternate/Emergency Brake System shall be precluded when the Normal Brake System is in use.	Design architecture decision
S18-WBS-R-2974	An emergency accumulator shall provide hydraulic power for EMERGENCY Wheel Brake System operating mode.	Design architecture decision
S18-WBS-R-2986	The BSCU shall be developed as FDAL A.	S18 WBS PSSA
S18-WBS-R-2997	The BSCU shall have two independent command channels.	Design architecture decision
S18-WBS-R-2999	The BSCU shall have two independent electrical power sources.	WBS Requirements Specification WBS-0001-103
S18-WBS-R-3011	Brake pedals shall be independent from the rudder control.	Design architecture decision
S18-WBS-R-3213	Normal Brake System shall be powered from the airplane HYD 1 hydraulic system.	Design architecture decision
S18-WBS-R-3224	Alternate/Emergency Brake System shall be powered from the airplane HYD 2 hydraulic system.	Design architecture decision
S18-WBS-R-3245	ALTERNATE Mode shall be automatically switched when the NORMAL Mode fails.	Design architecture decision

The higher-level requirements in [Table E16](#) required further decomposition into the item-level and interface requirements.

E.4.6.6 Interface Requirements

The development of interface requirements began with the identification of systems that support the WBS functions. It was important to ensure that the WBS was both receiving inputs required from other systems as well as providing outputs required by other systems functions. A thorough assessment of the associated functional failure conditions was performed during the WBS PSSA process to gain an accurate understanding of the contribution from each interfacing function.

E.4.6.6.1 Wheel Brake System/Airplane Interface Requirements (Inter-System)

The BSCU provides the primary interface for the WBS with other airplane systems. The BSCU interface to the airplane is comprised of communication signals needed for the wheel brake function and signals needed for health monitoring. The key system-to-system interfaces considered for the WBS are as follows:

- a. Hydraulics
- b. Electrical Power System
- c. Pilot controls
- d. Flight deck displays
- e. Propulsion
- f. Proximity sensing
- g. Earth Reference System
- h. Landing gear actuation
- i. Health Management System

[Table E17](#) shows the WBS interface requirements that were added.

Table E17 - Wheel Brake System interface requirements

Requirement Number	Requirement	Source
S18-WBS-ICD-0001	The BSCU shall receive the following inputs from pilot controls: - Brake pedal - Parking brake lever - Autobrake selector switch - Thrust lever resolver angle - Fuel cutoff switch	WBS architecture
S18-WBS-ICD-0002	The BSCU shall receive the following inputs from the Propulsion System: - Thrust lever resolver angle - Fuel cutoff switch	WBS architecture
S18-WBS-ICD-0003	The BSCU shall receive the following inputs from the Proximity Sensing System: - Truck tilt (on-ground) - Speed brake lever	WBS architecture
S18-WBS-ICD-0004	The BSCU shall receive the following inputs from Earth Reference System: - Ground speed - Pitch attitude - Longitudinal acceleration	WBS architecture
S18-WBS-ICD-0005	The BSCU shall receive the following inputs from landing gear actuation: - Landing gear actuation - Landing gear lever	WBS architecture
S18-WBS-ICD-0006	The BSCU shall receive power from Electrical Power System.	WBS architecture
S18-WBS-ICD-0101	The BSCU shall transmit the following outputs to flight deck displays: - Brake temperature - Tire pressure - Brake pedal information - Brake pedal - Wheel speed	WBS PSSA
S18-WBS-ICD-0102	The BSCU shall transmit the following outputs to Health Management System: - Fault reports	WBS PSSA

The WBS safety assessment and architecture trade studies were not the only sources of the requirements. Other airplane systems could impose new requirements to the WBS. For example, the hydraulics system was sized based on the required inputs for the flight control system and the WBS. In this example, the maximum hydraulic pressure was 2400 psi. This was a requirement allocated to the WBS. Moreover, the WBS also allocated requirements to other systems. For example, the following requirement has been allocated to the ground spoiler system: The spoilers shall increase the ground contact force xx% at speeds greater than XXX on ICAO PCN XX/... runway surface.

E.4.6.6.2 Interface Requirements (Intra-System)

[Table E18](#) shows the WBS intra-system interface requirements that were added for the interfaces between systems and items of the WBS:

Table E18 - Wheel Brake System intra-system interface requirements WBS-0001-103 - Version 4.0

Req No.	Requirement	Source
S18-WBS-ICD-1001	The BSCU shall transmit the following to the brake actuator: - Total commanded clamping force.	WBS architecture trade study
S18-WBS-ICD-1002	The brake actuators shall respond to a BSCU command within 5 ms of receiving it.	WBS timing analysis
S18-WBS-ICD-1101	The brake actuator shall transmit the following information to the BSCU: - Total measured clamping force. - Brake actuator power valid status. - Brake temperature. - Wheel speed.	WBS architecture trade study

(Editor's Note: An end-to-end timing analysis, coupled with simulations, was used to determine the interface timing constraints. This level of detail is left out for the sake of brevity.)

E.4.6.7 Wheel Brake System Requirements Specification

The final WBS Requirements Specification, [Table E19](#), contains the requirements allocated from aircraft level, derived requirements from the architecture development, safety requirements from the WBS PSSA, and the intra-system and inter-system interface requirements.

[Table E19](#) identifies the origin of these requirements (operational or safety-derived) and a traceability roadmap back to the airplane and/or WBS level requirements, as appropriate. These requirements were then allocated to the WBS and subsystem developers as design specifications.

(Editor's Note: Due to the complexity of the requirement set's integrity and size, not all requirements developed for the WBS are shown. In this example, the FDALs associated with functions/requirements were captured as separate requirements and/or documented as attributes as shown in the associated function column in [Table E19](#).)

Table E19 - Wheel Brake System Requirements Specification WBS-0001-103 - Version 5.0

Requirement Number	Safety Req	Requirement	Associated Function (if Applicable) and the FDAL		Traced from	Source
S18-WBS-R-0020	No	The Wheel Brake System shall have a means to decelerate the wheels on the ground.	Decelerate the wheels on ground	A	S18-ACFT-R-1100	S18 Airplane Requirements Specification
S18-WBS-R-0021	No	The Wheel Brake System shall be capable of decelerating the S18 airplane to a complete stop/to taxi speed in 2000 feet when wheel brakes, high lift speed brakes and reverse thrust are available, including when at maximum landing weight.	Decelerate the wheels on ground	A	Derived	S18 Airplane Trade Study
S18-WBS-R-0041	No	The Wheel Brake System shall provide directional control on ground by a differential braking function.	Decelerate the wheels differentially for steering	A	S18-ACFT-R-0182	S18 Airplane Requirements Specification
S18-WBS-R-0042	No	The Wheel Brake System shall provide a parking brake to prevent airplane motion on the ground.	Prevent airplane motion when parked	A	S18-ACFT-R-0183	S18 Airplane Requirements Specification
S18-WBS-R-0043	No	The Wheel Brake System shall provide autobraking capability during landing and RTO.	Decelerate the wheels on ground	A	S18-ACFT-R-1120	S18 Airplane Requirements Rpecification
S18-WBS-R-0044	No	The Wheel Brake System shall provide anti-skid braking.	Decelerate the wheels on ground	A	S18-ACFT-R-1130	S18 Airplane Requirements Specification
S18-WBS-R-0045	No	The Wheel Brake System shall provide hydraulic brake control.	Decelerate the wheels on ground	A	S18-ACFT-R-0184	S18 Airplane Requirements Specification
S18-WBS-R-0046	No	The Wheel Brake System shall override the autobrake when the commanded by the flight crew.	Decelerate the wheels on ground	A	S18-ACFT-R-0185	S18 Airplane Requirements Specification
S18-WBS-R-0047	No	The Wheel Brake System shall meet safety requirements while operating in an average atmospheric radiation environment per the IEC 62396 standard with an altitude of 40000 feet and a latitude of 45 degrees.	Decelerate the wheels on ground	A	S18-ACFT-R-0186	S18 Airplane Requirements Specification
S18-WBS-R-0049	No	The Wheel Brake System shall be controlled and monitored by a computer system called Brake System Control Unit.	Decelerate the wheels on ground	A	Derived	Design decision made to decrease the pilot workload and fulfill the brake functions (e.g., autobrake, annunciation, anti-skid).
S18-WBS-R-0050	No	Each wheel brake shall have separate hydraulic power supply lines.	Decelerate the wheels on ground	A	Derived	Design architecture decision
S18-WBS-R-0052	No	Each hydraulic Wheel Brake System circuit shall have meter valves, anti-skid valves and hydraulic fuses.	Decelerate the wheels on ground	A	Derived	Design architecture decision
S18-WBS-R-0055	No	Anti-skid system shall be capable to prevent skidding of the tires by reducing the pressure applied to the brakes.	Decelerate the wheels on ground	A	Derived	Design architecture decision
S18-WBS-R-0062	No	The Wheel Brake System shall include an emergency accumulator to supply hydraulic power to the wheel brakes.	Decelerate the wheels on ground	A	S18-ACFT-R-1551	S18 Airplane Requirements Specification
S18-WBS-R-0065	No	The emergency accumulator shall provide a minimum of 1800 psi.	Decelerate the wheels on ground	A	Derived	Design architecture decision (See problem report PR-WBS-0007)
S18-WBS-R-0100	Yes	The Wheel Brake System decelerate the wheels on the ground function shall be developed as FDAL A.	Decelerate the wheels on ground	A	S18-ACFT-R-0933	S18 Airplane Requirements Specification
S18-WBS-R-0066	Yes	The Alternate/Emergency Brake System hydraulic equipment and piping shall be installed aft of the engine 1 UERF trajectory envelope.	Decelerate the wheels on ground	A	S18-ACFT-R-1552	S18 Airplane Requirements Specification

Requirement Number	Safety Req	Requirement	Associated Function (if Applicable) and the FDAL		Traced from	Source
S18-WBS-R-0120	Yes	No single failure or event shall cause the complete loss of hydraulic power for both the WBS Normal and Alternate/Emergency Brake Systems.	Decelerate the wheels on ground	A	S18-ACFT-R-1550	S18 Airplane Requirements Specification
S18-WBS-R-0130	Yes	No single failure or event shall cause the complete loss of electrical power for both the WBS Normal and Alternate/Emergency Brake Systems.	Decelerate the wheels on ground	A	Derived	S18 WBS PSSA
S18-WBS-R-0150	Yes	Complete loss of decelerate the wheels on the ground function shall be less probable than 1.0E-07 for a landing.	Decelerate the wheels on ground	A	S18-ACFT-R-1385	S18 Airplane Requirements Specification
S18-WBS-R-0200	Yes	Two redundant control lanes shall be provided between the EBU and each of the two Alternate/Emergency Meter Valves.	Decelerate the wheels on ground	A	S18-ACFT-R-1600	S18 Airplane Requirements Specification
S18-WBS-R-0326	Yes	No single failure shall result in inadvertent wheel braking of all wheels during takeoff roll.	Decelerate the wheels on ground	A	Derived	S18 WBS PSSA
S18-WBS-R-0508	No	The Wheel Brake System shall have at least two independent hydraulic pressure sources.	Decelerate the wheels on ground	A	Derived	Coord Memo S18a-CM00XX
S18-WBS-R-0509	No	The Wheel Brake System shall have dual BSCU command functions.	Decelerate the wheels on ground	A	Derived	Coord Memo S18a-CM00XX
S18-WBS-R-0510	No	The rudder and nose wheel steering functions shall not be implemented by the Wheel Brake System.	Decelerate the wheels on ground	A	Derived	Coord Memo S18a-CM00XX
S18-WBS-R-0631	No	The Wheel Brake System shall indicate individual brake temperature.	Decelerate the wheels on ground	A	Derived	14 CFR/CS 25.735(j)
S18-WBS-R-0632	No	The Wheel Brake System shall control individual brake pressure.	Decelerate the wheels on ground	A	Derived	14 CFR/CS 25.735(j)
S18-WBS-R-1613	No	Hydraulic pressure shall be controlled for weight, autobrake mode, ground speed, wheel rotation, brake temperature and deceleration rate in accordance with graphs given in S18 brake force analysis AAS18-XXX.	Decelerate the wheels on ground	A	Derived	S18 brake force analysis AAS18-XXX
S18-WBS-R-2971	No	Wheel Brake System shall have one NORMAL operating mode.	Decelerate the wheels on ground	A	Derived	Design architecture decision
S18-WBS-R-2972	No	Wheel Brake System shall have one ALTERNATE operating mode.	Decelerate the wheels on ground	A	Derived	Design architecture decision
S18-WBS-R-2973	No	Operation of the Alternate/Emergency Brake System shall be precluded when the Normal Brake System is in use.	Decelerate the wheels on ground	A	Derived	Design architecture decision
S18-WBS-R-2974	No	An emergency accumulator shall provide hydraulic power for EMERGENCY Wheel Brake System operating mode.	Decelerate the wheels on ground	A	Derived	Design architecture decision
S18-WBS-R-2975	No	The emergency accumulator shall be attached to the HYD 2 hydraulic line between the Selector Valve and the Alternate/Emergency Meter Valve.	Decelerate the wheels on ground	A	Derived	Design architecture decision
S18-WBS-R-2986	Yes	The wheel brake command function of the BSCU shall be developed as FDAL A.	Decelerate the wheels on ground	A	Derived	S18 WBS PSSA
S18-WBS-R-2997	Yes	The BSCU shall have two independent command channels.	Decelerate the wheels on ground	A	Derived	Design architecture decision
S18-WBS-R-2999	Yes	The BSCU shall have two independent electrical power sources.	Decelerate the wheels on ground	A	S18-WBS-R-0130	WBS requirements specification WBS-0001-103
S18-WBS-R-3011	No	Brake pedals shall be independent from the rudder control.	Decelerate the wheels on ground	A	Derived	Design architecture decision
S18-WBS-R-3213	No	Normal Brake System shall be powered from the airplane's HYD 1 hydraulic system.	Decelerate the wheels on ground	A	Derived	Design architecture decision
S18-WBS-R-3224	No	Alternate/Emergency Brake System shall be powered from the airplane's HYD 2 hydraulic system.	Decelerate the wheels on ground	A	Derived	Design architecture decision

Requirement Number	Safety Req	Requirement	Associated Function (if Applicable) and the FDAL		Traced from	Source
S18-WBS-R-3245	No	ALTERNATE Mode shall be automatically selected when the NORMAL Mode fails.	Decelerate the wheels on ground	A	Derived	Design architecture decision
S18-WBS-R-6104	Yes	The probability of BSCU failure resulting in loss of a valid braking command output to the NMV shall not exceed 2.0E-04 per flight.	Decelerate the wheels on ground	A	Derived	S18 WBS PSSA
S18-WBS-R-6105	Yes	The probability of BSCU failure resulting in unannounced erroneous braking command to the NMV shall not exceed 2.0E-04 per flight.	Decelerate the wheels on ground	A	Derived	S18 WBS PSSA
S18-WBS-R-6106	Yes	The probability of BSCU failure resulting in the loss of command to open the SOV shall not exceed 2.0E-04 per flight.	Decelerate the wheels on ground	A	Derived	S18 WBS PSSA
S18-WBS-R-6107	Yes	The probability of BSCU failure resulting in unintended closure of the S/ASV shall not exceed 2.0E-04 per flight.	Decelerate the wheels on ground	A	Derived	S18 WBS PSSA
S18-WBS-R-6108	Yes	The SOV and NMV commands shall be provided by the BSCU upon loss of either airplane electrical power input.	Decelerate the wheels on ground	A	Derived	S18 WBS PSSA
S18-WBS-R-6109	Yes	When "HYD 1 Enable" output is enabled, then "Alt/Emer Ctrl" output shall be disabled.	Decelerate the wheels on ground	A	Derived	S18 WBS PSSA
S18-WBS-R-6110	Yes	No single failure shall cause erroneous NMV command and inhibit the SOV function.	Decelerate the wheels on ground	A	Derived	S18 WBS PSSA
S18-WBS-R-6111	Yes	The probability of "Loss of Normal Brake System Hydraulic Components" shall be less than 3.3E-05 per flight.	Decelerate the wheels on ground	A	Derived	S18 WBS PSSA
S18-WBS-R-6112	Yes	The probability of "Loss of Alternate/Emergency Brake System Hydraulic Components" shall be less than 3.3E-05 per flight.	Decelerate the wheels on ground	A	Derived	S18 WBS PSSA

(Editor's Note: Requirements that were designated as "safety" and "derived" (e.g., S18-WBS-R-2997) may or may not have been developed as part of the safety assessment process but have been communicated to the safety assessment process as existing requirements. Any requirements identified as necessary to achieve the safety objectives were identified and tagged as safety requirements.)

(Editor's Note: The rationale and specific safety reliance of the safety requirements were captured in the safety assessment data. An alternate solution could have been to capture similar rationale in the system requirements data.)

E.4.6.8 Allocation of System-Level Requirements

The following requirements in [Table E20](#) were provided to the BSCU supplier as the result of the final system architecture and requirements determined in [Table E19](#).

Table E20 - Wheel Brake System requirements allocated to BSCU development

Requirement Number	Safety	Requirement	FDAL
S18-WBS-R-0047	No	The Wheel Brake System shall meet safety requirements while operating in an average atmospheric radiation environment per the IEC 62396 standard with an altitude of 40000 feet and a latitude of 45 degrees.	A
S18-WBS-R-0509	No	The Wheel Brake System shall have dual BSCU command functions.	A
S18-WBS-R-1613	No	Hydraulic pressure shall be controlled for weight, autobrake mode, ground speed, wheel rotation, brake temperature and deceleration rate in accordance with graphs given in S18 brake force analysis AAS18-XXX.	A
S18-WBS-R-2986	Yes	The wheel brake command function of the BSCU shall be developed as FDAL A.	A
S18-WBS-R-2997	Yes	The BSCU shall have two independent command channels.	A
S18-WBS-R-6104	Yes	The probability of BSCU failure resulting in loss of a valid braking command output to the NMV shall not exceed 2.0E-04 per flight.	A
S18-WBS-R-6105	Yes	The probability of BSCU failure resulting in unannounced erroneous braking command to the NMV shall not exceed 2.0E-04 per flight.	A
S18-WBS-R-6106	Yes	The probability of BSCU failure resulting in the loss of command to open the SOV shall not exceed 2.0E-04 per flight.	A
S18-WBS-R-6107	Yes	The probability of BSCU failure resulting in unintended closure of the S/ASV shall not exceed 2.0E-04 per flight.	A
S18-WBS-R-6108	Yes	The SOV and NMV commands shall be provided by the BSCU upon loss of either airplane electrical power input.	A
S18-WBS-R-6109	Yes	When "HYD 1 Enable" output is enabled, then "Alt/Emer Ctrl" output shall be disabled.	A
S18-WBS-R-6110	Yes	No single failure shall cause erroneous NMV command and inhibit the SOV function.	A

E.4.6.9 BSCU Architecture Study

As noted in the WBS descriptions provided in [E.4.3](#) and [E.4.6.1.2](#), braking on ground could be commanded manually via brake pedals, or automatically via autobrake command without the input from the brake pedal. The autobrake functionality allows the pilot to pre-arm the deceleration rate prior to takeoff or landing. Preliminary safety analysis determined that a single command function without a monitor cannot meet the safety requirements. The development team worked with the safety team to define monitoring that allows the system to meet the safety requirements. Because the BSCU has two independent command channels, each command channel needs to monitor the signals. This results in the implementation of the monitor functions shown in [Figure E12](#).

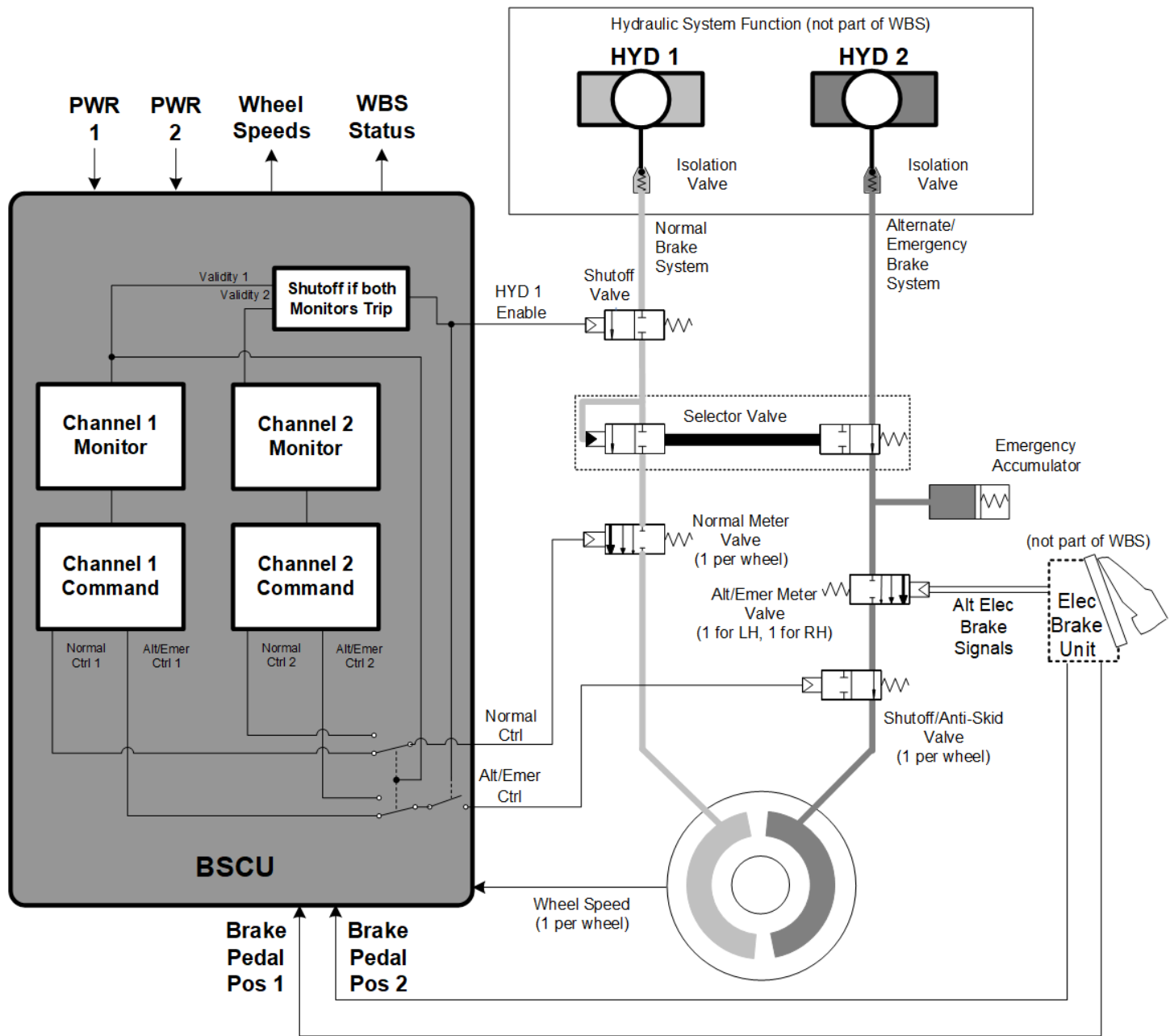


Figure E12 - BSCU architecture update iteration: addition of monitor functions

E.4.6.10 BSCU Description

The BSCU functions include, but are not limited to:

- a. Provide NORMAL Mode commands.
- b. Provide Shutoff Valve commands.

The BSCU has a dual command/monitor architecture (Channels #1 and #2). Channel #1 is the primary channel providing braking and anti-skid command to the wheel brakes. Channel #2 is the backup channel in case of Channel #1 failure.

When in NORMAL Mode, brake pedal inputs from the EBU (Brake Pedal Pos 1 and 2) are provided to each Command and Monitor of both channels. When HYD 1 is available, Command 1 provides braking and anti-skid commands by controlling the HYD 1 enable, Alt/Emer Ctrl, and Normal Ctrl outputs to the Shutoff Valve, Shutoff/Anti-Skid Valves, and Normal Meter Valves, respectively. When HYD 1 is not available, Command 1 provides anti-skid commands by controlling the HYD 1 Enable and Alt/Emer Ctrl outputs to the Shutoff Valve and Shutoff/Anti-Skid Valves, respectively. The Command 1 control outputs are wrapped back to Monitor 1 for comparison. A miscompare results in an invalid command detection by Monitor 1, which switches the Normal Ctrl and Alt/Emer Ctrl outputs from Command 1 to Command 2 and sets Validity 1 to invalid.

Similarly, Command 2 control outputs are wrapped back to Monitor 2 for comparison. A mismatch results in an invalid command detection by Monitor 2, which sets Validity 2 to invalid. When both monitors set their validity outputs to invalid, the control output (HYD 1 enable) power to the Shutoff Valve is removed (closing the valve) to isolate HYD 1 hydraulic system, the control output (alt/emerg ctrl) power to the Shutoff/Anti-Skid Valves is removed (opening the valves) to enable HYD 2 hydraulic system, i.e., switch to the Alternate/Emergency Brake System, and the braking command is transferred from the normal meter valves (controlled by the BSCU) to the alt/emerg meter valves (controlled by the EBU). The alt/emerg meter valves are then directly controlled by the EBU using HYD 2 hydraulic system and anti-skid is not available.

Channels #1 and #2 have similar command hardware and software, similar monitor hardware, but dissimilar monitor software. [Figure E13](#) illustrates an example of the functional allocation of these functions to the different hardware and software contained in the BSCU.

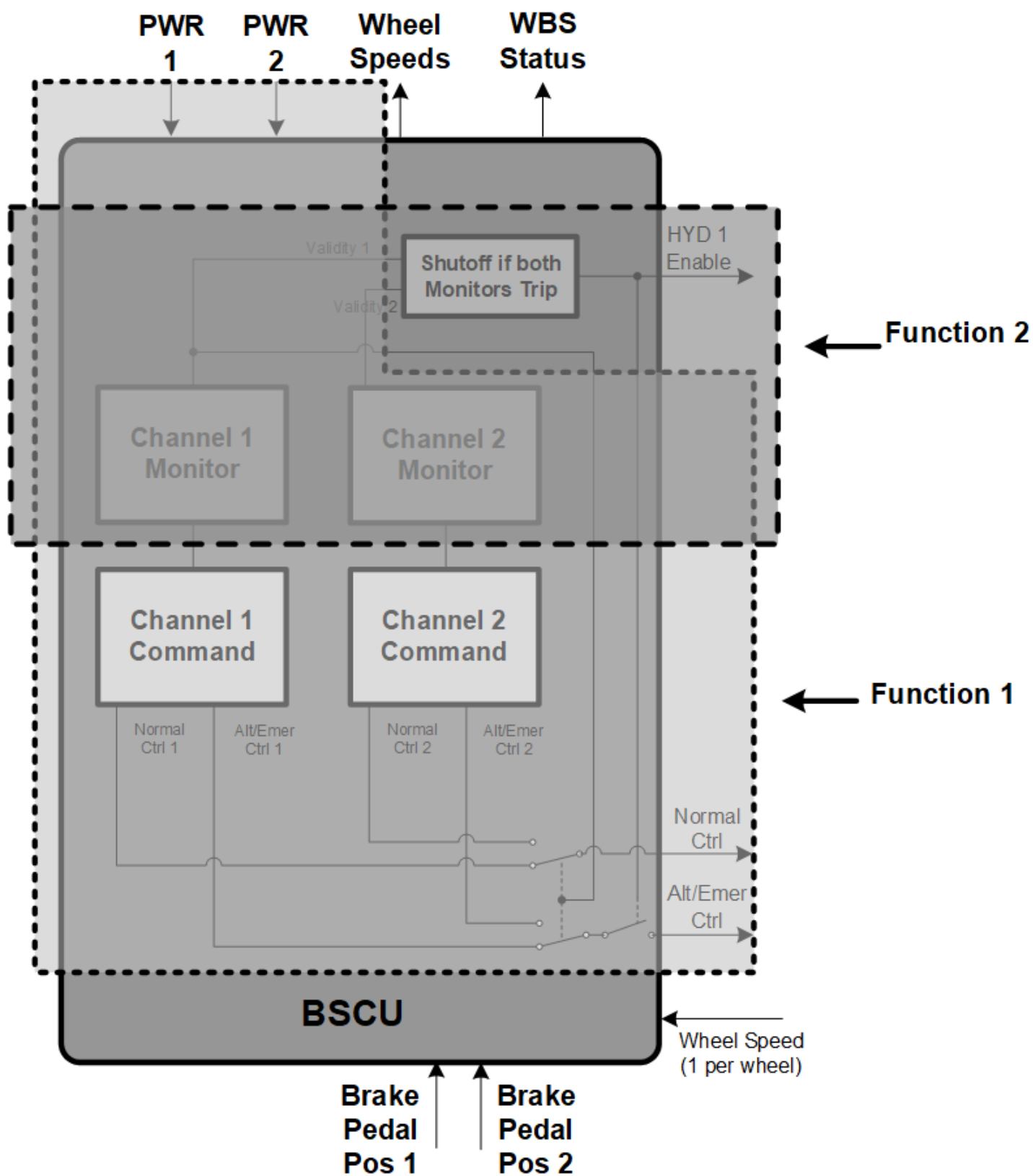


Figure E13 - BSCU functional mapping

E.4.6.11 BSCU Requirements Traceability

The S18 BSCU requirements traceability followed the specification tree as shown in [Figure E14](#). The traceability was bi-directional with the higher-level requirements being parent to the requirements below. Derived requirements had rationale since they do not trace to higher-level requirements.

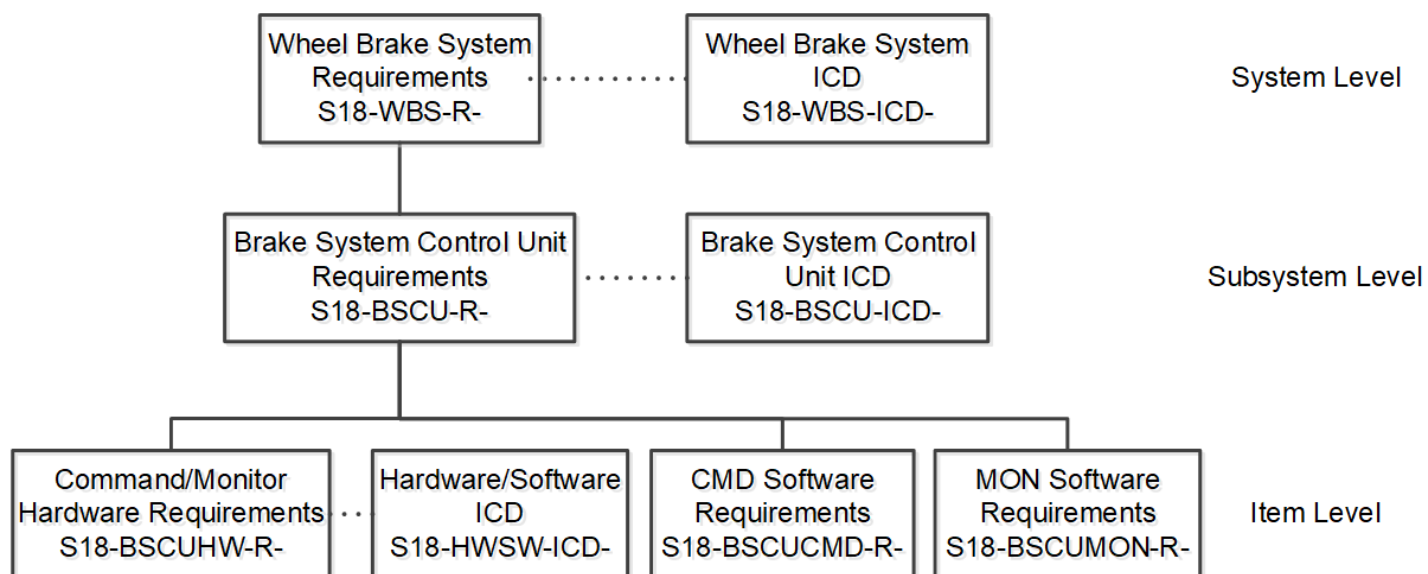


Figure E14 - BSCU requirements traceability

E.4.6.12 BSCU Hardware and Software IDAL Assignments from the BSCU PSSA Process

The BSCU PSSA process established the relevant item-level safety requirements and the Fault Tree Analysis (FTA) conducted confirmed that those requirements could be met with the architecture proposed. The FTA identified the set of Item Development Assurance Levels (IDALs), as shown in [Table E21](#), allocated to the hardware and software elements of the proposed BSCU in order to meet the overall safety objectives. Section [E.4.6.4](#) outlines these key safety requirements derived from the WBS-level assessment that are allocated to the complex hardware and software items of the BSCU.

Assignment of IDALs to functions within the WBS was done as part of the PSSA process in ARP4761A/ED-135 Appendix Q.

Table E21 - IDAL assignments (BSCU PSSA outputs)

Item	IDAL Assignment
Command 1 HW	A
Command 1 SW	B
Command 2 HW	A
Command 2 SW	B
Monitor 1 HW	A
Monitor 1 SW	B
Monitor 2 HW	A
Monitor 2 SW	B

E.4.6.13 Proposed Requirements from the BSCU PSSA Process

From the objectives and requirements identified at the WBS level and WBS FHAs, and through the analyses carried out during the BSCU PSSA process, proposed safety requirements were identified as shown in [Table E22](#).

Table E22 - Proposed safety requirements (BSCU PSSA outputs)

BSCU PSSA ID	Proposed Safety Requirement
BSCU-001	Channel #1 shall have physical independence from Channel #2.
BSCU-002	Within each channel, the command function shall have physical independence from the monitor function.
BSCU-003	The COMX software and MONX software shall be developed using different software requirement specifications.
BSCU-004	Within each channel, a power supply monitor shall shutoff the power supply when any voltage is detected to be out of specification.
BSCU-005	Within each channel, the operational state of the power supply shall have no effect on the operational state of the power supply monitor.
BSCU-006	Upon Channel #1 indicating an "invalid" state, Channel #2 shall command the "Normal Control" output.
BSCU-007	Upon both Channel #1 and Channel #2 indicating "invalid" states, the "HYD 1 Enable" output shall be disabled.
BSCU-008	Upon either Channel #1 or Channel #2 indicating a validity error, a WBS annunciation shall be output from the BSCU.
BSCU-009	At power-up the BSCU shall perform a self-test of Channel #2 including the ability to switch control from Channel #1 to Channel #2.
BSCU-010	At power-up each channel of the BSCU shall implement a test of its power supply monitor.
BSCU-011	If the power-up test fails, a WBS annunciation shall be output from the BSCU.

Assumptions identified by the BSCU PSSA were passed back to the WBS PSSA and communicated to the development team, as shown in [Table E23](#).

Table E23 - BSCU PSSA assumptions communicated to WBS level

Assumption to WBS Level
When "HYD 1 Enable" is disabled, the BSCU brake control outputs are not used.
A BSCU maintenance action will be initiated, if the WBS Status indicates a failure.

The first of these assumptions was considered as applicable to the BSCU operation as a functional requirement, and BSCU requirement S18-BSCU-R-0009 was created. The second was considered as a specification for a user maintenance action, which is outside the scope of WBS and BSCU development as captured in this appendix.

E.4.6.14 BSCU Requirements

[Table E24](#) is a sample of the requirements for the BSCU, which shows traceability back to the WBS requirements.

Table E24 - Brake System Control Unit Requirements Specification BSCU-0001-100 - Version 1.0

Requirement Number	Safety	Requirement	Associated Function (if Applicable) and the FDAL		Traced from
S18-BSCU-R-0001	Yes	The wheel brake command function of the BSCU shall be developed as FDAL A.	Provide WBS monitoring and commands	A	S18-WBS-R-2986
S18-BSCU-R-0002	Yes	The probability of BSCU failure resulting in loss of a valid braking command output to the NMV shall not exceed 2.0E-04 per flight.	Provide WBS monitoring and commands	A	S18-WBS-R-6104
S18-BSCU-R-0003	Yes	The probability of BSCU failure resulting in unannunciated erroneous braking command to the NMV shall not exceed 2.0E-04 per flight.	Provide WBS monitoring and commands	A	S18-WBS-R-6105
S18-BSCU-R-0004	Yes	The probability of BSCU failure resulting in the loss of command to open the SOV shall not exceed 2.0E-04 per flight.	Provide WBS monitoring and commands	A	S18-WBS-R-6106
S18-BSCU-R-0005	Yes	The probability of BSCU failure resulting in unintended closure of the S/ASV shall not exceed 2.0E-04 per flight.	Provide WBS monitoring and commands	A	S18-WBS-R-6107
S18-BSCU-R-0006	Yes	The SOV and NMV commands shall be provided by the BSCU upon loss of either airplane electrical power input.	Provide WBS monitoring and commands	A	S18-WBS-R-6108
S18-BSCU-R-0007	Yes	When “HYD 1 Enable” output is enabled, then “Alt/Emer Ctrl” output shall be disabled.	Provide WBS monitoring and commands	A	S18-WBS-R-6109
S18-BSCU-R-0008	Yes	No single failure shall cause erroneous NMV command and inhibit the SOV function.	Provide WBS monitoring and commands	A	S18-WBS-R-6110

Requirement Number	Safety	Requirement	Associated Function (if Applicable) and the FDAL		Traced from
S18-BSCU-R-0009	Yes	When “HYD 1 Enable” is disabled, the BSCU brake control outputs shall be disabled.	Provide WBS monitoring and commands	A	S18 BSCU PSSA
S18-BSCU-R-0010	No	BSCU shall calculate the required hydraulic pressure as a function of weight, autobrake mode, ground speed, wheel rotation, brake temperature and deceleration rate in accordance with graphs given in S18 brake force analysis AAS18-XXX.	Provide WBS monitoring and commands	A	S18-WBS-R-1613
S18-BSCU-R-0011	No	BSCU shall control the meter valves.	Provide WBS monitoring and commands	A	Derived
S18-BSCU-R-0012	No	BSCU shall control the anti-skid valves.	Provide WBS monitoring and commands	A	Derived
S18-BSCU-R-0013	No	BSCU shall control the shut off valves.	Provide WBS monitoring and commands	A	Derived
S18-BSCU-R-0027	No	Each BSCU shall have two command channels.	Provide WBS monitoring and commands	A	S18-WBS-R-0509
S18-BSCU-R-0051	No	Each BSCU channel shall have a monitoring unit.	Provide WBS monitoring and commands	A	S18-WBS-R-0509
S18-BSCU-R-0075	No	Each BSCU channel shall have a control unit.	Provide WBS monitoring and commands	A	S18-WBS-R-0509
S18-BSCU-R-0099	Yes	Each BSCU channel shall have independent power.	Provide WBS monitoring and commands	A	Derived
S18-BSCU-R-0123	Yes	Each BSCU channel shall have independent pedal input to both control and monitoring units.	Provide WBS monitoring and commands	A	Derived
S18-BSCU-R-0124	Yes	The command channels of each BSCU shall be independent of each other.	Provide WBS monitoring and commands	A	S18-WBS-R-2997
S18-BSCU-R-0126	Yes	The BSCU hardware shall be developed to IDAL A.	Provide WBS monitoring and commands	A	S18 BSCU PSSA
S18-BSCU-R-0127	Yes	The BSCU software shall be developed to IDAL B.	Provide WBS monitoring and commands	A	S18 BSCU PSSA
S18-BSCU-R-0128	No	BSCU shall meet safety requirements while operating in an average atmospheric radiation environment per the IEC 62396 standard with an altitude of 40000 feet and a latitude of 45 degrees.	Provide WBS monitoring and commands	A	S18-WBS-R-0047
S18-BSCU-R-0201	Yes	Channel #1 shall have physical independence from Channel #2.	Provide WBS monitoring and commands	A	S18 BSCU PSSA
S18-BSCU-R-0202	Yes	Within each channel, the command function shall have physical independence from the monitor function.	Provide WBS monitoring and commands	A	S18 BSCU PSSA
S18-BSCU-R-0203	Yes	The COMX software and MONX software shall be developed using different software requirement specifications.	Provide WBS monitoring and commands	A	S18 BSCU PSSA
S18-BSCU-R-0204	Yes	Within each channel, a power supply monitor shall shutoff the power supply when any voltage is detected to be out of specification.	Provide WBS monitoring and commands	A	S18 BSCU PSSA
S18-BSCU-R-0205	Yes	Within each channel, the operational state of the power supply shall have no effect on the operational state of the power supply monitor.	Provide WBS monitoring and commands	A	S18 BSCU PSSA
S18-BSCU-R-0206	Yes	Upon Channel #1 indicating an “invalid” state, Channel #2 shall command the “Normal Control” output.	Provide WBS monitoring and commands	A	S18 BSCU PSSA
S18-BSCU-R-0207	Yes	Upon both Channel #1 and Channel #2 indicating “invalid” states, the “HYD 1 Enable” output shall be disabled.	Provide WBS monitoring and commands	A	S18 BSCU PSSA
S18-BSCU-R-0208	Yes	Upon either Channel #1 or Channel #2 indicating a validity error, a WBS annunciation shall be output from the BSCU.	Provide WBS monitoring and commands	A	S18 BSCU PSSA
S18-BSCU-R-0209	Yes	At power-up, the BSCU shall perform a self-test of Channel #2 including the ability to switch control from Channel #1 to Channel #2.	Provide WBS monitoring and commands	A	S18 BSCU PSSA
S18-BSCU-R-0210	Yes	At power-up, each channel of the BSCU shall implement a test of its power supply monitor.	Provide WBS monitoring and commands	A	S18 BSCU PSSA
S18-BSCU-R-0211	Yes	If the power-up test fails, a WBS annunciation shall be output from the BSCU.	Provide WBS monitoring and commands	A	S18 BSCU PSSA

(Editor's Note: The rationale and specific safety reliance of the safety requirements were captured in the safety assessment data. An alternate solution could have been to capture similar rationale in the system requirements data.)

(Editor's Note: The generic wording of requirement S18-BSCU-R-0128 does not imply nor assume that the BSCU is actively operating during flight at high altitude. The expressed need is that the BSCU should withstand the specified environment without impacting compliance to its safety requirements, considering that atmospheric radiation encountered during flight may affect the BSCU, resulting in lasting effects on functionality and capability.)

(Editor's Note: Requirements that were designated as "safety" and "derived" (e.g., S18-BSCU-R-0099 and S18-BSCU-R-0123) may or may not have been developed as part of the safety assessment process but have been communicated to the safety assessment process as existing requirements. Any requirements identified as necessary to achieve the safety objectives were identified and tagged as safety requirements.)

E.4.7 Wheel Brake System Requirements Validation and Assumption Management

The WBS supplier used the validation process documented in their Development Assurance Plan to complete WBS validation activities. It is understood that similar activity was undertaken at other levels of the hierarchy by those teams responsible for those requirements, and are captured in life cycle artifacts at those levels.

The requirements validation involved performing activities which provided sufficient evidence that each requirement was correct and that the requirement set, as a whole, was complete. The methods listed in main body [5.4](#) assisted in determining the correctness and completeness of requirements and requirement sets.

E.4.7.1 Wheel Brake System Requirements Correctness Checks

Requirement correctness checks were performed on the requirements at each level of the requirements hierarchy for the WBS. The correctness checks were performed using the checklist in [Table E25](#), which was based on main body [5.4.3](#).

(Editor's Note: For brevity's sake, only a portion of the WBS correctness check output is shown in this example.)

Table E25 - Wheel Brake System requirement(s) correctness check (excerpt)

Correctness Check Criteria	Results (Yes/No)	Problem Report Number
Is each requirement correctly stated?	No	PR-WBS-0007
Is each requirement necessary for the set of requirements to be complete?	Yes	
Is each requirement appropriately written as a single requirements (rather than combined with others)?	Yes	
Does the requirement set correctly reflect the safety analyses?	Yes	
Does each requirement correctly state the FDAL of its associated function?	Yes	
Are all assumptions against the requirement captured?	Yes	

During the validation correctness checks, it was found that requirement S18-WBS-R-0065 was not correctly stated. Problem Report PR-WBS-0007 was created to address the deficiency. See Problem Report PR-WBS-0007 in [Figure E15](#).

Problem Report: PR-WBS-0007
Description: The requirement S18-WBS-R-0065: The emergency accumulator shall be capable of providing 2000 psi. Confirm that 2000 psi is needed to stop the airplane.
Action: In the analysis to confirm the 2000 psi value found the following: a. Simulation of the architecture b. Airframer provided the following information: - Need to be able to apply the brakes six times during an emergency landing deceleration (need to be able to press and release to avoid skidding when anti-skid is not available). - The design of the brake actuator uses approximately 300 psi per full depress of the mechanical pedal (source of information is brake actuator supplier and is based on volume of actuator versus volume of emergency accumulator plus hydraulic line). - Design margin of 50%.
Resolution/Conclusion: All information has been evaluated about emergency accumulator and an emergency accumulator capable of at least 1800 psi is required. The requirement is corrected as below; S18-WBS-R-0065: The emergency accumulator shall provide a minimum of 1800 psi.

Figure E15 - Example Problem Report during validation

E.4.7.1.1 Additional Checks

There were two WBS requirements that needed further analysis. Model-based analysis was used to finalize validation of these requirements.

Additional system requirements validated were:

S18-WBS-R-2973: Operation of the Alternate/Emergency Brake System shall be precluded when the Normal Brake System is in use.

S18-WBS-R-2975: The emergency accumulator shall be attached to the HYD 2 hydraulic line between the Selector Valve and the Alternate/Emergency Meter Valve.

The validation method used for these requirements was:

Model-based analysis (i.e., simulation).

The results of model-based analysis for these requirements were:

Architecture 3 has been simulated and the behavior model for each system component has been established. Results of the modeling are that the schematic of the WBS architecture does not work and there are some mistakes in the schematic. Risk of hydraulics supply to brakes is possible in NORMAL Mode by emergency accumulator. Therefore, emergency accumulator position needs to be changed to meet the requirement S18-WBS-R-2973. Derived system requirements from architecture 3 are not correct and need to be changed.

Problem Report PR-WBS-0008 was created to capture the problem. The proposed change to the requirement text was: The emergency accumulator shall be attached to the HYD 2 hydraulic line between the isolation valve and the Selector Valve.

This proposed requirement change was reviewed as part of the PSSA process, and it was determined that the change would create an unacceptable failure condition whereby the Selector Valve could make the ALTERNATE and EMERGENCY Modes unavailable. In response to that safety assessment, the design team decided not to move the emergency accumulator and instead created a derived requirement to charge the emergency accumulator during initialization of the WBS.

The new derived requirement S18-WBS-R-2976 was: Wheel Brake System shall charge the emergency accumulator prior to normal operation. This requirement was then allocated to the BSCU development, which created a new BSCU requirement S18-BSCU-R-0020 which states: "BSCU shall control the valves to charge the emergency accumulator prior to normal operation."

(Editor's Note: As part of the PSSA process, the safety engineer reviewed all derived requirements in order to evaluate any impact on the safety assessment.)

E.4.7.2 Wheel Brake System Requirements Completeness Checks

Completeness checks were performed by reviewing sets of WBS requirements that had been through correctness checks, to confirm that the set of requirements captured the functionality specified by the requirements. Checklists prepared in accordance with main body [5.4.4](#) were used for completeness checks. [Table E26](#) shows an excerpt of the checklist that was used by the system supplier to validate completeness of the WBS requirements.

Table E26 - Wheel Brake System requirements set completeness check (excerpt)

Completeness Check Criteria	Results (Yes/No)	Problem Report Number
Is traceability with supporting rationale established from airplane level to item level?	Yes	
Are all requirements (performance, operational, customer, etc.) from contract fully covered?	Yes	
Are all higher-level functions that are allocated to this system fully covered?	Yes	
Are all safety requirements represented in the requirements set (from FHA, previous safety assessments, in-service lessons learned, novelties, etc.)?	Yes	
Are all regulatory standards and guidelines represented in the requirements set?	Yes	
Industry and company design standards represented in the requirements set?	Yes	
Flight operations and maintenance scenarios represented in the requirements set?	Yes	
Are validation methods defined for each requirement (engineering review, traceability, analysis, test)?	Yes	
Does the set of requirements completely define the system/item?	No	PR-WBS-0008 PR-WBS-0009
Are all interface requirements to other systems, people and processes identified and agreed?	Yes	
Are the constraints associated with each interface defined in sufficient detail for the interface to be realized?	Yes	
Are the functional requirements set fully allocated and traced to the system architecture?	Yes	
Are safety requirements checked using experience from previous programs and novelties of the current program?	Yes	

In addition to the creation of a derived requirement that was discussed under PR-WBS-0008 in [E.4.7.1](#), the completeness check determined that the set of requirements defining brake temperature was not sufficient. Brake temperature is very important data to prevent the possibility of a fire hazard. Problem Report PR-WBS-0009 was created to resolve the issue and requirements were identified to be added to the Wheel Brake System Requirements Specification.

E.4.7.3 Wheel Brake System Requirements Validation Summary

(Editor's Note: Safety was part of the review of the validation summary to provide concurrence there was no additional validation related impacts to the safety assessment.)

At the conclusion of the validation activity (correctness and completeness checks), a summary was created to describe the activities that were performed and to provide an overall conclusion for the validation of the WBS requirements. This summary included the validation matrix that provided validation data for each requirement. The following is an excerpt from the WBS validation summary report:

"The applicable requirements were reviewed, analyzed and validated for the Wheel Brake System of the S18 airplane. The validation of the requirements has been performed in accordance with the Wheel Brake System Development Assurance Plan. The validation status of all requirements was raised to 'valid.' Model-based analysis has been used to support the validation of requirements."

(Editor's Note: See main body [5.4.7.3](#) for discussion of the contents of the validation summary.)

E.4.7.3.1 Wheel Brake System Validation Matrix

A validation matrix was included as part of the WBS validation summary to show the data resulting from the completion of validation activities, as shown in the matrix excerpt in [Table E26](#).

(Editor's Note: Typically this matrix should contain all of the system requirements, but only two are shown here for brevity.)

Classes of requirements, as defined in main body [5.3.1](#) (safety requirements, functional requirements, customer requirements, operational requirements, performance requirements, etc.), that are part of the WBS were validated by using the same validation process.

The last two columns in the WBS validation matrix excerpt were used to show independence. Validation data contained in the matrix was independently reviewed based on the FDAL assignments as part of correctness and completeness checks. In this matrix, the requirement owner (the person who wrote the requirement) was shown to be different from the validation owner (the person reviewing the requirement for correctness and completeness) in order to maintain process independence.

(Editor's Note: Validation matrix data was updated regularly during the development. The final matrix was generated when all validation conclusions become valid or captured in a problem report.)

(Editor's Note: While there are differences in the formats of the validation matrices shown in [Tables E27](#) and [E8](#), both matrices contain the information recommended in main body [5.4](#). Additional data could have been included based on the developer's policies or procedures.)

Table E27 - Wheel Brake System Validation Matrix (excerpt)

Requirement ID and Description	Rationale or Source of Reqmt	Associated Function(s) and Associated FDAL		Meet Correctness Criteria?	Meet Completeness Criteria?	Validation Method(s)	Validation Evidence	Validation Conclusion (Valid/Invalid)	Problem Report	Req Owner	Validation Owner
S18-WBS-R-0321 Loss of all wheel braking (unannunciated or annunciated) during landing or RTO shall be extremely remote as defined in 14 CFR/CS 25.1309	System FHA	Decelerate the wheels on the ground	A	Yes	Yes	Engineering review Traceability Analysis-Safety	Validation Checklist 3458 Traceability Matrix WBS-0001-103 Report number XXXX—System Preliminary FTA	Valid		JRW	RLC
S18-WBS-R-2975 The emergency accumulator shall be attached to the HYD 2 hydraulic line between the Selector Valve and the Alternate/Emergency Meter Valve.	S18-ACFT-R-0184	Decelerate the wheels on the ground	A	Yes	Yes	Engineering review Traceability Model-based analysis	Validation Checklist 3459 Traceability Matrix WBS-0001-103 Architecture simulation results	Valid	PR-WBS-0008	JRW	RLC

E.4.7.4 Wheel Brake System Assumption Management

For tracking purposes, the assumptions received from the SFHA, as shown in [Table E10](#), have had identifier prefix “SFHA” added. For tracking purposes, WBS assumptions were added to [Table E9](#) assumptions as shown in [Table E28](#) for completeness. The assumptions from the WBS PSSA, as shown in [Table E14](#), have had the prefix “WBS PSSA ASMP” added. The assumptions from the BSCU PSSA, as shown in [Table E23](#), have had the prefix “BSCU PSSA ASMP” added.

(Editor's Note: The communication of airplane-level assumptions to lower levels may not include assumptions for other levels or systems.)

Table E28 - WBS level safety assumptions

Assumption Identifier	Description	Confirmation
AFHA ASMP 3.2.2-1	Overrunning the runway length above “XYZ” knots is considered a high speed overrun.	Coord Memo S18a-CM00Y1
AFHA ASMP 3.2.2-2	The directional aspect of asymmetric failures of deceleration systems are functionally addressed by failure conditions of the “Control Direction on Ground” airplane function. (See function 2.3 AFHA.)	Coord Memo S18a-CM00Y1
AFHA ASMP 3.2.2-3	The flight crew will divert to a suitable airfield if aware of a condition that renders the airplane incapable of landing at the originally intended destination.	Coord Memo S18a-CM00Y1
AFHA ASMP 3.2.2-4	Crew awareness is not a factor for the identified malfunction, as these are immediately evident due to aircraft behavior and do not have their effects intensified or mitigated by crew awareness features.	Coord Memo S18a-CM00Y1
AFHA ASMP 3.2.2-5	The flight crew will not initiate an RTO in response to an annunciated failure of deceleration features during takeoff due to alert suppression.	Coord Memo S18a-CM00Y1
AFHA ASMP 3.2.2-6	Taxi is performed at groundspeeds below 30 knots.	Coord Memo S18a-CM00Y1
AFHA ASMP 3.2.2-7	Landings with failure condition in combination with environmental factors have been assessed.	Coord Memo S18a-CM00Y1
AFHA ASMP 3.2.2-8	Failures of deceleration capability will be detected and annunciated by on-board systems.	Coord Memo S18a-CM00Y1
PASA-ASMP-01	It is assumed that the high speed overrun is above 30 knots and low speed overrun is below (or equal) 30 knots. This assumption has been derived from the AFHA assumption ASMP 3.2.2-1 and ASMP 3.2.2-6 for the establishment of the criteria and terms of “high-speed overrun” and “low speed overrun.”	Coord Memo S18a-CM00Y2
PASA-ASMP-02	The degraded state of systems considered in the CoFFE analysis has been defined such as half functional capability.	Coord Memo S18a-CM00Y2
PASA-ASMP-03	Wheel Brake System uses ground detection information together with the wheel speed information. These functions are developed independently so that even if Erroneous Ground Detection Information (AGS.MF) is provided on ground, i.e., with false in-flight status, wheel braking function is available if there is correct wheel speed information.	Coord Memo S18a-CM00Y2
PRA-UERF-ASSUMPTION-01	The BSCU is installed in the avionics compartment located in the nose fuselage section of the airplane, forward of the UERF area.	PRA
PRA-UERF-ASSUMPTION-02	The electrical lines transmitting the brake control signals from the Electric Brake Unit to the BSCU are routed “to the shortest” from one end to the other in the nose fuselage section of the airplane, thus are kept forward of the UERF area.	PRA
PRA-UERF-ASSUMPTION-03	The hydraulic reservoirs and the high pressure manifolds of the hydraulic subsystems 1 and 2 are located in the fuselage, aft of the UERF area.	PRA
PRA-UERF-ASSUMPTION-04	The hydraulic power distribution lines from the engine driven pumps to the respective hydraulic reservoirs and high pressure manifolds exit the engine pylons aft of the wing rear spars, then run along, aft of, the wing rear spars to the fuselage, and are kept aft of the UERF area inside the fuselage.	PRA

Assumption Identifier	Description	Confirmation
PRA-UERF-ASSUMPTION-05	The hydraulic distribution lines routed to the forward part of the fuselage to supply hydraulic equipment located forward of the UERF area are fitted with appropriate isolation means located aft of the UERF area.	PRA
PRA-UERF-ASSUMPTION-06	The power supply lines from the electrical power center(s) to the EBU are routed "to the shortest" from one end to the other in the nose fuselage section of the airplane.	PRA
PRA-UERF-ASSUMPTION-07	In case of cabin depressurization the airplane altitude will be limited by procedure to 10000 feet.	PRA
SFHA SASP 1.1-1	Loss of 80% or more of deceleration capability is considered a total loss of the deceleration means.	Reply to Coord Memo S18a-CM00Z1
SFHA SASP 1.1-2	Failure Conditions are assessed on wet runway conditions.	Reply to Coord Memo S18a-CM00Z1
SFHA SASP 1.1-3	RTO is considered an operational condition caused either by an external event or by a system failure (annunciated or perceived by the flight crew) during takeoff run.	Reply to Coord Memo S18a-CM00Z1
SFHA SASP 1.1-4	The flight crew will not initiate an RTO due to an annunciated failure of the deceleration function after V1.	Reply to Coord Memo S18a-CM00Z1
SFHA SASP 1.1-5	Taxi is performed at groundspeeds below 30 knots.	Reply to Coord Memo S18a-CM00Z1
SFHA SASP 1.1-6	Overrunning the runway length at or above "XYZ" knots is considered a high speed overrun.	Reply to Coord Memo S18a-CM00Z1
SFHA SASP 1.1-7	Overrunning the runway length below "XYZ" knots is considered a low speed overrun.	Reply to Coord Memo S18a-CM00Z1
SFHA SASP 1.1-8	Failures of deceleration capability will be detected and annunciated by on-board systems.	Reply to Coord Memo S18a-CM00Z1
WBS PSSA ASMP 1	The probability of "Loss of Normal Braking System Hydraulic Equipment" will be less than 3.3E-05 per flight.	S18-WBS-R-6111
WBS PSSA ASMP 2	The probability of "Loss of Alternate Braking System Hydraulic Equipment" will be less than 3.3E-05 per flight.	S18-WBS-R-6112
WBS PSSA ASMP 3	The probability of 7 or more wheel speed sensors erroneous or inoperative will be less than 1.0E-07 per flight.	Reply to Coord Memo S18a-CM00ZZ
WBS PSSA ASMP 4	The failure rate for loss of an airplane electrical power bus will be less than 1.0E-04 per hour of flight.	Reply to Coord Memo S18a-CM00ZZ
WBS PSSA ASMP 5	The failure rate for loss of a left brake pedal position input will be less than 1.0E-04 per hour of flight.	Reply to Coord Memo S18a-CM00ZZ
WBS PSSA ASMP 6	Airplane electrical power bus 1 is independent from airplane electrical power bus 2.	Reply to Coord Memo S18a-CM00ZZ
WBS PSSA ASMP 7	HYD 1 hydraulic system is independent from HYD 2 hydraulic system.	Reply to Coord Memo S18a-CM00ZZ
BSCU PSSA ASMP 1	When "HYD 1 Enable" is disabled, the BSCU brake control outputs are not used.	S18-BSCU-R-0009
BSCU PSSA ASMP 2	A BSCU maintenance action will be initiated, if the WBS Status indicates a failure.	Reply to Coord Memo S18a-CM00ZZ

Several assumptions from the WBS and BSCU PSSAs have been written as requirements. SFHA assumptions were communicated to the airplane development process in Coord Memo S18a-CM00Z1, and a reply Coord Memo was received as confirmation of applicability to the development. WBS PSSA assumptions 3 through 7 and BSCU PSSA assumption 2 were communicated to the airplane development process in Coord Memo S18a-CM00ZZ, and a reply Coord Memo was received as confirmation of applicability to the development.

E.5 WHEEL BRAKE SYSTEM (WBS) IMPLEMENTATION AND VERIFICATION

E.5.1 Wheel Brake System Implementation

The outputs of implementation included software source code and breadboard or prototype hardware for the brake pedal control system, brake hydraulic system, anti-skid and autobrake systems, brakes/wheels/tires, brake temperature monitor system, brake cooling fans, tire pressure indication system, and tire and brake monitoring system. Initial item electronic hardware and software integration was on breadboards, prototypes, computer emulation, or lab or flight-worthy articles.

E.5.2 Brake System Control Unit Verification

When developing the BSCU requirements, the company tried to determine the method of verification at the time of generating the requirement. This was not possible in all cases, but the verification method was determined prior to the verification activities being performed. The BSCU verification matrix (excerpt) in [Table E29](#) shows some of the BSCU level requirements, their associated WBS level functions, the methods to be applied to verify each requirement, references to the verification procedures, verification results, any problem reports generated, and the verification conclusion (pass/fail).

Table E29 - Brake System Control Unit Verification Matrix (excerpt)

Requirement Number	Requirement	Associated Function(s)	Verification Method(s)	Verification Procedure/Results	Conclusion (Pass/Fail)
S18-BSCU-R-0013	BSCU shall control the shut off valves.	BSCU Wheel Brake Command	Analysis, System Integration Test	System Integration Test: S18-BSCU-SIT-0070 Software: BSCU Verification Procedure and Report	Pass
S18-BSCU-R-0126	The BSCU hardware shall be developed to IDAL A.	BSCU Wheel Brake Command	Analysis	BSCU Hardware Accomplishment Summary (HAS)	Pass
S18-BSCU-R-0127	The BSCU software shall be developed to IDAL B.	BSCU Wheel Brake Command	Analysis	BSCU Software Accomplishment Summary (SAS)	Pass
S18-BSCU-R-0020	BSCU shall control the valves to charge the emergency accumulator prior to normal operation.	BSCU Wheel Brake Command	System Integration Test	System Integration Test: S18-BSCU-SIT-0100	Fail; PR-BSCU-0010

(Editor's Note: The use of accomplishment summaries as verification data is included here to show that the development assurance processes for FDAL and IDAL as specified in the plans were followed.)

(Editor's Note: The BSCU verification matrix (excerpt) was based on the BSCU level requirements shown in [E.4.6.14](#). Since the FDAL of the BSCU wheel brake command function was A per S18-BSCU-R-0001, in accordance with main body [5.5.3](#), these requirements were verified with independence.

Requirements verification was preferred by test but included other methods when necessary to show complete verification of a requirement. Monitoring for unintended behavior was also required according to main body [4.6.4](#) and [5.5.5.3](#). Verification of safety requirements, particularly those dealing with failure rates or to show that a particular failure condition was extremely remote, was accomplished by a safety assessment.)

Once the verification method(s) for each requirement were reviewed and determined to be adequate for the FDAL and specifics of the requirement, the company began to develop verification procedures. In most cases, this involved formal test procedures. For formal test procedures, the company chose to develop a test setup and test procedures to run on that setup. The company called these tests their System Integration Tests (SITs). The formal SIT test procedures were documented on test procedure forms as shown in [Figure E16](#). Each test procedure was traced to one or more requirements depending on how many requirements the test exercised. After all of the identified test cases were identified, the test procedures were reviewed using the following checklist.

Test Procedure Inspection Checklist

- Does each test have a header that identifies the author, revision date, test objectives, required configuration, and initial setup?
- Is each test traceable to a specific requirement defined in the requirements?
- Does the test procedure define the exact sequence of steps required to execute the test?
- For each test, are the expected results clearly defined?
- Are the expected results consistent with the requirements?
- Are the test objectives achievable?
- Is test specimen configured and does it match data from manufacturer?
- Is test platform representative and calibrated?

After all of the identified verification procedures were written, the verification matrix was again reviewed. This time, each requirement was reviewed using the following checklist.

Requirement Verification Checklist

- Do the verification methods allocated to this requirement fully verify the requirement?
- Are any limits specified within the requirement accurately tested?

System Integration Test (SITs) Procedure																								
Test Number: S18-BSCU-SIT-0068		Date:																						
Author: T. Smith																								
Test Procedure Name: Maximum Braking Force Calculation																								
Requirements Coverage:	S18-BSCU-R-0006 S18-BSCU-R-0007 S18-BSCU-R-0010 S18-BSCU-R-0012 S18-WBS-R-1613 S18-WBS-R-0044 S18-WBS-R-0045 S18-WBS-R-0049																							
Test Description: This test demonstrates the proper determination of braking force required given varying wheel speed and air speed inputs. This test is performed for each condition defined S18 Brake Force Analysis AAS18-XXX.																								
Test Setup: Figure NN.N depicts the test setup for S18-BSCU-SIT-0068. The airplane simulation (located in the test station) provides simulated wheel speed and airspeed to the BSCU. The test station loops through the various conditions defined in S18 brake force analysis AAS18-XXX and provides the appropriate inputs for each test case. It also records the output of the BSCU to determine if the BSCU matches expected results.																								
Test Steps and Success Criteria: <table border="1" style="width: 100%; border-collapse: collapse; margin-top: 5px;"> <thead> <tr> <th style="width: 10%;">Test Step</th> <th style="width: 40%;">Test Step Description</th> <th style="width: 20%;">Expected Results</th> <th style="width: 20%;">Actual Results</th> <th style="width: 10%;">Pass/Fail</th> </tr> </thead> <tbody> <tr> <td style="text-align: center;">1</td> <td>Wt : XXX,XXX lbs Autobrake mode : AUTO Wheel speed: XX rpm CAS: XX knots Brake temperature: XX degrees F</td> <td>XXX psi HYD 1 Meter Valve actuation</td> <td></td> <td></td> </tr> <tr> <td style="text-align: center;">2</td> <td>Wt : XXX,XXX lbs Autobrake mode : AUTO Wheel speed: YY rpm CAS: XX knots Brake temperature: XX degrees F</td> <td>YYY psi HYD 1 Meter Valve actuation</td> <td></td> <td></td> </tr> <tr> <td style="text-align: center;">3</td> <td></td> <td></td> <td></td> <td></td> </tr> </tbody> </table>					Test Step	Test Step Description	Expected Results	Actual Results	Pass/Fail	1	Wt : XXX,XXX lbs Autobrake mode : AUTO Wheel speed: XX rpm CAS: XX knots Brake temperature: XX degrees F	XXX psi HYD 1 Meter Valve actuation			2	Wt : XXX,XXX lbs Autobrake mode : AUTO Wheel speed: YY rpm CAS: XX knots Brake temperature: XX degrees F	YYY psi HYD 1 Meter Valve actuation			3				
Test Step	Test Step Description	Expected Results	Actual Results	Pass/Fail																				
1	Wt : XXX,XXX lbs Autobrake mode : AUTO Wheel speed: XX rpm CAS: XX knots Brake temperature: XX degrees F	XXX psi HYD 1 Meter Valve actuation																						
2	Wt : XXX,XXX lbs Autobrake mode : AUTO Wheel speed: YY rpm CAS: XX knots Brake temperature: XX degrees F	YYY psi HYD 1 Meter Valve actuation																						
3																								
NOTE: the test station records the inputs from the BSCU and flags a test failure if it does not meet these criteria. The recorded data is stored on the test station for post processing after the test is complete.																								
Test Results:		PASS		FAIL																				
Failure Description:																								
Failure Resolution:																								

Figure E16 - BSCU System Integration Test procedure

E.5.2.1 Brake System Control Unit Open Problem Report

[Figure E17](#) is an example of an open BSCU verification Problem Report description.

PR-BSCU-0010

The system integration test of requirement "S18-BSCU-R-0020: BSCU shall control the valves to charge the emergency accumulator prior to normal operation" failed because the BSCU completed power up BIT, entered the NORMAL mode, and didn't charge the emergency accumulator until the parking brake was released. The intent of the requirement was to charge the emergency accumulator before entering the NORMAL mode.

The BSCU emergency accumulator charging logic was inhibited by the parking brake being set.

The emergency accumulator was being charged before taxiing and takeoff, but not before push back from the gate.

This PR was discussed with the customer, and they agree to delay the implementation of this fix leaving it open for the baseline 1.0 of the BSCU.

Figure E17 - Open BSCU verification Problem Report description (example)

This Open Problem Report, PR-BSCU-0010, was reviewed as part of the SSA process, and it was determined that the safety assessment was not impacted by this Open Problem Report (OPR).

(Editor's Note: As part of the SSA process the safety engineer reviewed all OPRs in order to verify their impact on the safety assessment.)

E.5.3 Wheel Brake System Verification

The WBS verification matrix (excerpt) in [Table E30](#) shows the WBS requirements, their associated system-level functions, the methods to be applied to verify each requirement, references to the verification procedures, verification results, any problem reports generated, and the verification conclusion (pass/fail).

Table E30 - Wheel Brake System Verification Matrix (excerpt)

Requirement Number	Requirement	Associated Function(s)	Verification Method(s)	Verification Procedure/Results	Conclusion (Pass/Fail)
S18-WBS-R-0020	The Wheel Brake System shall have a means to decelerate the wheels on the ground.	Decelerate the wheels on the ground	Test, analysis	See airplane-level verification results	To be verified at the airplane level
S18-WBS-R-0043	The Wheel Brake System shall provide autobraking capability during landing and RTO.	Decelerate the wheels on the ground	Test	System Integration Test: S18-WBS-SIT-0099	Pass
S18-WBS-R-2986	The wheel brake command function of the BSCU shall be developed as FDAL A.	Decelerate the wheels on the ground	Inspection	WBS Accomplishment Summary	Pass
S18-WBS-R-6104	The probability of BSCU failure resulting in loss of a valid braking command output to the NMV shall not exceed 2.0E-04 per flight.	Decelerate the wheels on the ground	Analysis	WBS SSA (FTA)	Pass
S18-WBS-R-6105	The probability of BSCU failure resulting in unannunciated erroneous braking command to the NMV shall not exceed 2.0E-04 per flight.	Decelerate the wheels on the ground	Analysis	WBS SSA (FTA)	Pass
S18-WBS-R-6106	The probability of BSCU failure resulting in the loss of command to open the SOV shall not exceed 2.0E-04 per flight.	Decelerate the wheels on the ground	Analysis	WBS SSA (FTA)	Pass

Requirement Number	Requirement	Associated Function(s)	Verification Method(s)	Verification Procedure/Results	Conclusion (Pass/Fail)
S18-WBS-R-6107	The probability of BSCU failure resulting in unintended closure of the S/ASV shall not exceed 2.0E-04 per flight.	Decelerate the wheels on the ground	Analysis	WBS SSA (FTA)	Pass
S18-WBS-R-6108	The SOV and NMV commands shall be provided by the BSCU upon loss of either airplane electrical power input.	Decelerate the wheels on the ground	Test	System Integration Test: S18-WBS-SIT-0100	Pass
S18-WBS-R-6109	When "HYD 1 enable" output is enabled, then "alt/emerg ctrl" output shall be disabled.	Decelerate the wheels on the ground	Modeling, test	Modeling: Modeling/Simulation Analysis Report XXX System Integration Tests: S18-WBS-SIT-0050 S18-WBS-SIT-0102	Pass Pass
S18-WBS-R-6110	No single failure shall cause erroneous NMV command and inhibit the SOV function.	Decelerate the wheels on the ground	Analysis	WBS SSA (FTA) WBS CMA	Pass
S18-WBS-R-6111	The probability of "Loss of Normal Brake System Hydraulic Components" shall be less than 3.3E-05 per flight.	Decelerate the wheels on the ground	Analysis	WBS SSA (FTA)	Pass
S18-WBS-R-6112	The probability of "Loss of Alternate/Emergency Brake System Hydraulic Components" shall be less than 3.3E-05 per flight.	Decelerate the wheels on the ground	Analysis	WBS SSA (FTA)	Pass

(Editor's Note: The use of an accomplishment summary as verification data is included here to show that the development assurance process for FDAL A as specified in the plans was followed. It is understood that the verification status of the applicable requirement(s) will not be considered complete until the Accomplishment Summary is finalized.)

(Editor's Note: The WBS verification matrix (excerpt) is based on the system-level requirements shown in [E.4.6.7](#). Since the FDAL of the WBS "Decelerate the Wheels on the Ground" function was A per S18-WBS-R-0100, in accordance with main body [5.5.3](#), these requirements were verified with independence.

Requirements verification was preferred by test but included other methods when necessary to show complete verification of a requirement. Monitoring for unintended behavior was also required according to main body [4.6.4](#) and [5.5.5.3](#). Verification of safety requirements, particularly those dealing with failure rates or to show that a particular failure condition was extremely remote, was performed by a safety assessment. Verification of certain system-level requirements was deferred to the airplane level (e.g., S18-WBS-R-0020).)

[Figure E18](#) is an example of a system integration test procedure form.

System Integration Test (SITs) Procedure				
Test Number: S18-WBS-SIT-0099		Date:		
Author: J. Sample				
Test Procedure Name: Autobrake				
Requirements Coverage:	S18-WBS-R-0043			
Test Description: This test demonstrates the proper operation of the autobrake functionality. This functionality automatically engages pressurized wheel braking at the pilot-selected pressure level upon touchdown to the landing surface. The airplane automatically decelerates at the selected level regardless of other factors, such as airplane drag and other deceleration methods (such as deployment of thrust reversers or spoilers)....				
Test Setup: Figure NN.N depicts the test setup for S18-WBS-SIT-0099. The airplane simulation (located in the test station) provides the touchdown indication, the pressure level selected by the pilot, simulated wheel speed and airspeed to the BSCU. The test station receives...				
Test Success Criteria: 1) For each condition listed above, the WBS shall 2) 3) NOTE: The test station records the inputs from the WBS and flags a test failure if it does not meet these criteria. The recorded data is stored on the test station for post processing after the test is complete.				
Test Results:		PASS		FAIL
Failure Description:				
Failure Resolution:				

Figure E18 - Wheel Brake System integration test procedure form

E.5.4 S18 Airplane Verification

When developing the airplane-level wheel brake requirements, the company tried to determine the method of the verification at the time of generating the requirement. This was not possible in all cases, but the verification method was determined prior to the verification activities being performed.

The excerpt of the S18 Airplane verification matrix in [Table E31](#) shows the airplane-level requirements associated with braking, their associated airplane-level functions, the methods to be applied to verify each requirement, references to the verification procedures, verification results, and any problem reports generated, and the verification conclusion (pass/fail).

Table E31 - S18 Airplane Verification Matrix (excerpt)

Requirement Number	Requirement	Associated Function(s)	Verification Method(s)	Verification Procedure/Results	Conclusion (Pass/Fail)
S18-ACFT-R-1000	The S18 airplane shall have a means to decelerate on ground.	Decelerate on Ground	Analysis	AAS18-XX1 analysis report	Pass
S18-ACFT-R-1100	The S18 airplane shall have a means to decelerate the wheels on ground.	Decelerate the Wheels on Ground			
S18-ACFT-R-1110	The S18 airplane shall have pilot-controlled wheel braking capability.	Decelerate the Wheels on Ground			
S18-ACFT-R-1120	The S18 airplane shall have an autobraking capability during landing and RTO.	Decelerate the Wheels on Ground	Analysis	AAS18-XX1 analysis report	Pass
S18-ACFT-R-1130	The S18 airplane shall have anti-skid braking.	Decelerate the Wheels on Ground	Analysis	AAS18-XX1 analysis report	Pass
S18-ACFT-R-1140	The S18 airplane shall provide interface for Wheel Brake System status and annunciations.	Decelerate the Wheels on Ground			
S18-ACFT-R-0181	The S18 airplane shall decelerate the wheels on gear retraction.	Decelerate the Wheels on Gear Retraction			
S18-ACFT-R-0182	The S18 airplane shall be capable of decelerating the wheels differentially.	Decelerate the Wheels Differentially for Directional Control			
S18-ACFT-R-0183	The S18 airplane shall have a parking brake to prevent airplane motion when parked.	Control Airplane Energy	Analysis	AAS18-XX1 analysis report	Pass
S18-ACFT-R-0184	The S18 airplane shall have hydraulically driven braking.	Decelerate the Wheels on Ground	Analysis	AAS18-XX2 analysis report	Pass
S18-ACFT-R-0185	The S18 airplane shall have an autobrake override that can be initiated by the flight crew.	Decelerate the Wheels on Ground	Test	TTS18-425 test procedure TTS18-426 test result	Pass
S18-ACFT-R-0186	The S18 airplane shall meet safety requirements while operating in an average atmospheric radiation environment per the IEC 62396 standard with an altitude of 40000 feet and a latitude of 45 degrees.	Protect against Single Event Effects (SEE)	Analysis	AAS18-5552 analysis report	Pass
S18-ACFT-R-0933	The Wheel Brake System decelerate the wheels on the ground function shall be developed as FDAL A.	Decelerate the Wheels on Ground	Analysis	S18 Airplane Accomplishment Summary	
S18-ACFT-R-1322	No single failure or event shall result in the complete loss of wheel brake and the loss of one thrust reverser.	Decelerate on Ground	Analysis	ASA	
S18-ACFT-R-1385	Complete loss of wheel braking shall be less than 1.0E-07 for a landing.	Decelerate the Wheels on Ground	Analysis	ASA	
S18-ACFT-R-1550	Loss of power from both hydraulic subsystems powered by the engines shall not lead to complete loss of wheel braking.	Decelerate on Ground	Analysis	ASA	
S18-ACFT-R-1551	Wheel Brake System shall include an emergency accumulator to supply hydraulic power to the wheel brakes.	Decelerate on Ground	Analysis	ASA	
S18-ACFT-R-1552	The Alternate/Emergency Brake System hydraulic equipment and piping shall be installed aft of the engine 1 UERF trajectory envelope.	Decelerate on Ground	Analysis	ASA	
S18-ACFT-R-1600	Two redundant control lanes shall be provided between the Electric Brake Unit (EBU) and each of the two Alternate/Emergency Meter Valves.	Decelerate on Ground	Analysis	ASA	

(Editor's Note: For this example, there were no Open Problem Reports at the airplane level.)

(Editor's Note: The verification matrix was based on the airplane-level requirements shown in [E.3.12.1](#). Since the FDAL of the airplane's "Decelerate on Ground" function was A per S18-ACFT-R-0933, in accordance with main body [5.5.3](#), these requirements were verified with independence. It is understood that the verification status of the applicable requirement(s) will not be considered complete until the Accomplishment Summary is finalized.

Requirements verification was preferred by test but included other methods when necessary to show complete verification of a requirement. Monitoring for unintended behavior was also required according to main body [4.6.4](#) and [5.5.5.3](#). Verification of safety requirements, particularly those dealing with failure rates or to show that a particular failure condition was extremely remote, was accomplished by a safety assessment.)

[Figure E19](#) is an example of an airplane-level test procedure form.

Airplane Test Procedure				
Test Number: AAS18-Test-007		Date:		
Author: S. Example				
Test Procedure Name: Maximum Weight Landing				
Requirements Coverage:	S18-ACFT-R-1000			
Test Description: This test demonstrates the proper maximum weight airplane deceleration rates and landing distances.				
Test Setup: Figure NN.N depicts the test setup for AAS18-Test-007.				
Test Success Criteria 1) For each condition listed above, the airplane shall 2) 3) NOTE: The test station records the inputs from the airplane and flags a test failure if it does not meet these criteria. The recorded data is stored on the test station for post processing after the test is complete.				
Test Results:		PASS		FAIL
Failure Description:				
Failure Resolution:				

Figure E19 - Airplane-level test procedure form

E.6 WHEEL BRAKE SYSTEM CONFIGURATION MANAGEMENT

E.6.1 Configuration Identification and Baseline Establishment

The WBS itself, along with its associated data items, was at the top level of configuration. [Tables E32](#) and [E33](#) show a list of the WBS and BSCU deliverables for configuration baselines. A configuration baseline was a snapshot of the system configuration index consisting of versions of each configuration item used to support milestones, design reviews and process assurance audits.

In the configuration indices shown in [Tables E32](#) and [E33](#), System Control category represented the applicable level of configuration control. SC1 meant that the version of the document was released or locked and could only be changed with a problem report and accepted after a review of the change. SC1 also meant that all versions were required to be archived in an environment that allowed for versions to be recovered. SC2 meant that the versions could roll without any authoritative reviews or problem reporting activities, as long as previous versions could be recovered or restored if needed.

Problem reports listed on the Configuration Index represented issues that were still in work. They were identified to get the complete scope of the deliverable and its potential changes. Problem reports listed on the configuration index let auditors know that there was an issue with the deliverable. When there was an issue with the deliverable that the auditor identified, then the audit created a problem report.

Table E32 - Wheel Brake System Configuration Index WBS-0001-001 - Version 1.0

Configuration Identification Number	CI Identification Number	Version	Associated Configuration Item Names	System Control Category	Open PRs
WBS-0001-001 Version 1.0	WBS-0001-101	9.0	Diagram—Interface - WBS	SC1	
	WBS-0001-102	11.0	BOM - WBS	SC1	
	WBS-0001-103	5.0	Wheel Brake System Requirements Specification	SC1	PR-WBS-0007 PR-WBS-0008
	WBS-0001-104	5.0	WBS System Design Document (SDD)	SC1	
	WBS-0001-105	3.0	WBS Development Assurance Plan	SC2	
	WBS-0001-110	2.0	PSSA	SC1	
	WBS-0001-111	2.0	SSA	SC1	
	WBS-0001-112	2.0	SFHA	SC1	
	WBS-0001-113	1.0	Verification matrix	SC2	
	WBS-0001-114	2.0	Validation matrix	SC2	
	WBS-1000-001	4.0	BSCU COM (hardware) Configuration Index	SC1	
	WBS-1010-001	4.0	BSCU MON (hardware) Configuration Index	SC1	
	WBS-1100-001	4.0	BSCU COM (software) Configuration Index	SC1	
	WBS-1110-001	4.0	BSCU MON (software) Configuration Index	SC1	
	WBS-2000-001	3.0	Shutoff Valve Configuration Index	SC1	
	WBS-2100-001	3.0	Selector Valve Configuration Index	SC1	
	WBS-2200-001	3.0	Normal Meter Valve Configuration Index	SC1	
	WBS-2300-001	3.0	Alternate/Emergency Meter Valve Configuration Index	SC1	
	WBS-2400-001	3.0	Shutoff/Anti-Skid Valve Configuration Index	SC1	
	WBS-3000-001	3.0	Brake Pedal Position Sensor Configuration Index	SC1	
	WBS-4000-001	3.0	Emergency Accumulator Configuration Index	SC1	

Configuration ID numbers called out in the 1st column of [Table E33](#) were considered part of the configuration index captured in the higher-level matrix, [Table E32](#).

Table E33 - BSCU COM Configuration Index - Version 4.0

Configuration Identification Number	CI Identification Number	Version	Associated Data Item Names	System Control Category	Open PRs
WBS-1000-001 Version 4.0	WBS-1000-101	4.0	Drawing—BSCU COM	SC1	
	WBS-1000-102	4.0	BOM—BSCU COM	SC1	
	WBS-1000-103	4.0	BSCU COM Plan for Hardware Aspects of Certification (PHAC)	SC1	
	WBS-1000-104	4.0	BSCU COM Hardware Requirements Document (HRD)	SC1	
	WBS-1000-105	4.0	BSCU COM Hardware Design Document (HDD)	SC2	
	WBS-1000-106	4.0	BSCU COM Hardware Verification Plan (HVP)	SC2	
	WBS-1000-107	4.0	BSCU COM Hardware Verification Results (HVR)	SC2	
	WBS-1000-108	4.0	BSCU COM Hardware Accomplishment Summary (HAS)	SC1	
WBS-1010-001 Version 4.0	WBS-1010-101	4.0	Drawing—BSCU MON	SC1	
	WBS-1010-102	4.0	BOM—BSCU MON	SC1	
	WBS-1010-103	4.0	BSCU MON Plan for Hardware Aspects of Certification (PHAC)	SC1	
	WBS-1010-104	4.0	BSCU MON Hardware Requirements Document (HRD)	SC1	
	WBS-1010-105	4.0	BSCU MON Hardware Design Document (HDD)	SC2	
	WBS-1010-106	4.0	BSCU MON Hardware Verification Plan (HVP)	SC2	
	WBS-1010-107	4.0	BSCU MON Hardware Verification Results (HVR)	SC2	
	WBS-1010-108	4.0	BSCU MON Hardware Accomplishment Summary (HAS)	SC1	
WBS-1100-001 Version 4.0	WBS-1100-101	4.0	BSCU COM Plan for Software Aspects of Certification (PSAC)	SC1	
	WBS-1100-102	4.0	BSCU COM Software Requirements Document (SRD)	SC1	
	WBS-1100-103	4.0	BSCU COM Software Design Document (SDD)	SC1	
	WBS-1100-104	4.0	BSCU COM source code	SC1	
	WBS-1100-105	4.0	BSCU COM executable object code	SC1	
	WBS-1100-106	4.0	BSCU COM Software Verification Plan (SVP)	SC2	
	WBS-1100-107	4.0	BSCU COM Software Verification Results (SVR)	SC2	
	WBS-1100-108	4.0	BSCU COM Software Accomplishment Summary (SAS)	SC1	
WBS-1110-001 Version 4.0	WBS-1110-101	4.0	BSCU MON Plan for Software Aspects of Certification (PSAC)	SC1	
	WBS-1110-102	4.0	BSCU MON Software Requirements Document (SRD)	SC1	
	WBS-1110-103	4.0	BSCU MON Software Design Document (SDD)	SC1	
	WBS-1110-104	4.0	BSCU MON source code	SC1	
	WBS-1110-105	4.0	BSCU MON executable object code	SC1	
	WBS-1110-106	4.0	BSCU MON Software Verification Plan (SVP)	SC2	
	WBS-1110-107	4.0	BSCU MON Software Verification Results (SVR)	SC2	
	WBS-1110-108	4.0	BSCU MON Software Accomplishment Summary (SAS)	SC1	
WBS-2000-001 Version 3.0	WBS-2000-101	3.0	Drawing—Shutoff Valve	SC1	
	WBS-2000-102	3.0	BOM—Shutoff Valve	SC1	
WBS-2100-001 Version 3.0	WBS-2100-101	3.0	Drawing—Selector Valve	SC1	
	WBS-2100-102	3.0	BOM—Selector Valve	SC1	
WBS-2200-001 Version 3.0	WBS-2200-101	3.0	Drawing—Normal Meter Valve	SC1	
	WBS-2200-102	3.0	BOM—Normal Meter Valve	SC1	
WBS-2300-001 Version 3.0	WBS-2300-101	3.0	Drawing—Alternate/Emergency Meter Valve	SC1	
	WBS-2300-102	3.0	BOM—Alternate/Emergency Meter Valve	SC1	
WBS-2400-001 Version 3.0	WBS-2400-101	3.0	Drawing—Shutoff/Anti-Skid Valve	SC1	
	WBS-2400-102	3.0	BOM—Shutoff/Anti-Skid Valve	SC1	
WBS-3000-001 Version 3.0	WBS-3000-101	3.0	Drawing—Sensor —Position	SC1	
	WBS-3000-102	3.0	BOM—Sensor—Position	SC1	
WBS-4000-001 Version 3.0	WBS-4000-101	3.0	Drawing—Emergency Accumulator—Hydraulic	SC1	
	WBS-4000-102	3.0	BOM—Emergency Accumulator—Hydraulic	SC1	

E.6.2 Change Control and Problem Reporting

Problem report PR-WBS-0007 shown in [Figure E15](#) illustrates the WBS change control and problem reporting procedures. This problem report was written against WBS requirement S18-WBS-R-0065. As a result of the recommendation of this problem report, a change notice was raised against the baseline 1.0 of WBS-0001-103, the WBS Requirements Specification. This and other changes resulting from the requirements validation process were incorporated into a minor baseline 1.1 of WBS-0001-103. The example engineering change notice (ECN) is shown in [Figure E20](#).

ECN #: 12345-67		Class: I		Date: 03-DEC-2008	
Entered by: JSMITH					
Eng Resp: 01		Urgent: N		Program: 12345	
Released by: JSMITH					
Customer Approval Req'd: Y				Cust. Approval Date: 03-DEC-2008	
Ref	Op Code	Item Number	E/C Rev	QTY	Requested by
1	100	WBS-0001-103 REV 5.0	New	1	JSMITH
Description: Modify requirement S18-WBS-R-0065. From: "The emergency accumulator shall capable of providing 2000 psi." To: "The emergency accumulator shall provide a minimum 1800 psi."					
Reason: The requirement has been adjusted based on additional information provided by the airframer.					
Substantiation: See Problem Report PR-WBS-0007.					

Figure E20 - Wheel Brake System engineering change notice

E.7 PROCESS ASSURANCE

The first objective for process assurance was to ensure the necessary plan or plans were developed and maintained for all aspects of airplane, system, and item development. This means that once the plans were created and mature, they were subject to review to ensure there was not a lack of key process aspects and plan content, and that the plans available were in a controlled environment as designated by the company.

The second objective was to ensure development activities and processes were conducted in accordance with the plans identified in the first objective. This means that process assurance included review and audit of the evidence from the output of the activities described in the planning sections of the Development Assurance Plan. This evidence was proof that the activities were performed according to the plan.

(Editor's Note: The next sections contain an excerpt from the process assurance section and portions of the configuration management section of the Development Assurance Plan used during the process assurance audit, and the audit of evidence (WBS tables and figures) from this appendix. This section also reflects what the next steps could be when the plan allowed the program to progress after a finding or an issue. The examples used in [E.7.1](#), [E.7.2](#), and [E.7.3](#) are intended to give the reader an example of what process assurance may look like. See main body [5.7](#) for additional process assurance guidance.)

E.7.1 S18 Process Assurance Section of the Development Assurance Plan

The process assurance section of the Development Assurance Plan described how and when process assurance was to be conducted in relation to development activities and transition activities. The plan included how an auditor was supposed to respond to discrepancies and gaps between the plan and the evidence, and how auditor responses were dispositioned for program need.

(Editor's Note: Some plans offer a way to carry findings, issues, and action items over to other points of a program when process assurance occurs again, and some plans require that all findings, issues, and action items be closed before proceeding to the next phase of a program. Steps and procedures for closure and disposition of auditor responses are entirely up to the program and the company and will not be discussed in detail in this section.)

The following is an excerpt from the process assurance section of the Development Assurance Plan:

- a. At a point after the Preliminary Design Review, a process assurance review is conducted against the configuration management plan.
- b. Issues, Findings, and Action Items responses will be given by the auditor.
 - Issue: An issue is given when the evidence is insufficient to meet the intent of the activity.
 - Finding: A finding is an issue that has been determined to be a result of a process violation.
 - Action Item: An action item is assigned when a finding or an issue is given. An action item explains what needs to happen to mitigate the finding or issue.
- c. Audit sampling is when an auditor reviews a certain percentage or a certain amount of evidence collected as determined by the auditor or the plan.

E.7.2 Process Assurance Audit of Configuration Management Activities

During process assurance activities, auditors reviewed aspects of the Development Assurance Plan to garner evidence. For this example, [Figure E21](#) is a description of how changes to requirements were managed per the configuration management section of the Development Assurance Plan for the WBS. Section G.2.2.2 of the Development Assurance Plan shown in [Figure E21](#) outlines the configuration management team's activities.

G.2.2.2 Configuration Management (CM)

The braking system Configuration Management (CM) team will be responsible for:

- Maintaining a traceability matrix of all customer-approved requirements
- Oversight of the requirements change control process
- Applying changes to the requirements matrix
- Maintaining the modification history of requirements

Figure E21 - Requirements change management

Process assurance auditors reviewed the plan's details and audited the evidence to ensure that the configuration management team acted accordingly and produced artifacts that show:

- a. Evidence of a trace matrix.
- b. Evidence of a change control process.
- c. Evidence that the trace matrix was approved by customers when requirement changes were made to the matrix.
- d. Evidence of matrix changes were version controlled.
- e. Evidence that all requirement changes were intentional.
- f. Evidence of version control and historical data for requirements.

E.7.3 Configuration Management Audit Results

The auditor looked for the following evidence based on the planned activities in [Figure E21](#) and found the annotated results:

- a. Evidence of a trace matrix—**Sufficient** (see [Table E19](#)).
- b. Evidence of a change control process—**Sufficient** (see [Figure E20](#)).
- c. Evidence that the trace matrix was approved by customers when requirement changes were made to the matrix—**Insufficient**.
 - Evidence that trace matrix was approved by customers at each version is insufficient. The auditor reviewed the trace matrix and determined that there were changes made as stated in Coord Memo S18a-CM00XX. After the change was implemented, there was no approval for the trace matrix by the customer.
 - The auditor determined this is a finding and wrote an action item for the team to review the current trace matrix with the customer.
- d. Evidence of matrix changes were version controlled—**Insufficient**.
 - Evidence of matrix changes and version control has been found insufficient. The historical records for each version to the matrix is missing. No historical data available to compare the recent version of the matrix to the previous versions. This is a process violation of the Development Assurance Plan requiring archival data to be retrievable.
 - If there was evidence of changes and matrix versions available, the auditor would compare the historical version and the current version and check to see if each change had a problem report that allowed the change to occur. Another option would be for the auditor to ask for a comparison and the list of changes. While reviewing the comparison, the auditor can sample the list of changes and review the problem reports associated with each change.
 - The auditor determined this is a finding and wrote an action item for the team to recover past versions of the matrix and archive them. If no past versions are available, the action is to review every problem report applied to the trace matrix and only allow those updates to be shown on the trace matrix.
- e. Evidence that all requirement changes were intentional—**Insufficient**.
 - Evidence that all requirement changes were intentional has been found insufficient. There are problem reports for some requirement changes, but after review of the changes, there are requirement changes that do not have an associated problem report. Some of these changes may be unintended. Change control was not established for every requirement.
 - The auditor determined the lack of completeness in documenting all changes in the problem reports as an issue and wrote an action item for the team to update the OPRs to include all changes to the given requirement set.

f. Evidence of version control and historical data for requirements—**Insufficient**.

- Version control and historical data for requirements has been found to be insufficient. Version control was not discovered for all requirements and their changes. While problem reports were discovered for requirements changes, the problem reports did not identify the requirement version or the version of the document in which the change was implemented. The problem reports did not identify if the problem report is versioning the requirement or removing and replacing the requirement. See PR-WBS-0007, [Figure E15](#).
- The auditor determined that the lack of version control was an issue and wrote an action item for the team to update the OPRs to determine versions of requirements or versions of documents containing the requirements.